UPSC ISS · STATISTICS PAPER-I · OBJECTIVE

Computer Application and Data Processing

Every previous-year question from 2018 to 2026, with answer, exam shortcut, tips & tricks and a step-by-step solution.

Questions in this volume	**180** of 720 across the four units
Papers covered	9 — 2018 to 2026
Per paper	2018: 20 · 2019: 20 · 2020: 20 · 2021: 20 · 2022: 20 · 2023: 20 · 2024: 20 · 2025: 20 · 2026: 20
Syllabus topics	10
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Layout of every entry	QUESTION → OPTIONS → ANSWER → EXAM SHORTCUT → TIPS & TRICKS → STEP-BY-STEP SOLUTION
Dataset version	1.0

What is authentic and what is not. The questions and their four options are reproduced word for word from the original UPSC ISS Statistics Paper-I booklets. Wording, option text, option order and question numbers are unaltered.

None of the source booklets carried an official answer key. Every ANSWER, EXAM SHORTCUT, TIPS & TRICKS note and STEP-BY-STEP SOLUTION in this volume was worked out for this project and is marked *AI-derived explanation* where it appears. They are a study aid, not an official UPSC solution.

Source defects. 3 question(s) in this volume have a defect in the printed paper. None has been silently corrected: the original text is reproduced exactly and the problem is described under the question. 1 of them has no valid option at all and is therefore left unkeyed rather than guessed.

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C1 · COMPUTER APPLICATION AND DATA PROCESSING

Computer Organisation: CPU, ALU, Control & Memory Units, I/O Units

11 questions · 2018: 1 2019: 1 2020: 0 2021: 0 2022: 3 2023: 2 2024: 2 2025: 1 2026: 1

Weight. 11 questions (6.1% of Computer Application and Data Processing) · appeared in 7 of 9 papers · peak year 2022 (3)

Examiner's pattern. Fetch-decode-execute and the CPU's internal division of labour. Recurring: which unit does what (ALU vs control unit), the main operations of the CPU, BIOS functions, the machine cycle phases, the system bus, and RISC vs CISC.

Must-know. Machine cycle = fetch, decode, execute, store. ALU performs arithmetic AND logic (AND/OR live here). The control unit directs but does not compute. BIOS runs POST and boots the machine.

1. 2018 · Q61

Units of a Computer System: CPU, ALU, Control, I/O · Conceptual

QUESTION

Consider the following statements: 1. ALU is a component of the Control Unit. 2. CPU is a component of the Control Unit. 3. The Control Unit is not a component of the ALU. 4. ALU is a component of the CPU.

OPTIONS

(a) 4 only

(b) 3 and 4 only

(c) 1 and 3 only

(d) 2, 3 and 4

ANSWER (AI-derived — no official key exists for this paper)

(b) 3 and 4 only

EXAM SHORTCUT (AI-derived explanation)

One hierarchy settles everything: CPU contains BOTH the ALU and the Control Unit as siblings. So statements 1 and 2 invert the hierarchy and fail; 3 and 4 are correct.

TIPS & TRICKS (AI-derived explanation)

- Draw the box: CPU = Control Unit + ALU + registers. The ALU and CU are peers inside the CPU; neither contains the other.
- Statements 1 and 2 are both 'containment reversed' traps. Whenever an option asserts that a bigger unit sits inside a smaller one, reject it on sight.
- Statement 3 is phrased negatively ('is NOT a component of'), which makes it TRUE. Read negations carefully - they flip the truth value of an otherwise false-looking claim.
- The CU fetches and decodes instructions and issues control signals; the ALU performs arithmetic and logic. Keeping those two job descriptions apart answers most questions in this sub-topic.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The Central Processing Unit (CPU) consists of the Control Unit (CU), the Arithmetic and Logic Unit (ALU) and a set of registers.
2. Statement 1: 'ALU is a component of the Control Unit' - false; the ALU and CU are separate peer units within the CPU.
3. Statement 2: 'CPU is a component of the Control Unit' - false; the containment is the other way round, the CU is a component of the CPU.
4. Statement 3: 'The Control Unit is not a component of the ALU' - true, since the CU sits alongside the ALU, not inside it.
5. Statement 4: 'ALU is a component of the CPU' - true by the standard architecture.
6. Only statements 3 and 4 are correct, so option (b) is right.

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2. 2019 · Q41 Units of a Computer System: CPU, ALU, Control, I/O · Definition

QUESTION

A bus that manages the transfer of data and addresses among various components is

OPTIONS

(a) address bus

(b) control bus

(c) data bus

(d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(b) control bus

EXAM SHORTCUT (AI-derived explanation)

The address bus CARRIES addresses and the data bus CARRIES data; it is the control bus that issues the read/write and timing signals which MANAGE those transfers.

TIPS & TRICKS (AI-derived explanation)

- Split the system bus into its three jobs: address bus = where, data bus = what, control bus = when and how. The verb in the question tells you which one is wanted.
- The word 'manages' is the discriminator here. A bus that merely carries something is not managing it - management means control signals.
- The address bus is unidirectional (CPU to memory) while the data bus is bidirectional; the control bus carries individual signals in both directions.
- Bus widths matter elsewhere: an n-line address bus addresses 2^n locations, and the data-bus width equals the word size.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A computer's system bus has three functional parts.

2. The address bus carries the address of the memory location or I/O port being accessed; it transports addresses but does not manage anything.
3. The data bus carries the actual data being transferred between components.
4. The control bus carries the control and timing signals - memory read, memory write, I/O read, I/O write, bus request, bus grant, interrupt acknowledge - which coordinate and manage the transfer of data and addresses among all components.
5. Since the question asks which bus MANAGES the transfers, the answer is the control bus.
6. Option (b) is correct.

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3. 2022 · Q31 Memory - RAM, ROM & Units of Computer Memory · Factual

QUESTION

Which one of the following registers keeps track of the next instruction to be executed?

OPTIONS

(a) Program counter

(b) Instruction register

(c) Accumulator RAM

(d) Memory Address Register

ANSWER (AI-derived — no official key exists for this paper)

(a) Program counter

EXAM SHORTCUT (AI-derived explanation)

The program counter holds the ADDRESS of the next instruction to be executed and is incremented after each fetch.

TIPS & TRICKS (AI-derived explanation)

- Register roles: PC holds the address of the NEXT instruction; IR holds the CURRENT instruction being decoded; MAR holds the address being accessed on the bus; accumulator holds arithmetic results.
- The PC is what makes sequential execution automatic - incrementing it is part of every fetch cycle, and a jump instruction simply overwrites it.
- Option (b) instruction register is the classic near-miss: it stores the instruction itself, not the address of the next one.
- MAR (option d) is used for every memory access, whether for instructions or data, so it does not specifically track program flow.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. During the fetch-decode-execute cycle the processor must always know where the next instruction lies.
2. The Program Counter (PC), also called the instruction pointer, holds the memory address of the next instruction to be fetched, and is automatically incremented once the current instruction has been fetched.

3. The Instruction Register (IR) holds the instruction currently being decoded and executed, not the address of the next one.
4. The Memory Address Register (MAR) holds the address of whichever memory location is currently being read or written.
5. The accumulator holds operands and results of arithmetic and logic operations.
6. Hence the program counter tracks the next instruction, option (a).

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4. 2022 · Q73 Memory - RAM, ROM & Units of Computer Memory · Conceptual

QUESTION

By using which units is data processing performed by a computer system? 1. Monitor, 2. Memory unit, 3. CPU

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(b) 2 and 3 only

EXAM SHORTCUT (AI-derived explanation)

Processing is done by the CPU using data held in the memory unit. The monitor is an OUTPUT device and takes no part in processing.

TIPS & TRICKS (AI-derived explanation)

- Functional units: input, memory, ALU, control, output. Processing is the ALU and control unit (i.e. the CPU) acting on data in memory.
- The monitor merely DISPLAYS results after processing, so it belongs to the output stage - any option containing item 1 is out.
- Data must be loaded into main memory before the CPU can operate on it, which is why the memory unit is genuinely part of processing rather than mere storage.
- Input-process-output is the standard model: keyboard/mouse in, CPU plus memory processing, monitor/printer out.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A computer system is organised into input units, a memory unit, a central processing unit (comprising the ALU and control unit) and output units.
2. Item 3, the CPU: performs all arithmetic and logical operations and controls the sequence of execution - the processing element itself.
3. Item 2, the memory unit: holds the program and the data being operated on; the CPU cannot process data that is not in memory, so it is an essential part of the processing stage.

4. Item 1, the monitor: is an output device that displays results once processing is complete, and plays no part in the processing itself.
5. Hence data processing is performed by the memory unit and the CPU.
6. Option (b) is correct.

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5. 2022 · Q74 Low & High Level Languages, Compiler, Assembler · Conceptual

QUESTION

Consider statements on machine cycle: 1. It consists of four phases — fetching, decoding, executing, storing. 2. It is a cycle in which machine language instructions are executed. 3. A program of 6 machine language instructions runs in 6 separate machine cycles.

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

The machine cycle has the four phases fetch, decode, execute and store; it is the cycle in which machine instructions are executed; and each instruction consumes its own cycle, so six instructions take six cycles. All three statements hold.

TIPS & TRICKS (AI-derived explanation)

- Four phases to memorise: FETCH the instruction, DECODE it, EXECUTE it, STORE the result. Some texts merge the last two, but four is the standard exam answer.
- One instruction per machine cycle is the defining relationship - hence a program of n instructions requires n machine cycles in a simple non-pipelined processor.
- Distinguish the machine cycle from the CLOCK cycle: one machine cycle spans several clock cycles, which is why clock speed alone does not determine instruction throughput.
- When all three statements restate parts of the same standard definition, 'all of them' is the natural key.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: the machine cycle is conventionally described as having four phases - fetching the instruction from memory, decoding it in the control unit, executing it in the ALU, and storing the result. TRUE.
2. Statement 2: the machine cycle is precisely the repeating sequence by which the processor carries out machine-language instructions one after another. TRUE.

3. Statement 3: in a simple non-pipelined processor each instruction occupies one complete machine cycle, so a program of 6 machine-language instructions requires 6 machine cycles. TRUE.
4. Note that one machine cycle spans several clock cycles; the two must not be confused.
5. All three statements are correct.
6. Hence option (d) is correct.

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6. **2023 · Q61** Variables, Control Structures, Arrays, Functions, Loops · Factual

QUESTION

Consider the following in respect of two kinds of microprocessors CISC (Complex Instruction Set Computer) and RISC (Reduced Instruction Set Computer): 1. The instructions of CISC are of variable length. 2. RISC supports parallel processing. 3. ARM is based on RISC. Which of the statements given above are correct?

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER **(AI-derived — no official key exists for this paper)**

(d) 1, 2 and 3

EXAM SHORTCUT **(AI-derived explanation)**

CISC instructions are indeed variable length, RISC's simple fixed-length instructions are what make pipelining and parallel execution practical, and ARM is the best-known RISC architecture. All three hold.

TIPS & TRICKS **(AI-derived explanation)**

- Core contrast: CISC has many complex, VARIABLE-length instructions taking several cycles; RISC has few simple, FIXED-length instructions executing in one cycle.
- Fixed-length instructions are precisely what make decoding uniform and pipelining efficient, so statement 2 follows from the RISC design philosophy.
- Named examples worth knowing: RISC - ARM, MIPS, SPARC, PowerPC, RISC-V; CISC - Intel x86, Motorola 68000.
- ARM chips dominate mobile devices because the RISC design gives high performance per watt - a real-world anchor for statement 3.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Statement 1: CISC architectures provide a large set of complex instructions of differing formats and lengths, so instruction length is variable. TRUE.
2. Statement 2: RISC uses a small set of simple, fixed-length instructions with uniform decoding, which makes deep pipelining and superscalar (parallel) execution practical. TRUE.

3. Statement 3: ARM - originally the Acorn RISC Machine - is the most widely deployed RISC architecture, used in the great majority of mobile devices. TRUE.
4. All three statements describe standard properties of the two architectures.
5. Hence option (d) is correct.

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7. **2023 · Q76** Units of a Computer System: CPU, ALU, Control, I/O · Conceptual

QUESTION

RISC processors are always preferred to CISC processors because of: 1. Compact size 2. Small instruction set 3. Executing multiple clock cycles Which of the above are correct?

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(a) 1 and 2 only

EXAM SHORTCUT (AI-derived explanation)

RISC's advantages are its compact, simple design and its small instruction set. Executing in MULTIPLE clock cycles is a CISC trait, not a RISC advantage - so statement 3 fails.

TIPS & TRICKS (AI-derived explanation)

- RISC executes most instructions in a SINGLE clock cycle; multi-cycle instructions are the CISC characteristic. Statement 3 attributes the wrong property.
- Fewer and simpler instructions mean less decoding logic, hence a smaller die area and lower power - which is statement 1.
- The simplicity also frees chip area for more registers and deeper pipelines, the real source of RISC's speed.
- Read such statements for the direction of the claim: 'executing multiple clock cycles' would be a DISADVANTAGE even if true, so it cannot be a reason for preference.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: because RISC needs far less instruction-decoding and microcode logic, the processor core is simpler and physically more compact, with lower power consumption. A genuine advantage. TRUE.
2. Statement 2: RISC is defined by its small set of simple, fixed-format instructions, which simplifies the hardware and enables efficient pipelining. TRUE.
3. Statement 3: RISC is designed so that most instructions complete in a SINGLE clock cycle; it is CISC whose complex instructions take several cycles.

4. Attributing multi-cycle execution to RISC therefore reverses the distinction, and in any case would be a drawback rather than a reason for preference. FALSE.
5. Only statements 1 and 2 are correct.
6. Hence option (a) is correct.

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8. 2024 · Q49 Memory - RAM, ROM & Units of Computer Memory · Conceptual

QUESTION

Which of the following are main operations of CPU? I. Fetching instructions from the memory II. Decoding the instructions III. Executing the instructions IV. Storing the results back in the memory Select the correct answer using the code given below:

OPTIONS

(a) I, III and IV only

(b) I, II and IV only

(c) II and III only

(d) I, II, III and IV

ANSWER (AI-derived — no official key exists for this paper)

(d) I, II, III and IV

EXAM SHORTCUT (AI-derived explanation)

Fetch, decode, execute and store are the four phases of the machine cycle, so all four listed operations are CPU operations.

TIPS & TRICKS (AI-derived explanation)

- The machine cycle is fetch - decode - execute - store; memorising these four words answers a family of questions.
- The control unit handles fetching and decoding while the ALU performs execution, but both are parts of the CPU, so all four remain CPU operations.
- Storing the result back to memory is genuinely part of the cycle (the write-back phase), so item IV must be included.
- This repeats $2022Q'_{74}s$ content in list form - the machine cycle is clearly a recurring theme.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The CPU repeatedly executes the machine cycle, which consists of four phases.
2. Item I, fetch: the control unit uses the program counter to read the next instruction from memory into the instruction register.
3. Item II, decode: the control unit interprets the instruction and determines the operation and operands required.
4. Item III, execute: the arithmetic and logic unit performs the required arithmetic, logical or data-movement operation.
5. Item IV, store: the result is written back to a register or to memory, completing the cycle.

6. All four are main operations of the CPU, option (d).

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9. 2024 · Q51 Operations of a Computer & Basics · Conceptual

QUESTION

What are the functions of BIOS in computer system? I. It performs power-on self-test. II. It boots the computer system. III. It provides hardware-independent access to the physical devices. Select the correct answer using the code given below:

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(d) I, II and III

EXAM SHORTCUT (AI-derived explanation)

The BIOS runs the power-on self-test, bootstraps the operating system, and supplies a standard low-level interface to the hardware - all three functions are genuine.

TIPS & TRICKS (AI-derived explanation)

- BIOS duties in order of execution: POST (test the hardware), then bootstrap (find and load the OS), then provide run-time hardware services.
- Statement III captures the abstraction role: BIOS routines let software access disks, keyboard and display without knowing the specific hardware details.
- BIOS resides in ROM or flash so that it is available before any disk has been read - a key reason it can bootstrap the machine.
- Modern machines use UEFI, the successor to BIOS, performing the same three functions with a richer interface.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. BIOS (Basic Input Output System) is firmware held in ROM or flash memory on the motherboard, available as soon as power is applied.
2. Statement I: on power-up the BIOS runs the Power-On Self-Test, checking the processor, memory and essential peripherals before anything else happens. TRUE.
3. Statement II: it then locates the boot device, loads the bootstrap loader and hands control to the operating system. TRUE.
4. Statement III: the BIOS provides a standard set of low-level service routines through which software can access disks, the keyboard and the display without needing to know the specific hardware details - a hardware-independent access layer. TRUE.
5. All three are genuine BIOS functions.

6. Hence option (d) is correct.

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10. 2025 · Q80 Memory - RAM, ROM & Units of Computer Memory · Factual

QUESTION

AND and OR operations are part of which unit of the CPU?

OPTIONS

(a) Arithmetic Unit

(b) Logic Unit

(c) Control Unit

(d) Main Memory Unit

ANSWER (AI-derived — no official key exists for this paper)

(b) Logic Unit

EXAM SHORTCUT (AI-derived explanation)

AND and OR are logical, not arithmetic, operations - so they belong to the logic unit of the ALU. Option (b).

TIPS & TRICKS (AI-derived explanation)

- The ALU has two functional halves: the arithmetic unit (add, subtract, multiply, divide, increment) and the logic unit (AND, OR, NOT, XOR, shifts, comparisons).
- The control unit fetches, decodes and sequences instructions; it does not perform data operations.
- Main memory stores data and instructions; it performs no operations at all.
- Comparison instructions are also logic-unit work, since they are implemented by subtraction followed by flag testing.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The CPU consists of the arithmetic and logic unit (ALU), the control unit, and registers, with main memory external to it.
2. Within the ALU, the arithmetic unit performs addition, subtraction, multiplication and division, while the logic unit performs Boolean operations such as AND, OR, NOT and XOR, along with shifts and comparisons.
3. AND and OR are Boolean operations, hence they are carried out by the logic unit.
4. The control unit (option c) only decodes instructions and generates control signals; the main memory unit (option d) only stores data.
5. Hence option (b) is correct.

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11. 2026 · Q80 Memory - RAM, ROM & Units of Computer Memory · Factual

QUESTION

Which is used for storing results produced by the arithmetic and logic unit?

OPTIONS

(a) Program counter

(b) Accumulator

(c) Memory address register

(d) Instruction register

ANSWER (AI-derived — no official key exists for this paper)

(b) Accumulator

EXAM SHORTCUT (AI-derived explanation)

The accumulator is the register that holds the result of ALU operations. Answer (b).

TIPS & TRICKS (AI-derived explanation)

- Register roles: PC holds the address of the next instruction; IR holds the current instruction; MAR holds the address being accessed; MBR/MDR holds the data in transit; accumulator holds ALU operands and results.
- In an accumulator-based machine, one operand and the result of every arithmetic operation implicitly use this register - hence the name.
- Mnemonic: the accumulator 'accumulates' successive results, as in repeated addition.
- Modern RISC processors replace the single accumulator with a general-purpose register file, but the exam expects the classical accumulator answer.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The arithmetic and logic unit performs arithmetic and Boolean operations and must deposit its output in a register.
2. The accumulator is the special-purpose register directly connected to the ALU that supplies one operand and receives the result of each operation.
3. The program counter (option a) holds the address of the next instruction to be fetched; the memory address register (option c) holds the address of the memory location currently being accessed; the instruction register (option d) holds the instruction currently being decoded.
4. None of these stores ALU results.
5. Hence option (b) is correct.

C2 · COMPUTER APPLICATION AND DATA PROCESSING

Hardware: Input, Output & Peripheral Devices

11 questions · 2018: 0 2019: 0 2020: 4 2021: 1 2022: 0 2023: 1 2024: 2 2025: 0 2026: 3

Weight. 11 questions (6.1% of Computer Application and Data Processing) · appeared in 5 of 9 papers · peak year 2020 (4)

Examiner's pattern. Device classification and one arithmetic item. Know input vs output vs both, the OCR/OMR/MICR trio, CRT internals, and plotter/printer behaviour. The dpi calculation (dots = dpi^2 x page area) has appeared as a numerical question.

Must-know. Total dots = dpi^2 width height in inches. MICR reads magnetic ink characters (banking); OMR reads marks; OCR reads printed characters. A webcam and a light pen are input devices.

1. 2020 · Q41 Hardware: Input, Output & Peripheral Devices · Factual

QUESTION

Which devices are paired to form a terminal?

OPTIONS

(a) Keyboard and display monitor

(b) Keyboard and printer

(c) Printer and display monitor

(d) Projector and printer

ANSWER (AI-derived — no official key exists for this paper)

(a) Keyboard and display monitor

EXAM SHORTCUT (AI-derived explanation)

A terminal needs one input device and one output device for interactive use: a keyboard to type on and a monitor to read from.

TIPS & TRICKS (AI-derived explanation)

- A terminal is by definition an input/output station, so the pair must contain exactly one input device and one display/output device.
- Options (c) and (d) contain two OUTPUT devices and no way to enter data, so they cannot form a terminal.
- Historically a keyboard plus printer was a 'hard-copy terminal' (teletype), but the standard modern answer is the VDU: keyboard plus display monitor.
- Terminal types worth knowing: dumb terminal (no processing), smart terminal (limited editing), intelligent terminal (its own processor).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A terminal is a device through which a user both enters data into a computer and receives output from it, so it must combine an input device with an output device.
2. A keyboard is the standard input device for entering commands and data.

3. A display monitor (VDU) is the standard output device for viewing responses.
4. Together they form the familiar video display terminal.
5. Printer plus monitor, and projector plus printer, provide only output; keyboard plus printer is the obsolete hard-copy terminal rather than the standard pairing.
6. Hence the answer is keyboard and display monitor, option (a).

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2. 2020 · Q46 Hardware: Input, Output & Peripheral Devices · Factual

QUESTION

Which are optical character recognition devices? 1. OCR 2. OMR 3. MICR

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(a) 1 and 2 only

EXAM SHORTCUT (AI-derived explanation)

OCR and OMR both read marks or characters using LIGHT, so both are optical. MICR reads magnetic ink and is therefore magnetic, not optical.

TIPS & TRICKS (AI-derived explanation)

- Read the expansions: Optical Character Recognition, Optical Mark Recognition, MAGNETIC Ink Character Recognition. The first letter of each acronym answers the question.
- MICR is the technology used on bank cheques, where the characters are printed in iron-oxide ink and read by magnetic heads.
- OMR is the technology behind OMR answer sheets - it detects the presence or absence of a pencil mark in a defined position.
- OCR goes further and recognises the SHAPE of printed or handwritten characters, converting an image into editable text.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. OCR (Optical Character Recognition) scans printed or handwritten characters with light and converts their images into machine-readable text. Optical.
2. OMR (Optical Mark Recognition) detects the presence of pencil or ink marks in predefined positions by reflecting light off the sheet. Optical.
3. MICR stands for Magnetic Ink Character Recognition. It reads characters printed in magnetic ink using magnetic read heads, and is used on bank cheques.
4. Because MICR relies on magnetism rather than light, it is not an optical recognition device.

5. Hence only items 1 and 2 are optical, option (a).

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3. 2020 · Q48 Variables, Control Structures, Arrays, Functions, Loops · Definition

QUESTION

What is the main function of an input device in a computer?

OPTIONS

(a) Receiving data from a computer

(b) Providing data to a computer

(c) Storing data for processing

(d) Processing the data

ANSWER (AI-derived — no official key exists for this paper)

(b) Providing data to a computer

EXAM SHORTCUT (AI-derived explanation)

An input device feeds data INTO the computer; receiving data from the computer is the job of an output device.

TIPS & TRICKS (AI-derived explanation)

- Fix the direction: input = user to computer, output = computer to user. Option (a) describes an output device and is the deliberate reversal.
- Storing data (option c) is memory's role and processing it (option d) is the CPU's - both are different units of the system.
- Input devices also convert human-usable form into machine-readable form: keyboard, mouse, scanner, microphone, joystick, barcode reader.
- Some devices are BOTH, such as a touchscreen or a modem; but the defining function asked about here is the input direction.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A computer system consists of input units, a central processing unit, memory, and output units.
2. The function of an input device is to accept data and instructions from the outside world, convert them into machine-readable form, and supply them to the computer.
3. Receiving data FROM the computer and presenting it to the user is the function of an output device such as a monitor or printer.
4. Storing data for processing is the function of memory, and processing the data is the function of the CPU.
5. Hence the main function of an input device is providing data to a computer, option (b).

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4. 2020 · Q49 Hardware: Input, Output & Peripheral Devices · Factual

QUESTION

Which components are internal to a Cathode Ray Tube monitor? 1. Electron gun 2. Electromagnetic coils 3. Screen

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

A CRT monitor contains an electron gun, deflection (electromagnetic) coils and a phosphor-coated screen - all three are internal components of the unit.

TIPS & TRICKS (AI-derived explanation)

- Trace the beam to recall the parts in order: electron GUN emits, deflection COILS steer, phosphor SCREEN glows. Three stages, three components.
- 'Internal to the monitor' means inside its casing; the deflection yoke sits around the neck of the tube, still within the monitor housing.
- The screen's inner surface carries the phosphor coating that emits light when struck by electrons - it is part of the tube, not an external accessory.
- Related CRT terms: anode, control grid, focusing coils, shadow mask (colour CRTs), raster scan and refresh rate.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A cathode ray tube monitor works by firing a beam of electrons at a phosphor-coated screen.
2. The electron gun heats a cathode and accelerates the emitted electrons into a narrow beam - an internal component.
3. Electromagnetic deflection coils (the yoke) surround the neck of the tube and steer the beam horizontally and vertically to scan the raster - an internal component of the monitor unit.
4. The screen is the front face of the tube, coated on its inner surface with phosphor that glows where the beam strikes it - also part of the tube.
5. All three listed items are therefore components of a CRT monitor.
6. Hence option (d) is correct.

5. 2021 · Q59 Hardware: Input, Output & Peripheral Devices · Factual

QUESTION

Which of the following are output devices? 1. Printer 2. Speaker 3. Plotter 4. Webcam

OPTIONS

(a) 1 and 2 only

(b) 1, 2 and 3 only

(c) 3 and 4 only

(d) 1, 2, 3 and 4

ANSWER (AI-derived — no official key exists for this paper)

(b) 1, 2 and 3 only

EXAM SHORTCUT (AI-derived explanation)

Printer, speaker and plotter all deliver information FROM the computer; a webcam captures images INTO it, so it is an input device.

TIPS & TRICKS (AI-derived explanation)

- Direction test: output devices convert machine data into human-usable form; input devices do the reverse. Apply it to each item in turn.
- A plotter is simply a printer for line drawings and engineering diagrams - unfamiliar name, familiar category.
- A webcam is the video analogue of a microphone: both capture the outside world and are inputs.
- Devices that are BOTH include touchscreens, modems and network cards - worth remembering for 'which is both' variants of this question.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. An output device presents processed data from the computer to the user; an input device supplies data to the computer.
2. Item 1, printer: produces hard copy of the computer's output. OUTPUT.
3. Item 2, speaker: converts digital audio signals into sound for the user. OUTPUT.
4. Item 3, plotter: draws vector graphics such as engineering drawings on paper. OUTPUT.
5. Item 4, webcam: captures live images and video and sends them into the computer. INPUT.
6. Hence items 1, 2 and 3 are output devices, option (b).

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6. 2023 · Q75 Hardware: Input, Output & Peripheral Devices · Factual

QUESTION

Which of the following devices are used for recognizing the characters in the supermarkets?

OPTIONS

(a) OCR device only

(b) OMR device only

(c) MICR device

(d) Both OCR and OMR devices

ANSWER (AI-derived — no official key exists for this paper)

(a) OCR device only

EXAM SHORTCUT (AI-derived explanation)

Supermarket point-of-sale systems read printed characters and codes optically, which is OCR; OMR only detects marks and MICR reads magnetic ink on cheques.

TIPS & TRICKS (AI-derived explanation)

- Match the technology to the medium: OCR recognises printed or written CHARACTERS, OMR detects the presence of MARKS in fixed positions, MICR reads magnetic-ink characters on bank cheques.
- The question says 'recognizing the characters', which is precisely OCR's job - OMR cannot identify a character at all.
- MICR is confined to banking because it requires special iron-oxide ink, so it has no supermarket application.
- OMR's domain is answer sheets and survey forms, where only the position of a mark matters - not retail character reading.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The task described is recognising printed characters at a supermarket point of sale.
2. OCR (Optical Character Recognition) scans printed or handwritten characters with light and converts their images into machine-readable text - exactly the required capability.
3. OMR (Optical Mark Recognition) only detects whether a mark is present in a predefined position; it cannot identify which character has been printed.
4. MICR (Magnetic Ink Character Recognition) reads characters printed in magnetic ink and is used almost exclusively for bank cheques.
5. Since only OCR recognises characters in this setting, the answer is OCR alone.
6. Option (a) is correct.

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7. **2024 · Q48** Hardware: Input, Output & Peripheral Devices · Conceptual

QUESTION

Consider the following statements in respect of Webcam: I. It is an input device. II. The quality of the video taken through the Webcam largely depends on frame rate and resolution. III. For a video to be smooth and clear, the frame rate is maximum and resolution is minimum. Which of the statements given above is/are correct?

OPTIONS

(a) II only

(b) I and II only

(c) I and III only

(d) I, II and III

ANSWER **(AI-derived — no official key exists for this paper)**

(b) I and II only

EXAM SHORTCUT **(AI-derived explanation)**

A webcam is an input device and its quality does depend on frame rate and resolution, so I and II hold. Statement III is wrong: BOTH frame rate and resolution should be high for smooth, clear video.

TIPS & TRICKS **(AI-derived explanation)**

- Separate the two quality dimensions: FRAME RATE governs smoothness of motion, RESOLUTION governs sharpness of detail. Good video needs both to be high.
- Statement III asks for minimum resolution, which would give smooth but BLURRED video - internally contradictory with the word 'clear'.
- A webcam captures images and sends them into the computer, so it is unambiguously an input device.
- The trade-off in practice is bandwidth: raising both frame rate and resolution increases the data rate, which is why compression is used - but neither is deliberately minimised.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Statement I: a webcam captures live images and video and transmits them into the computer, so it is an input device. TRUE.
2. Statement II: video quality is governed principally by the frame rate (frames per second, controlling smoothness of motion) and the resolution (pixels per frame, controlling detail). TRUE.
3. Statement III: smooth motion requires a HIGH frame rate, and clear, detailed images require a HIGH resolution.
4. Minimising the resolution would make the picture blurred, contradicting the requirement that the video be 'clear'. FALSE.
5. Only statements I and II are correct.
6. Hence option (b) is correct.

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8. **2024 · Q58** Hardware: Input, Output & Peripheral Devices · Conceptual

QUESTION

Consider the following statements: I. Light pen is an electro-optical pointing device which is generally connected to the Visual Display Unit (VDU) of the computer. II. Touchscreen is a pointing device that enables us to enter data such as text, pictures and images by directly touching the screen. III. Joystick is a pointing device which controls the movement of the cursor on the screen by pointing in a particular direction only. Which of the statements given above are correct?

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(d) I, II and III

EXAM SHORTCUT (AI-derived explanation)

All three are standard textbook descriptions: the light pen is an electro-optical device used against the VDU, the touchscreen accepts input by direct touch, and the joystick moves the cursor in the direction it is pushed.

TIPS & TRICKS (AI-derived explanation)

- Pointing devices to know: mouse, trackball, joystick, light pen, touchpad, touchscreen, stylus - all convert physical motion or contact into cursor movement or coordinates.
- The light pen contains a photodetector that senses the screen's refresh beam, which is why it works only against a CRT-style VDU - hence 'electro-optical'.
- The classic distinction for statement III: with a mouse the cursor stops when the mouse stops, whereas with a joystick the cursor keeps moving in the direction the stick is held.
- Touchscreens are unusual in being both an input and an output device, since the display and the sensor share the same surface.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement I: a light pen contains a photodetector and is held against the screen of the visual display unit, sensing the display's own light to determine its position. It is therefore an electro-optical pointing device connected to the VDU. TRUE.
2. Statement II: a touchscreen senses the position of a finger or stylus on the display surface, allowing text, drawings and selections to be entered by touching the screen directly. TRUE.
3. Statement III: a joystick moves the cursor in whichever direction the stick is pushed, and - unlike a mouse - the cursor continues to move while the stick is held in that direction. TRUE.
4. All three descriptions match the standard textbook accounts of these pointing devices.
5. Hence option (d) is correct.

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9. **2026 · Q72** Hardware: Input, Output & Peripheral Devices · Conceptual

QUESTION

A flatbed plotter moves its pin in the X-Y direction. If the pen speed is increased too much, what is the effect on drawing quality?

OPTIONS

(a) Better resolution

(b) No effect

(c) Better accuracy

(d) Lines may be distorted

ANSWER (AI-derived — no official key exists for this paper)

(d) Lines may be distorted

EXAM SHORTCUT (AI-derived explanation)

Moving the pen faster than the mechanism and ink can follow causes overshoot, skipping and wavering, so lines get distorted. Answer (d).

TIPS & TRICKS (AI-derived explanation)

- Plotters are mechanical devices: inertia in the pen carriage means acceleration and deceleration lag behind the commanded path at high speed.
- Resolution and accuracy are set by the stepper-motor step size and the drive mechanics, not by how fast the pen is driven - so options (a) and (c) confuse speed with precision.
- Physical symptoms of excessive pen speed: rounded or overshot corners, uneven ink density, broken lines and vibration-induced wobble.
- There is always a speed-versus-quality trade-off in any mechanical output device, so 'no effect' (option b) is implausible on principle.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A flatbed plotter draws by physically moving a pen across the paper under X-Y stepper-motor control.
2. If the commanded pen speed is raised too far, the mechanical system cannot track the intended path: the carriage overshoots on direction changes, vibration sets in, and the pen may skip or lift momentarily.
3. In addition, ink flow is rate-limited, so at high speed the line becomes faint or broken.
4. The result is distorted lines - wavering, overshot corners and uneven density.
5. Resolution and accuracy (options a and c) are determined by the stepper step size and mechanical precision and are not improved by higher speed; and a purely mechanical process cannot be unaffected (option b).
6. Hence option (d) is correct.

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10. **2026 · Q75** Memory - RAM, ROM & Units of Computer Memory · Definition

QUESTION

Which technique lets a peripheral device transfer data continuously without constant CPU intervention?

OPTIONS

(a) Program-controlled data transfer

(b) Direct memory access

(c) Polling

(d) Interrupt-driven data transfer

ANSWER (AI-derived — no official key exists for this paper)

(b) Direct memory access

EXAM SHORTCUT (AI-derived explanation)

Transferring a block of data straight between a device and memory without the CPU handling each word is Direct Memory Access. Answer (b).

TIPS & TRICKS (AI-derived explanation)

- Three I/O techniques in increasing CPU independence: programmed I/O (CPU polls and moves every word) -> interrupt-driven I/O (device signals, CPU still moves the data) -> DMA (a controller moves the whole block).
- The key phrase in the question is 'continuously ... without constant CPU intervention' - only DMA removes the CPU from the per-word data path.
- Under DMA the CPU is involved only twice: to program the DMA controller with source, destination and count, and to service the completion interrupt.
- Polling (option c) is the most CPU-intensive scheme of all, and interrupt-driven transfer (option d) still requires the CPU to execute the transfer instructions for each item.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. In programmed (program-controlled) I/O and in polling, the CPU executes instructions to move every individual data item, so it is fully occupied during the transfer.
2. In interrupt-driven I/O the device signals readiness, sparing the CPU from busy-waiting, but the CPU still runs the interrupt service routine to move each data item.
3. Direct Memory Access (DMA) uses a dedicated DMA controller that takes over the system bus and transfers an entire block directly between the peripheral and main memory.
4. The CPU only initialises the controller (source address, destination address, byte count) and is interrupted once at the end of the block, so the transfer proceeds continuously without constant CPU intervention.
5. Hence option (b) is correct.

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11. 2026 · Q78 Hardware: Input, Output & Peripheral Devices · Numerical

QUESTION

A laser printer has 1200 dpi resolution. For an 8×10-inch page, how many total dots can theoretically be printed?

OPTIONS

(a) 0.096 million

(b) 96 million

(c) 115.2 million

(d) 120 million

ANSWER (AI-derived — no official key exists for this paper)

(c) 115.2 million

EXAM SHORTCUT (AI-derived explanation)

Dots per side: $1200 \times 8 = 9600$ and $1200 \times 10 = 12000$. Product = $9600 \times 12000 = 115{,}200{,}000 = 115.2$ million.

TIPS & TRICKS (AI-derived explanation)

- Resolution is dots per linear inch, so it must be applied to EACH dimension - the total is dpi^2 times the area.
- Shortcut: total = $1200^2 x(8 \times 10) = 1{,}440{,}000 \times 80 = 115{,}200{,}000$. Squaring the dpi first is the fastest route.
- Multiplying 1200 by the area 80 just once gives 96,000, which inflates to the distractor 96 million - that is exactly the error the option list is testing.
- Order-of-magnitude check: a 1200 dpi page has over 100 million dots, which is why high-resolution printing demands so much memory.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The printer resolution is 1200 dots per inch in each direction.
2. Along the 8-inch side: $1200 \times 8 = 9600$ dots.
3. Along the 10-inch side: $1200 \times 10 = 12000$ dots.
4. Total dots = $9600 \times 12000 = 115{,}200{,}000$.
5. Equivalently, total = $(1200)^2 x(8 \times 10) = 1{,}440{,}000 \times 80 = 115{,}200{,}000 = 115.2$ million.
6. Hence option (c) is correct.

C3 · COMPUTER APPLICATION AND DATA PROCESSING

Memory: RAM, ROM, Cache; Units of Memory & Storage Capacity

23 questions · 2018: 3 2019: 2 2020: 4 2021: 4 2022: 4 2023: 2 2024: 0 2025: 2 2026: 2

Weight. 23 questions (12.8% of Computer Application and Data Processing) · appeared in 8 of 9 papers · peak year 2020 (4)

Examiner's pattern. Memory hierarchy plus address arithmetic. Recurring: RAM vs ROM vs PROM/EPROM, volatile vs non-volatile, where the CPU looks first (cache), registers such as the program counter and accumulator, virtual memory, and computing either address lines or total disk capacity.

Must-know. Address lines = log2(number of locations); disk capacity = surfaces x tracks x sectors x bytes-per-sector; 1 GB = 2^30 bytes; ROM/EPROM/flash are non-volatile, DRAM/SRAM are volatile; the accumulator stores ALU results.

1. 2018 · Q63 Memory - RAM, ROM & Units of Computer Memory · Numerical

QUESTION

A system has 1024 memory locations, each storing 8 bits. How many address lines are required?

OPTIONS

(a) 8

(b) 10

(c) 16

(d) 1024

ANSWER (AI-derived — no official key exists for this paper)

(b) 10

EXAM SHORTCUT (AI-derived explanation)

Address lines depend only on the NUMBER of locations: $1024 = 2^{10}$, so 10 address lines. The 8 bits per location fix the data-bus width, not the address bus.

TIPS & TRICKS (AI-derived explanation)

- Formula: number of locations = 2^(address lines). Take $\log$ base 2 of the location count and ignore the word size entirely.
- The '8 bits' is a deliberate distractor and is planted as option (a) - it is the data width, not the address width.
- Memorise the powers: $2^{10} = 1K, 2^{16} = 64K, 2^{20} = 1M$. Then 1024 locations means 10 lines without any calculation.
- Conversely, m address lines and n data lines give a capacity of $2^m x n$ bits - the formula behind most memory-sizing questions.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The number of distinct memory locations that can be addressed by k address lines is 2^k.

2. Here the system has 1024 memory locations, so $2^k = 1024$.
3. Since $1024 = 2^{10}$, we get $k = 10$.
4. The word size of 8 bits determines the width of the DATA bus and has no bearing on the number of address lines.
5. Hence 10 address lines are required, option (b).

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2. 2018 · Q69 Memory - RAM, ROM & Units of Computer Memory · Factual

QUESTION

Which type of memory is the auxiliary memory?

OPTIONS

(a) SRAM

(b) Magnetic tape

(c) Flash ROM

(d) Cache memory

ANSWER (AI-derived — no official key exists for this paper)

(b) Magnetic tape

EXAM SHORTCUT (AI-derived explanation)

Auxiliary (secondary) memory is non-volatile external storage. Of the four, only magnetic tape is external backing store; SRAM and cache are primary, and flash ROM is firmware memory.

TIPS & TRICKS (AI-derived explanation)

- Memory hierarchy by speed and role: registers, cache, main memory (RAM), then auxiliary/secondary storage (disk, tape, optical). 'Auxiliary' always means the bottom layer.
- Auxiliary memory is characterised by being non-volatile, high capacity, low cost per bit and NOT directly addressable by the CPU - data must be transferred to main memory first.
- SRAM (option a) is the technology used to build cache, and cache (option d) is primary memory - both sit near the top of the hierarchy.
- Magnetic tape is sequential-access, which is why it survives as archival/backup auxiliary storage rather than as working memory.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Auxiliary memory (also called secondary or backing storage) is non-volatile storage that lies outside the CPU's directly addressable main memory.
2. SRAM is a fast volatile technology used for cache and registers - primary memory.
3. Cache memory is a small, very fast buffer between the CPU and main memory - primary memory.
4. Flash ROM is used to hold firmware/BIOS and is treated as read-mostly primary memory rather than auxiliary storage.
5. Magnetic tape is a high-capacity, non-volatile, sequential-access external medium used for backup and archiving - the classic auxiliary memory.

6. Hence option (b) is correct.

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3. 2018 · Q79 Memory - RAM, ROM & Units of Computer Memory · Numerical

QUESTION

A disc storage has 16 data-recording surfaces, 256 tracks/surface, 16 sectors/track, 512 bytes/sector. What is the storage capacity?

OPTIONS

(a) 3325652 bytes

(b) 3256662 bytes

(c) 33554432 bytes

(d) 335444332 bytes

ANSWER (AI-derived — no official key exists for this paper)

(c) 33554432 bytes

EXAM SHORTCUT (AI-derived explanation)

Work in powers of two: $16 \times 256 \times 16 \times 512 = 2^4 x 2^8 x 2^4 x 2^9 = 2^{25} =$ 33,554,432 bytes.

TIPS & TRICKS (AI-derived explanation)

- Convert every factor to a power of 2 and simply ADD the exponents - $4 + 8 + 4 + 9 = 25$ - instead of multiplying four numbers.
- $2^{25} = 32$ MB exactly, since $2^{20} = 1$ MB and $2^5 = 32$. Recognising the answer as a round 32 MB is the fastest confirmation.
- Capacity chain: surfaces x tracks per surface x sectors per track x bytes per sector. Multiply the whole chain; omitting any link is the standard error.
- Only option (c) is a power of two, and disk capacities built from powers of two must be too - that alone identifies the answer.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Total capacity = (number of surfaces) x (tracks per surface) x (sectors per track) x (bytes per sector).
2. Substituting: 16 x 256 x 16 x 512.
3. Writing each factor as a power of two: $2^4 x 2^8 x 2^4 x 2^9$.
4. Adding the exponents: $2^{4+8+4+9} = 2^{25}$.
5. $2^{25} =$ 33,554,432 bytes, i.e. exactly 32 MB.
6. Hence option (c) is correct.

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4. 2019 · Q62 Memory - RAM, ROM & Units of Computer Memory · Definition

QUESTION

Which one of the following is content-based addressable memory?

OPTIONS

(a) ROM

(b) Cache memory

(c) Associative memory

(d) SRAM

ANSWER (AI-derived — no official key exists for this paper)

(c) Associative memory

EXAM SHORTCUT (AI-derived explanation)

Content-addressable memory is another name for ASSOCIATIVE memory: it is searched by data content rather than by address.

TIPS & TRICKS (AI-derived explanation)

- CAM = Content Addressable Memory = Associative Memory. Learn the three names as one item; they appear interchangeably in question papers.
- Ordinary RAM/ROM is accessed by supplying an ADDRESS; associative memory is accessed by supplying the DATA and asking where (or whether) it is stored.
- Cache memory often USES associative mapping internally, but 'cache' names a role in the hierarchy while 'associative memory' names the addressing scheme - do not confuse the two.
- Typical applications: translation lookaside buffers, network routing tables and database search - all places where a parallel content search is needed.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Conventional memories (ROM, SRAM, DRAM) are location-addressed: the CPU supplies an address and the memory returns the contents of that location.
2. Associative memory, also called content-addressable memory (CAM), is searched by content: a data word is presented and the memory reports, in parallel, whether and where a match exists.
3. This parallel search makes it fast but expensive, so it is used for small, search-intensive structures.
4. Cache memory is defined by its position in the hierarchy rather than by its addressing scheme, even though it commonly employs associative mapping.
5. Hence the content-based addressable memory is associative memory, option (c).

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5. 2019 · Q70 Memory - RAM, ROM & Units of Computer Memory · Factual

QUESTION

Which type of memory is suitable for mobile phones, digital cameras and iPods?

OPTIONS

(a) Programmable ROM

(b) Flash memory

(c) RAM

(d) Optical disk

ANSWER **(AI-derived — no official key exists for this paper)**

(b) Flash memory

EXAM SHORTCUT **(AI-derived explanation)**

Portable devices need memory that is non-volatile, compact, low-power and electrically rewritable - that is flash memory.

TIPS & TRICKS **(AI-derived explanation)**

- Flash is electrically erasable and rewritable in blocks, non-volatile, solid state (no moving parts) and low power - exactly the requirement list for phones, cameras and music players.
- PROM (option a) can be written only ONCE, so it cannot store photographs or music that change constantly.
- RAM (option c) is volatile: everything would be lost when the device is switched off.
- Optical disks (option d) need a mechanical drive and are far too bulky and fragile for handheld devices.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Mobile phones, digital cameras and portable music players need storage that retains data with the power off, tolerates being carried about, and can be rewritten many times.
2. Programmable ROM can be programmed only once and cannot be updated, so it is unsuitable for user data.
3. RAM is volatile - its contents vanish when power is removed - so it cannot serve as the main storage of a portable device.
4. Optical disks require a mechanical drive with moving parts, making them unsuitable for compact handheld devices.
5. Flash memory is non-volatile, electrically erasable and rewritable, compact, shock-resistant and low-power, which is why it is used in SD cards, USB drives and phone storage.
6. Hence the answer is flash memory, option (b).

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6. **2020 · Q54** Operating Systems, Packages & Utilities · Factual

QUESTION

Programs stored in which memory can *not* be erased?

OPTIONS

(a) RAM

(b) ROM

(c) Virtual memory

(d) Cache memory

ANSWER (AI-derived — no official key exists for this paper)

(b) ROM

EXAM SHORTCUT (AI-derived explanation)

ROM is Read-Only Memory: its contents are written once at manufacture and cannot be erased in normal operation.

TIPS & TRICKS (AI-derived explanation)

- The acronym is the answer: Read-Only Memory can be read but not written or erased during normal use.
- RAM, cache and virtual memory are all read-write and volatile or reusable, so none of them holds data permanently.
- ROM variants worth knowing: PROM (programmable once), EPROM (erasable by ultraviolet light), EEPROM and flash (electrically erasable). Strict ROM is the non-erasable one asked about here.
- ROM typically holds the BIOS/bootstrap loader - code that must survive power loss and must not be alterable by ordinary programs.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. RAM is volatile read-write memory; its contents are lost when power is removed and can be overwritten at will.
2. Cache memory is a small, fast volatile buffer whose contents are constantly replaced as the CPU works.
3. Virtual memory is an area of secondary storage used as an extension of main memory, and it is continually overwritten.
4. ROM (Read-Only Memory) is non-volatile and its contents are fixed at the time of manufacture, so programs stored in it - such as the bootstrap loader and BIOS - cannot be erased in normal operation.
5. Hence the answer is ROM, option (b).

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7. **2020 · Q55** Memory - RAM, ROM & Units of Computer Memory · Numerical

QUESTION

One Gigabyte (GB) equals

OPTIONS

(a) 1024×1024 Megabytes

(b) 1024×1024 Terabytes

(c) 1024×1024 Bytes

(d) 1024×1024 Kilobytes

ANSWER (AI-derived — no official key exists for this paper)

(d) 1024×1024 Kilobytes

EXAM SHORTCUT (AI-derived explanation)

1 GB = 1024 MB = 1024 x 1024 KB. Two steps down the ladder from GB lands on kilobytes.

TIPS & TRICKS (AI-derived explanation)

- Memory ladder: 1 KB = 1024 B, 1 MB = 1024 KB, 1 GB = 1024 MB, 1 TB = 1024 GB. Each rung is a factor of $1024 = 2^{10}$.
- $1024 \times 1024 = 2^{20}$, so the question is asking which unit is 2^{20} below a gigabyte - two rungs down, i.e. kilobytes.
- Option (b) goes UP the ladder to terabytes, which would make a gigabyte larger than a terabyte - impossible.
- In powers of two: 1 GB = 2^{30} bytes, 1 MB = 2^{20} bytes, 1 KB = 2^{10} bytes. Working in exponents removes all ambiguity.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The standard binary memory units are 1 KB = 1024 bytes, 1 MB = 1024 KB, 1 GB = 1024 MB and 1 TB = 1024 GB.
2. So 1 GB = 1024 MB.
3. Substituting 1 MB = 1024 KB gives 1 GB = 1024 x 1024 KB.
4. In powers of two, 1 GB = 2^{30} bytes and 1 KB = 2^{10} bytes, so the ratio is $2^{20} = 1024 \times 1024$, confirming the result.
5. Option (a) would make a gigabyte equal to a terabyte, and option (b) would make it larger than a terabyte; option (c) understates it by a factor of 1024.
6. Hence 1 GB = 1024 x 1024 kilobytes, option (d).

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8. 2020 · Q58 Number Systems & Binary Arithmetic · Definition

QUESTION

A sequential electronic circuit used to store 1 bit of information is called a

OPTIONS

(a) Register

(b) Flip-flop

(c) Counter

(d) Semiconductor

ANSWER (AI-derived — no official key exists for this paper)

(b) Flip-flop

EXAM SHORTCUT (AI-derived explanation)

The basic one-bit sequential storage element is the flip-flop; registers and counters are built by combining several flip-flops.

TIPS & TRICKS (AI-derived explanation)

- Hierarchy: flip-flop stores 1 bit, a register is n flip-flops storing n bits, a counter is a register that increments. The question asks for ONE bit, so flip-flop.
- 'Sequential' means the output depends on past inputs as well as present ones - the defining property of memory elements as opposed to combinational gates.
- Flip-flop types to know: SR, JK, D and T. The D flip-flop is the simplest one-bit storage cell.
- A latch is the level-triggered cousin of the edge-triggered flip-flop; both store one bit but differ in how they respond to the clock.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A sequential circuit has memory: its output depends on the current inputs and on the stored past state.
2. A flip-flop is the elementary sequential circuit, holding exactly one bit of information and changing state on a clock edge.
3. A register is formed by grouping several flip-flops together to store a multi-bit word.
4. A counter is a register with additional logic that advances through a sequence of states.
5. A semiconductor is a material, not a circuit at all.
6. Hence the one-bit sequential storage element is the flip-flop, option (b).

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9. **2020 · Q60** Memory - RAM, ROM & Units of Computer Memory · Factual

QUESTION

Which is used to store data permanently?

OPTIONS

(a) Primary memory

(b) Secondary memory

(c) Cache memory

(d) Register

ANSWER (AI-derived — no official key exists for this paper)

(b) Secondary memory

EXAM SHORTCUT (AI-derived explanation)

Permanent (non-volatile) storage is the role of secondary memory - hard disks, SSDs, optical media and tape.

TIPS & TRICKS (AI-derived explanation)

- Volatility is the discriminator: registers, cache and primary memory (RAM) all lose their contents when power is removed; secondary storage does not.
- Secondary memory is also characterised by large capacity, low cost per bit, and the fact that the CPU cannot address it directly - data must be loaded into main memory first.

- Note that ROM is technically non-volatile primary memory, but it holds firmware, not general user data - the question is about storing data permanently.
- Speed ranking (fastest to slowest): register, cache, primary memory, secondary memory. Capacity and permanence run in the opposite direction.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Registers inside the CPU hold operands for the current instruction and are volatile.
2. Cache memory is a small fast volatile buffer whose contents are continuously replaced.
3. Primary memory (RAM) is the working store of running programs and is volatile - its contents are lost when the power is switched off.
4. Secondary memory - hard disks, solid-state drives, optical disks and magnetic tape - is non-volatile, retaining data indefinitely without power, and offers large capacity at low cost per bit.
5. Hence data are stored permanently in secondary memory, option (b).

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10. 2021 · Q56 Operating Systems, Packages & Utilities · Conceptual

QUESTION

In which technique is a virtual address used to map the physical address of data?

OPTIONS

(a) Segmentation

(b) Swapping

(c) Scheduling

(d) Paging

ANSWER (AI-derived — no official key exists for this paper)

(d) Paging

EXAM SHORTCUT (AI-derived explanation)

Paging is the technique in which a virtual (logical) address is split into a page number and offset and translated through a page table into a physical frame address.

TIPS & TRICKS (AI-derived explanation)

- Address translation in paging: virtual address = page number + offset; the page table maps the page number to a frame number, and the offset passes through unchanged.
- Segmentation also maps logical to physical addresses but does so by variable-length SEGMENTS with base and limit registers; paging with its fixed-size pages is the standard answer for virtual-address mapping.
- Swapping moves whole processes between memory and disk and scheduling allocates CPU time - neither performs address translation, so options (b) and (c) are out of category.
- The TLB (translation lookaside buffer) is the associative cache that speeds up this page-table lookup - a natural follow-up fact.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Under paging, the logical (virtual) address space of a process is divided into fixed-size pages and physical memory into frames of the same size.
2. A virtual address is interpreted as a page number plus an offset within the page.
3. The page number indexes the process's page table, which returns the physical frame number; concatenating the frame number with the unchanged offset gives the physical address.
4. This mapping is exactly the use of a virtual address to locate the physical address of data.
5. Swapping transfers entire processes between memory and backing store, and scheduling allocates processor time; neither performs address translation.
6. Hence the technique is paging, option (d).

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11. 2021 · Q72 Memory - RAM, ROM & Units of Computer Memory · Numerical

QUESTION

With 4 chips, how many data lines are required for Read/Write of a memory chip with 2048 locations, each of 2 bytes?

OPTIONS

(a) 11

(b) 16

(c) 2

(d) 12

ANSWER (AI-derived — no official key exists for this paper)

(b) 16

EXAM SHORTCUT (AI-derived explanation)

Data lines are fixed by the WORD size, not the number of locations: 2 bytes = 16 bits, so 16 data lines are needed.

TIPS & TRICKS (AI-derived explanation)

- Separate the two buses: ADDRESS lines are determined by the number of locations ($2048 = 2^{11}$ gives 11), while DATA lines are determined by the word size (2 bytes = 16 bits).
- Option (a) 11 is the address-line count, planted for candidates who answer the wrong half of the specification.
- Chips placed in parallel to widen the word would multiply the data lines, but chips stacked to add capacity share the same data bus; here the 16-bit word governs.
- General formula: a memory of 2^m locations by n bits needs m address lines and n data lines, giving a capacity of $2^m xn$ bits.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The number of data lines of a memory is equal to the width of one addressable location, since all bits of a word are read or written in parallel.

2. Each location here holds 2 bytes.
3. One byte is 8 bits, so 2 bytes = 16 bits.
4. Hence 16 data lines are required for read/write operations.
5. The figure $2048 = 2^{11}$ determines the ADDRESS lines (11 of them), not the data lines, and is included as a distractor.
6. Option (b) is correct.

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12. 2021 · Q74 Operating Systems, Packages & Utilities · Conceptual

QUESTION

Types of memory: 1. Cache 2. ROM 3. Optical disks 4. Registers. Which are internal process memory?

OPTIONS

(a) 1 and 2 only

(b) 1, 2 and 4 only

(c) 1 and 4 only

(d) 2, 3 and 4 only

ANSWER (AI-derived — no official key exists for this paper)

(c) 1 and 4 only

EXAM SHORTCUT (AI-derived explanation)

Internal process memory means storage inside the processor itself: registers and cache. ROM and optical disks lie outside the CPU.

TIPS & TRICKS (AI-derived explanation)

- Hierarchy by location: registers and cache sit INSIDE the CPU; main memory (RAM/ROM) is external primary memory; disks and optical media are secondary storage.
- 'Internal process memory' is the standard term for the fastest tier used during instruction execution - registers first, then cache.
- ROM is non-volatile primary memory holding firmware; it is accessed over the memory bus and is not internal to the processor.
- Optical disks are removable secondary storage, several orders of magnitude slower, and can be excluded immediately.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Internal process memory refers to the storage located within the processor and used directly during instruction execution.
2. Item 4, registers: the fastest storage, located inside the CPU and holding operands and results of the current instruction. INTERNAL.
3. Item 1, cache: a small, very fast memory built into or immediately adjacent to the CPU to hold recently used instructions and data. INTERNAL.

4. Item 2, ROM: non-volatile primary memory holding firmware, accessed over the system bus and external to the processor. NOT internal process memory.
5. Item 3, optical disks: removable secondary storage, far slower and entirely external.
6. Hence items 1 and 4 are internal process memory, option (c).

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13. 2021 · Q77 Variables, Control Structures, Arrays, Functions, Loops · Factual

QUESTION

Which memory location does the CPU refer to first while searching for data?

OPTIONS

(a) ROM

(b) Secondary memory

(c) Main memory

(d) Cache memory

ANSWER (AI-derived — no official key exists for this paper)

(d) Cache memory

EXAM SHORTCUT (AI-derived explanation)

The CPU always checks the fastest tier first: cache memory, before going to main memory and then to secondary storage.

TIPS & TRICKS (AI-derived explanation)

- Search order follows the hierarchy top-down: registers, then cache (L_1, L_2, L_3), then main memory, then secondary storage. Registers hold current operands, so the first SEARCH is in cache.
- A cache hit avoids the far slower main-memory access; a miss triggers a fetch from RAM and the block is then cached for next time.
- The principle behind this design is LOCALITY OF REFERENCE - programs reuse recently accessed data (temporal) and nearby data (spatial).
- ROM holds firmware and is not part of the data-search path; secondary memory is the last resort, reached only through the operating system.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The memory hierarchy runs from fastest and smallest to slowest and largest: registers, cache, main memory, secondary storage.
2. When the CPU needs a data item it first looks in cache memory, the small high-speed store that holds recently and frequently used data.
3. If the item is present (a cache hit) it is supplied immediately, avoiding the much slower main-memory access.
4. If it is absent (a cache miss) the block is fetched from main memory and placed in cache for future use; only if it is not in main memory is secondary storage consulted.
5. This ordering exploits locality of reference and is the reason cache exists.

6. Hence the CPU refers first to cache memory, option (d).

---

14. 2022 · Q34 Memory - RAM, ROM & Units of Computer Memory · Definition

QUESTION

Which one of the following refers to associative memory?

OPTIONS

(a) The address of the data is generated by the CPU

(b) The address of the data is supplied by the users

(c) The data are accessed sequentially

(d) There is no need for an address because the data is used as an address

ANSWER (AI-derived — no official key exists for this paper)

(d) There is no need for an address because the data is used as an address

EXAM SHORTCUT (AI-derived explanation)

Associative memory is content-addressable: the data itself is presented and searched for, so no separate address is needed.

TIPS & TRICKS (AI-derived explanation)

- The defining feature is content addressing - you supply the DATA and the memory tells you whether and where it is stored, the reverse of ordinary addressing.
- Options (a) and (b) describe ordinary location-addressed memory; option (c) describes sequential-access media such as tape.
- Because every cell compares its contents in parallel, associative memory is fast but expensive, so it is used only for small structures such as the TLB and cache tag store.
- Same concept appears as 2019 Q62 in reverse phrasing - 'content-based addressable memory' is the exact synonym.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. In conventional memory the CPU supplies an address and the memory returns the contents of that location.
2. Associative memory, also called content-addressable memory, works in the opposite direction: a data word is presented and every cell compares its stored contents with it in parallel.
3. The memory then reports whether a match exists and, if required, where. No address needs to be supplied because the data itself serves as the search key.
4. Options (a) and (b) describe address-based access, and option (c) describes sequential access as on magnetic tape.
5. Hence associative memory is characterised by there being no need for an address, the data being used as the address, option (d).

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15. 2022 · Q35 Memory - RAM, ROM & Units of Computer Memory · Numerical

QUESTION

The 2's complement representation of -59 in an 8-bit word size computer system is:

OPTIONS

(a) 11000100

(b) 11000101

(c) 00111011

(d) 11011010

ANSWER (AI-derived — no official key exists for this paper)

(b) 11000101

EXAM SHORTCUT (AI-derived explanation)

$59 = 00111011$; invert to 11000100; add 1 to get 11000101.

TIPS & TRICKS (AI-derived explanation)

- Two's complement in two steps: write the magnitude in binary, invert every bit (one's complement), then add 1. Skipping the +1 leaves option (a).
- $59 = 32 + 16 + 8 + 2 + 1 = 00111011$. Building from powers of two is faster and safer than repeated division.
- The most significant bit of the answer must be 1 for a negative number, which eliminates option (c) - that is simply +59.
- Verify by adding: $00111011 + 11000101 = 100000000$, and discarding the ninth-bit carry leaves 00000000, confirming the pair sums to zero.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. First write the magnitude 59 in 8-bit binary: $59 = 32 + 16 + 8 + 2 + 1$, so $59 = 00111011$.
2. Take the one's complement by inverting every bit: 11000100.
3. Add 1 to obtain the two's complement: $11000100 + 1 = 11000101$.
4. The leading bit is 1, correctly indicating a negative value.
5. Check: $00111011 + 11000101 = 100000000$; discarding the carry out of the 8-bit word leaves 00000000, so the two representations are indeed additive inverses.
6. Hence -59 is 11000101, option (b).

---

16. 2022 · Q71 Memory - RAM, ROM & Units of Computer Memory · Factual

QUESTION

On which type of ROM can data be written only once?

OPTIONS

(a) PROM

(b) EPROM

(c) EEPROM

(d) EROM

ANSWER (AI-derived — no official key exists for this paper)

(a) PROM

EXAM SHORTCUT (AI-derived explanation)

PROM - Programmable ROM - is blank when manufactured and can be programmed exactly ONCE by blowing internal fuses.

TIPS & TRICKS (AI-derived explanation)

- ROM family by erasability: ROM (fixed at manufacture), PROM (write once), EPROM (erasable by ultraviolet light), EEPROM/flash (electrically erasable, many times).
- The 'E' prefixes signal ERASABLE, so any option beginning with E can be rewritten and cannot be the write-once answer.
- PROM is programmed by permanently blowing fuse links, which is why the process is irreversible - a one-line justification worth quoting.
- 'EROM' in option (d) is not a standard device name; the genuine family members are ROM, PROM, EPROM, EEPROM and flash.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Plain ROM has its contents fixed during manufacture and cannot be written by the user at all.
2. PROM (Programmable Read-Only Memory) is supplied blank and can be programmed once by the user, typically by blowing internal fuse links. Because the fuses are destroyed permanently, the process cannot be reversed or repeated.
3. EPROM (Erasable PROM) can be erased by exposure to ultraviolet light and reprogrammed many times.
4. EEPROM (Electrically Erasable PROM) can be erased and rewritten electrically, byte by byte, without removal from the circuit.
5. 'EROM' is not a standard device designation.
6. Hence data can be written only once on PROM, option (a).

---

17. 2022 · Q76 Memory - RAM, ROM & Units of Computer Memory · Factual

QUESTION

Which of the following is/are non-volatile? 1. DRAM, 2. SRAM, 3. EPROM

OPTIONS

(a) 1 only

(b) 3 only

(c) 1 and 3

(d) 2 and 3

ANSWER **(AI-derived — no official key exists for this paper)**

(b) 3 only

EXAM SHORTCUT **(AI-derived explanation)**

DRAM and SRAM are both volatile forms of RAM; only EPROM, a member of the ROM family, retains its contents without power.

TIPS & TRICKS **(AI-derived explanation)**

- Volatility rule of thumb: anything called RAM is volatile; anything in the ROM family (ROM, PROM, EPROM, EEPROM, flash) is non-volatile.
- DRAM additionally needs periodic REFRESH even while powered, and SRAM holds its value only while power is applied - both lose everything at power-off.
- EPROM stores charge on floating gates that persists for years without power, which is exactly what non-volatility means.
- Practical anchor: main memory (RAM) is wiped on reboot, whereas the BIOS held in EPROM/flash survives - the everyday demonstration of the distinction.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Volatile memory loses its contents when the power supply is removed; non-volatile memory retains them.
2. Item 1, DRAM (Dynamic RAM): stores each bit as charge on a capacitor, requiring periodic refresh even while powered, and loses everything when power is removed. VOLATILE.
3. Item 2, SRAM (Static RAM): stores each bit in a flip-flop; no refresh is needed but power is, so the contents vanish at power-off. VOLATILE.
4. Item 3, EPROM (Erasable Programmable ROM): stores bits as trapped charge on floating gates, which persists for years without power and is removed only by ultraviolet exposure. NON-VOLATILE.
5. Hence only item 3 is non-volatile.
6. Option (b) is correct.

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18. 2023 · Q63 Operating Systems, Packages & Utilities · Definition

QUESTION

The technique of dividing the physical memory space into multiple blocks is known as:

OPTIONS

(a) Paging

(b) Segmentation

(c) Fragmentation

(d) Swapping

ANSWER (AI-derived — no official key exists for this paper)

(a) Paging

EXAM SHORTCUT (AI-derived explanation)

Paging divides PHYSICAL memory into fixed-size blocks called frames (and logical memory into equal-sized pages).

TIPS & TRICKS (AI-derived explanation)

- Key distinction: paging divides PHYSICAL memory into equal-sized frames; segmentation divides the LOGICAL address space into variable-sized, logically meaningful segments.
- The phrase 'physical memory space into multiple blocks' points squarely at frames, hence paging.
- Fragmentation is a PROBLEM (wasted space), not a technique, and swapping moves whole processes between memory and disk - neither divides memory into blocks.
- Paging eliminates external fragmentation but introduces a little internal fragmentation in the last frame of each process - a useful follow-up fact.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Under paging, physical memory is divided into fixed-size blocks called FRAMES, and the logical address space of each process into pages of the same size.
2. A page table maps each page to whatever frame currently holds it, so a process need not occupy contiguous physical memory.
3. Segmentation instead divides the LOGICAL address space into variable-length segments corresponding to program units such as code, stack and data.
4. Fragmentation is not a technique at all but the wastage that arises from allocation, and swapping transfers whole processes between memory and backing store.
5. Hence the division of physical memory into multiple blocks is paging, option (a).

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19. 2023 · Q65 Memory - RAM, ROM & Units of Computer Memory · Conceptual

QUESTION

Consider the following statements regarding cache memory in the computer system: 1. The size of cache memory is always kept higher than the size of RAM. 2. Cache memory is faster to access than RAM but slower to access than CPU registers. 3. The effectiveness of cache memory is measured by hit rate. Which of the statements given above are correct?

OPTIONS

(a) 1 and 3 only

(b) 2 and 3 only

(c) 1 and 2 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(b) 2 and 3 only

EXAM SHORTCUT (AI-derived explanation)

Cache is much SMALLER than RAM, so statement 1 fails. Statements 2 (registers > cache > RAM in speed) and 3 (hit rate measures effectiveness) are both correct.

TIPS & TRICKS (AI-derived explanation)

- Hierarchy rule: as you move towards the CPU, memory gets faster, smaller and more expensive per bit. Cache is therefore smaller than RAM, never larger.
- Speed order: registers, then cache (L_1, L_2, L_3), then main memory, then secondary storage. Statement 2 states exactly this.
- Hit rate = (number of cache hits)/(total accesses); the effective access time is $hxt_{cache} + (1 - h)xt_{memory}$, so a high hit rate is the whole point of cache.
- Any statement containing 'always' deserves scrutiny - here it is the giveaway that statement 1 is the false one.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: cache memory is built from fast, expensive SRAM and is deliberately kept SMALL - typically a few kilobytes to a few megabytes against gigabytes of RAM. The claim that it is always larger than RAM is FALSE.
2. Statement 2: the memory hierarchy in decreasing order of speed is registers, cache, main memory, secondary storage. Cache is therefore faster than RAM but slower than CPU registers. TRUE.
3. Statement 3: the effectiveness of cache is measured by the hit rate h, the proportion of accesses satisfied from cache; the effective access time is $ht_{cache} + (1 - h)t_{memory}$. TRUE.
4. Only statements 2 and 3 are correct.
5. Hence option (b) is correct.

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20. 2025 · Q34 Memory - RAM, ROM & Units of Computer Memory · Conceptual

QUESTION

Consider: I. Better performance II. Low price III. Low power and heat IV. High reliability. Which are advantages of solid-state storage devices over hard disk drives?

OPTIONS

(a) I, II and III

(b) I, II and IV

(c) I, III and IV

(d) II, III and IV

ANSWER (AI-derived — no official key exists for this paper)

(c) I, III and IV

EXAM SHORTCUT (AI-derived explanation)

SSDs beat hard disks on speed, power consumption and reliability, but they cost MORE per gigabyte - so items I, III and IV qualify and II does not.

TIPS & TRICKS (AI-derived explanation)

- SSDs have no moving parts, which explains all three genuine advantages at once: faster access, lower power and heat, and greater shock resistance.
- Price per gigabyte is the classical DISADVANTAGE of solid-state storage - any option containing 'low price' is suspect.
- Access-time contrast: an SSD responds in microseconds while a hard disk needs milliseconds of seek and rotational latency.
- SSDs do have a finite write-endurance, but for the comparison asked here reliability still favours them over mechanical drives.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A solid-state drive stores data in flash memory with no moving parts, unlike a hard disk drive with its spinning platters and moving heads.
2. Item I, better performance: with no seek or rotational latency, access times fall from milliseconds to microseconds. ADVANTAGE.
3. Item III, low power and heat: no motor or actuator has to be driven, so power draw and heat generation are much lower. ADVANTAGE.
4. Item IV, high reliability: the absence of mechanical parts makes SSDs far more resistant to shock and mechanical failure. ADVANTAGE.
5. Item II, low price: solid-state storage costs substantially more per gigabyte than magnetic disk, so price is a DISADVANTAGE, not an advantage.
6. Hence the advantages are I, III and IV, option (c).

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21. 2025 · Q77 Operating Systems, Packages & Utilities · Conceptual

QUESTION

About virtual memory in a computer system: I. It provides large addressable space without worrying about physical memory size limits. II. It provides efficient sharing of main memory among users in a multiprogrammed OS. III. It provides a large secondary memory. Which are correct?

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(a) I and II only

EXAM SHORTCUT (AI-derived explanation)

Virtual memory enlarges the apparent address space and lets many processes share physical RAM - I and II. It does not create secondary memory; it merely uses the disk that already exists, so III is false. Answer (a).

TIPS & TRICKS (AI-derived explanation)

- Virtual memory is an illusion of large main memory, not an enlargement of disk - statement III inverts cause and effect.
- Two textbook benefits: (i) programs may be larger than physical RAM, (ii) more processes fit in memory simultaneously, raising multiprogramming and CPU utilisation.
- Mechanisms: paging with a page table and MMU address translation, demand paging, page faults and replacement policies (FIFO, LRU, Optimal).
- Thrashing is the failure mode - excessive page faults when the working set exceeds available frames.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Virtual memory is a memory-management technique in which the logical address space presented to a program is decoupled from the physical main memory, with pages moved between RAM and a backing store on disk on demand.
2. Statement I: a process can use a large logical address space regardless of how much physical memory is installed - this is the defining benefit. True.
3. Statement II: because only the active pages of each process occupy frames, main memory is shared efficiently among many users in a multiprogrammed operating system. True.
4. Statement III: virtual memory does not provide or enlarge secondary memory. It consumes part of the existing secondary storage as a swap area to create the illusion of large main memory. False.
5. Only statements I and II are correct.
6. Hence option (a) is correct.

---

22. 2026 · Q66

Memory - RAM, ROM & Units of Computer Memory · Factual

QUESTION

What is the fixed length of the instruction format used in RISC processors?

OPTIONS

(a) 30 bits

(b) 16 bits

(c) 32 bits

(d) 24 bits

ANSWER (AI-derived — no official key exists for this paper)

(c) 32 bits

EXAM SHORTCUT (AI-derived explanation)

RISC architectures use a uniform 32-bit instruction word - that fixed width is what makes the pipeline simple. Answer (c).

TIPS & TRICKS (AI-derived explanation)

- Defining RISC features: fixed-length 32-bit instructions, load/store architecture, large uniform register file, few addressing modes, hardwired control, one instruction per cycle target.
- CISC (for example x86) uses variable-length instructions, from 1 byte to 15 bytes - the direct contrast that the question is testing.
- Fixed 32-bit width means the fetch stage always knows where the next instruction starts, which is exactly what enables clean pipelining.
- Classic RISC families - MIPS, SPARC, early ARM, RISC-$V's$ base RV32I - all use 32-bit instruction encodings.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A reduced instruction set computer (RISC) is designed around a simple, regular instruction encoding to support efficient pipelining.
2. All instructions in a classical RISC design occupy a single fixed-length word of 32 bits, regardless of the operation.
3. This uniformity lets the instruction fetch and decode stages operate at a constant rate, since the address of the next instruction is always the current one plus 4 bytes.
4. By contrast, CISC processors use variable-length instructions, which complicates decoding and pipelining.
5. Hence option (c) is correct.

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23. 2026 · Q71 Memory - RAM, ROM & Units of Computer Memory · Factual

QUESTION

What medium is used in optical storage systems for reading and recording data?

OPTIONS

(a) Ultraviolet light

(b) High-energy visible light

(c) Laser light

(d) Infrared light

ANSWER (AI-derived — no official key exists for this paper)

(c) Laser light

EXAM SHORTCUT (AI-derived explanation)

Optical storage - CD, DVD, Blu-ray - reads and writes with a laser beam. The name 'optical' plus 'laser' is the whole answer. Answer (c).

TIPS & TRICKS (AI-derived explanation)

- Wavelength ladder worth knowing: CD uses a 780 nm infrared laser, DVD a 650 nm red laser, Blu-ray a 405 nm blue-violet laser. Shorter wavelength means a tighter spot and more capacity.
- Note the distinction the question is drawing: the medium of reading/recording is LASER light, a coherent beam - not merely light of some colour band, which is what options (a), (b) and (d) name.
- Data are stored as pits and lands whose transitions change the reflected laser intensity, which a photodetector converts into bits.
- Contrast with magnetic storage (hard disks, tape), which uses magnetised regions, and solid-state storage, which uses trapped charge in flash cells.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Optical storage devices - CD, DVD and Blu-ray drives - store data as a spiral track of pits and lands on a reflective layer.
2. A focused laser beam is directed at the disc surface; the difference in reflected intensity between pits and lands is detected by a photodiode and decoded into bits.
3. For recordable media, a higher-power pulse from the same laser alters the dye or phase-change layer to write data.
4. The essential medium is therefore laser light - a coherent, precisely focusable beam. Options (a), (b) and (d) name broad light bands rather than the laser itself; while a CD laser happens to be infrared and Blu-ray blue-violet, what defines optical storage is the use of a laser.
5. Hence option (c) is correct.

C4 · COMPUTER APPLICATION AND DATA PROCESSING

Number Systems & Binary Arithmetic

24 questions · 2018: 6 2019: 2 2020: 5 2021: 3 2022: 2 2023: 2 2024: 2 2025: 1 2026: 1

Weight. 24 questions (13.3% of Computer Application and Data Processing) · appeared in 9 of 9 papers · peak year 2018 (6)

Examiner's pattern. Base conversion is guaranteed marks every year. The octal 6251 -> hexadecimal CA9 item was asked in both 2021 and 2023; 2's complement of -59 in an 8-bit word appeared in both 2022 and 2024. Also: weighted vs non-weighted codes, the maximum integer in an n-bit word, and normalised floating-point form.

Must-know. Octal -> hex via binary: group in 3s, regroup in 4s. 2's complement = 1's complement + 1. Max unsigned in n bits = 2^n - 1; max signed = 2^(n-1) - 1. BCD and 8421 are weighted; Gray and ASCII are not.

1. 2018 · Q62 Number Systems & Binary Arithmetic · Numerical

QUESTION

The decimal equivalent of octal 325.12 is

OPTIONS

(a) 532.123

(b) 213.15625

(c) 213.56121

(d) 124.123

ANSWER (AI-derived — no official key exists for this paper)

(b) 213.15625

EXAM SHORTCUT (AI-derived explanation)

Integer part: $3(64) + 2(8) + 5 = 213$. Fractional part: $1/8 + 2/64 = 0.125 + 0.03125 = 0.15625$. Answer 213.15625.

TIPS & TRICKS (AI-derived explanation)

- Octal place values run 64, 8, 1 to the left of the point and $1/8, 1/64, 1/512$ to the right. Write the weights above the digits before multiplying.
- Every octal fraction terminates in decimal because 8 is a power of 2, so the answer must be an exact decimal - options (a), (c) and (d) with their ragged tails are implausible on that ground alone.
- Do the integer part first; 213 alone already isolates options (b) and (c), and only then does the fraction decide between them.
- Alternative route: convert 325.12 (octal) to binary as 011 010 101 . 001 010 and then group in fours - useful when the question asks for a hexadecimal answer instead.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Octal 325.12 has digits 3, 2, 5 before the point and 1, 2 after it.

2. Integer part: $3 \times 8^2 + 2 \times 8^1 + 5 \times 8^0 = 3(64) + 2(8) + 5 = 192 + 16 + 5 = 213.$
3. Fractional part: $1 \times 8^{-1} + 2 \times 8^{-2} = 1/8 + 2/64 = 0.125 + 0.03125 = 0.15625.$
4. Adding: $213 + 0.15625 = 213.15625.$
5. Hence option (b) is correct.

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2. **2018 · Q65** Number Systems & Binary Arithmetic · Conceptual

QUESTION

Regarding representations of negative integers, 2's complement is generally used because: 1. More numbers can be stored in 2's complement. 2. Arithmetic operations can be performed faster. 3. It is easy to represent negative numbers in 2's complement.

OPTIONS

(a) 1 and 3 only

(b) 2 and 3 only

(c) 1 and 2 only

(d) 1, 2 and 3

ANSWER **(AI-derived — no official key exists for this paper)**

(d) 1, 2 and 3

EXAM SHORTCUT **(AI-derived explanation)**

All three advantages are standard: only one zero representation means one extra value can be stored, subtraction becomes plain addition with no end-around carry (faster), and negation is a simple invert-and-add-one.

TIPS & TRICKS **(AI-derived explanation)**

- Range comparison for 8 bits: sign-magnitude and $1's$ complement give -127 to +127 (two zeros), while $2's$ complement gives -128 to +127 - one extra number stored.
- Speed advantage: in $2's$ complement A - B is computed as A + ($2's$ complement of B) with the carry simply discarded; $1's$ complement needs an end-around carry, an extra cycle.
- The elimination of negative zero is the single fact behind advantages 1 and 2 - remember it and both follow.
- When every listed statement is a textbook advantage of the same design choice, 'all of them' is usually the key; verify only the one that sounds weakest.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Statement 1: in sign-magnitude and $1's$ complement representations, zero has two encodings (+0 and -0), wasting one pattern. In $2's$ complement zero has a single encoding, so an n-bit word represents 2^n distinct values from -2^{n-1} to $2^{n-1} - 1$, one more than the alternatives. TRUE.
2. Statement 2: $2's$ complement lets subtraction be carried out by the ADDER alone - A - B becomes A + (complement of B) with the final carry simply discarded, whereas $1's$ complement requires an end-around carry. Arithmetic is therefore faster and the hardware simpler. TRUE.

3. Statement 3: negating a number needs only a bitwise inversion followed by adding 1, a cheap and uniform operation, and the sign is read directly from the most significant bit. TRUE.
4. All three are the standard reasons for preferring $2's$ complement.
5. Hence option (d) is correct.

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3. 2018 · Q66 Database, Information & DBMS · Conceptual

QUESTION

The correct normalised form for 11.234 is

OPTIONS

(a) 11.234

(b) $.11234E+2$

(c) $.011234E+1$

(d) $1.1234E \times 10$

ANSWER (AI-derived — no official key exists for this paper)

(b) $.11234E+2$

EXAM SHORTCUT (AI-derived explanation)

Normalised form places the mantissa in [0.1, 1) with a leading zero before the point: $11.234 = 0.11234 \times 10^2$, written .11234E+2.

TIPS & TRICKS (AI-derived explanation)

- In the classical (FORTRAN-style) normalised form the mantissa must satisfy 0.1 <= |m| < 1 - i.e. the first digit AFTER the decimal point is non-zero.
- Option (c) .011234E+1 has a leading zero in the mantissa, so it is un-normalised even though its value is correct. 'Correct value' and 'normalised' are different tests.
- Option (d) mixes two notations (E and x10) and is syntactically invalid; option (a) is simply the unconverted number.
- Count the shift: moving the point two places left multiplies by 10^2, so the exponent is +2. Direction rule: point moves left, exponent goes up.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A floating-point number is in normalised form when it is written as +-0.d1 d2 d3 ... $x10^e$ with the leading digit d1 non-zero, so that the mantissa lies in [0.1, 1).
2. Take 11.234 and shift the decimal point two places to the left: $11.234 = 0.11234 \times 10^2$.
3. The mantissa 0.11234 lies in [0.1, 1) and its first digit after the point is 1, which is non-zero, so the form is normalised.
4. In E-notation this is written .11234E+2.
5. Option (c) has value $0.011234 \times 10 = 0.11234$, which is not even equal to 11.234, and its mantissa is un-normalised; options (a) and (d) are not in normalised E-notation at all.

6. Hence option (b) is correct.

---

4. **2018 · Q67** Memory - RAM, ROM & Units of Computer Memory · Numerical

QUESTION

The maximum decimal integer that can be stored in an 8-bit word processor is

OPTIONS

(a) $(128)_{10}$

(b) $(127)_{10}$

(c) $(129)_{10}$

(d) $(130)_{10}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $(127)_{10}$

EXAM SHORTCUT (AI-derived explanation)

An 8-bit signed word uses 1 bit for the sign and 7 for magnitude, so the largest integer is $2^7 - 1 = 127$.

TIPS & TRICKS (AI-derived explanation)

- Signed n-bit range in $2's$ complement: -2^{n-1} to +$2^{n-1} - 1$. For $n = 8$ that is -128 to +127.
- 128 (option a) is the magnitude of the most NEGATIVE value, not the largest positive one - the classic off-by-one trap.
- If the question had said 'unsigned', the answer would be $2^8 - 1 = 255$. Always check whether a sign bit is implied; 'integer' in these papers means signed.
- Powers of two to have ready: $2^7 = 128, 2^{15} = 32768, 2^{31} = 2147483648$ - the three standard maxima plus/minus one.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. An 8-bit word holds $2^8 = 256$ distinct bit patterns.
2. For signed integers, one bit is used for the sign and the remaining 7 bits carry the magnitude.
3. In $2's$ complement the representable range is -2^{8-1} to $2^{8-1} - 1$, that is -128 to +127.
4. Hence the maximum decimal integer that can be stored is $2^7 - 1 = 127$.
5. Option (b) is correct.

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5. **2018 · Q73** Number Systems & Binary Arithmetic · Factual

QUESTION

Consider the following statements: 1. Weighted 4-bit BCD code is known as the 1246-weighted code. 2. The octal system is also known as the base-8 system. 3. The octal system is known as the base-2 system. 4. Weighted 4-bit BCD code is known as the 8421-weighted code.

OPTIONS

(a) 2 only

(b) 3 and 4 only

(c) 1 and 2 only

(d) 2 and 4 only

ANSWER (AI-derived — no official key exists for this paper)

(d) 2 and 4 only

EXAM SHORTCUT (AI-derived explanation)

The standard weighted 4-bit BCD code is 8421, so statement 4 is true and statement $1(1246)$ is invented. Octal is base 8, so statement 2 is true and statement 3 is false. Answer: 2 and 4.

TIPS & TRICKS (AI-derived explanation)

- 8421 BCD: the four bit positions carry weights 8, 4, 2, 1 - the same as ordinary binary, which is why it is called the 'natural' BCD code.
- Other weighted codes worth knowing: 2421 (Aiken) and 5211; the non-weighted ones are Excess-3 and Gray. '1246' is not a code at all.
- Statements 1 and 4 are a true/false PAIR about the same fact, as are 2 and 3. Spot the pairs, decide each fact once, and the option follows automatically.
- Base names: binary = 2, octal = 8, decimal = 10, hexadecimal = 16. Octal digits run 0-7, which is the quickest way to remember base 8.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: the weighted 4-bit BCD code has weights 8, 4, 2, 1 - there is no '1246-weighted' code. FALSE.
2. Statement 2: the octal number system uses eight digits, 0 to 7, and is therefore the base-8 system. TRUE.
3. Statement 3: base-2 is the binary system, not octal. FALSE.
4. Statement 4: the standard weighted 4-bit BCD code is indeed the 8421 code. TRUE.
5. Only statements 2 and 4 are correct.
6. Hence option (d) is correct.

---

6. 2018 · Q74 Memory - RAM, ROM & Units of Computer Memory · Conceptual

QUESTION

The maximum permissible integer in a computer with an n-bit word and one word per integer is

OPTIONS

(a) $2^n - 1$

(b) $2^{n-1} - 1$

(c) $2^{n-1} + 1$

(d) $2^n + 1$

ANSWER (AI-derived — no official key exists for this paper)

(b) $2^{n-1} - 1$

EXAM SHORTCUT (AI-derived explanation)

One bit of the n-bit word carries the sign, leaving n - 1 magnitude bits, so the largest integer is $2^{n-1} - 1$.

TIPS & TRICKS (AI-derived explanation)

- This is the general form of the 8-bit result $127 = 2^7 - 1$; check any general formula against the concrete case you already know.
- Option $(a) 2^n - 1$ is the UNSIGNED maximum. The word 'integer' in these papers implies a signed representation, so one bit is lost to the sign.
- Options (c) and (d) add rather than subtract 1 and are impossible: the count of non-negative patterns is 2^{n-1}, and the largest of them (starting from 0) must be $2^{n-1} - 1$.
- Symmetric fact worth pairing: the minimum is -2^{n-1} in $2's$ complement, which is one further from zero than the maximum.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. An n-bit word provides 2^n distinct bit patterns.
2. For signed integers the most significant bit is the sign bit, so 2^{n-1} patterns are available for non-negative values.
3. Those non-negative values run from 0 up to $2^{n-1} - 1$.
4. Hence the maximum permissible integer is $2^{n-1} - 1$.
5. For $n = 8$ this gives 127, agreeing with the standard 8-bit result.
6. Option (b) is correct.

---

7. 2019 · Q42

Number Systems & Binary Arithmetic · Factual

QUESTION

Which of the following codes is/are positional weighted code(s)? 1. Gray code 2. BCD code 3. ASCII code 4. Excess-3 code

OPTIONS

(a) 1 and 4

(b) 1 and 2 only

(c) 1, 2 and 3

(d) 2 only

ANSWER (AI-derived — no official key exists for this paper)

(d) 2 only

EXAM SHORTCUT (AI-derived explanation)

Only BCD is positionally weighted (weights 8, 4, 2, 1). Gray, ASCII and Excess-3 are all non-weighted codes.

TIPS & TRICKS (AI-derived explanation)

- Weighted codes: 8421 (BCD), 2421 (Aiken), $5211, 84-2-1$. Non-weighted codes: Gray, Excess-3, ASCII, EBCDIC.
- Excess-3 is derived FROM 8421 BCD by adding 3, but the shift destroys positional weighting - a favourite trap because it looks like BCD.
- Gray code is designed so that successive values differ in exactly one bit; that property is incompatible with fixed positional weights.
- ASCII is an alphanumeric character code, not a numeric code at all, so weighting is meaningless for it.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A positional weighted code assigns a fixed weight to each bit position, so the decimal value is the weighted sum of the bits.
2. Item 1, Gray code: consecutive numbers differ in a single bit and no fixed weights can be assigned. NOT weighted.
3. Item 2, BCD (8421) code: the four bit positions carry weights 8, 4, 2 and 1, and the digit is their weighted sum. WEIGHTED.
4. Item 3, ASCII: a 7-bit alphanumeric character code with no numeric weighting. NOT weighted.
5. Item 4, Excess-3 code: formed by adding 3 to the 8421 BCD value, which destroys the positional weighting. NOT weighted.
6. Only item 2 is a positional weighted code, option (d).

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8. 2019 · Q43 Number Systems & Binary Arithmetic · Numerical

QUESTION

When $(217)_{10}$ is divided by $(12)_{10}$, the quotient and remainder in binary system will be respectively

OPTIONS

(a) 11010 and 1001

(b) 10100 and 0001

(c) 10010 and 0001

(d) 10010 and 0010

ANSWER (AI-derived — no official key exists for this paper)

(c) 10010 and 0001

EXAM SHORTCUT (AI-derived explanation)

$217 = 12 \times 18 + 1$, so the quotient is 18 and the remainder 1. In binary, $18 = 10010$ and $1 = 0001$.

TIPS & TRICKS (AI-derived explanation)

- Do the division in DECIMAL first, then convert both results. Converting to binary before dividing wastes time and invites errors.
- $18 = 16 + 2 = 2^4 + 2^1$ gives 10010 directly; recognising the powers of two beats repeated division by 2.
- Check by reconstruction: $12 \times 18 = 216$, plus the remainder 1, gives 217 exactly.
- Option (d) has remainder $0010 = 2$ and option (a) has quotient $11010 = 26$ - both fail the reconstruction test at once.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Divide in decimal: $217/12 = 18$ with a remainder, since $12 \times 18 = 216$ and $217 - 216 = 1$.
2. So the quotient is 18 and the remainder is 1.
3. Convert the quotient: $18 = 16 + 2 = 2^4 + 2^1$, so 18 in binary is 10010.
4. Convert the remainder: 1 in binary, written to four places, is 0001.
5. Hence the quotient and remainder are 10010 and 0001.
6. Option (c) is correct.

---

9. **2020 · Q43** Number Systems & Binary Arithmetic · Factual

QUESTION

Which are weighted codes? 1. BCD code 2. GRAY code 3. ASCII code 4. 8421 code

OPTIONS

(a) 1 and 3 only

(b) 1 and 4 only

(c) 3 and 4 only

(d) 2 and 3 only

ANSWER (AI-derived — no official key exists for this paper)

(b) 1 and 4 only

EXAM SHORTCUT (AI-derived explanation)

BCD and 8421 are two names for the SAME weighted code, so items 1 and 4 are both weighted. Gray and ASCII are non-weighted.

TIPS & TRICKS (AI-derived explanation)

- Recognise the synonym: 'BCD code' and '8421 code' denote the same thing, so they must stand or fall together - only option (b) keeps them together.
- Weighted codes: 8421 (BCD), 2421 (Aiken), $5211, 84-2-1$. Non-weighted: Gray, Excess-3, ASCII, EBCDIC.
- Gray code's defining property - consecutive values differ in exactly one bit - is incompatible with fixed positional weights.
- ASCII is an alphanumeric character code, so positional weighting does not even apply to it.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A weighted code assigns a fixed weight to each bit position, so the value is the weighted sum of the bits.
2. Item 1, BCD code: the standard binary-coded decimal code has weights 8, 4, 2, 1. WEIGHTED.
3. Item 4, 8421 code: this is simply the explicit name of the same 8421 BCD code. WEIGHTED.
4. Item 2, Gray code: designed so that successive values differ in one bit only; no fixed weights can be assigned. NOT weighted.
5. Item 3, ASCII: a 7-bit alphanumeric character code with no numeric positional weighting. NOT weighted.
6. Hence items 1 and 4 are the weighted codes, option (b).

---

10. 2020 · Q45 Number Systems & Binary Arithmetic · Factual

QUESTION

Which is the most popular method for representing negative numbers in a computer system?

OPTIONS

(a) One's complement

(b) Signed magnitude

(c) Two's complement

(d) Floating-point system

ANSWER (AI-derived — no official key exists for this paper)

(c) Two's complement

EXAM SHORTCUT (AI-derived explanation)

Two's complement is the universal choice: one representation of zero, one extra representable value, and subtraction performed by the adder alone.

TIPS & TRICKS (AI-derived explanation)

- Three reasons to remember: unique zero, range -2^{n-1} to $2^{n-1} - 1$, and subtraction implemented as addition with the carry discarded.
- One's complement suffers from negative zero and needs an end-around carry, which costs an extra cycle - the reason it lost.
- Signed magnitude also has two zeros and requires separate add and subtract logic depending on the signs.
- Floating point (option d) represents real numbers generally; it is not a scheme for representing negative INTEGERS and is out of category.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. In signed-magnitude representation the sign bit is separate from the magnitude, giving two encodings of zero (+0 and -0) and requiring different hardware paths for addition and subtraction.

2. One's complement also has two zeros and needs an end-around carry when adding, which slows arithmetic.
3. Two's complement has a single representation of zero, gives one extra representable value (range - 2^{n-1} to $2^{n-1} - 1$), and allows A - B to be computed as A plus the two's complement of B using the ordinary adder with the final carry discarded.
4. These advantages make two's complement the standard in virtually all modern processors.
5. Floating point is a scheme for representing real numbers, not specifically negative integers.
6. Hence the answer is two's complement, option (c).

---

11. 2020 · Q50 Number Systems & Binary Arithmetic · Numerical

QUESTION

In the hexadecimal system, the binary number $(111110111.110101)_2$ is equivalent to

OPTIONS

(a) $(2F7.C4)_{16}$

(b) $(1F7.D5)_{16}$

(c) $(1F7.D4)_{16}$

(d) $(2F7.D4)_{16}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $(1F7.D4)_{16}$

EXAM SHORTCUT (AI-derived explanation)

Group the bits in fours: $000111110111 = 1F7$ for the integer part, and $11010100 = D_4$ for the fraction. Answer $(1F7.D_4)16$.

TIPS & TRICKS (AI-derived explanation)

- Binary to hex is pure grouping: four bits per hex digit. Group the INTEGER part leftwards from the point and the FRACTION rightwards from the point.
- Pad with zeros in the right direction: leading zeros for the integer part (1 becomes 0001) and TRAILING zeros for the fraction (01 becomes 0100).
- Padding the fraction on the wrong side is the single most common error and turns D4 into D5 - exactly the difference between options (b) and (c).
- Verify the integer digit count: nine bits give three hex digits, so the answer must start with 1, not 2. That eliminates options (a) and (d) immediately.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Take the integer part 111110111 and group it in fours from the binary point leftwards: $1|1110|1111$... more carefully, from the right: 0111, then 1111, then the remaining 1 padded to 0001.
2. So the integer part is $000111110111 = 1, F, 7 = 1F7$ in hexadecimal.
3. Take the fractional part 110101 and group it in fours from the binary point rightwards: 1101 | 01, padding the last group with trailing zeros to give 0100.

4. So the fraction is $11010100 = D, 4 = D_4$ in hexadecimal.
5. Combining, $(111110111.110101)_2 = (1F7.D_4)_{16}$.
6. Option (c) is correct.

---

12. 2020 · Q51 Number Systems & Binary Arithmetic · Numerical

QUESTION

Which decimal number corresponds to 8421-BCD representation 010100100001.0110?

OPTIONS

(a) 573.4

(b) 531.6

(c) 521.6

(d) 421.6

ANSWER (AI-derived — no official key exists for this paper)

(c) 521.6

EXAM SHORTCUT (AI-derived explanation)

Split into nibbles: $0101 = 5, 0010 = 2, 0001 = 1$, and after the point $0110 = 6$. So the decimal number is 521.6.

TIPS & TRICKS (AI-derived explanation)

- In 8421 BCD each DECIMAL digit occupies its own group of four bits, so simply cut the string into nibbles and translate each one separately.
- Do not convert the whole string as a binary number - BCD is a digit-by-digit encoding, and treating it as pure binary gives a completely different value.
- Check the digit count: twelve bits before the point give three decimal digits, and four bits after give one - matching a three-digit integer with one decimal place.
- Valid BCD nibbles run only from 0000 to 1001; anything from 1010 to 1111 would be an invalid BCD group and a sign of a misread.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. In the 8421 BCD system each decimal digit is represented by its own group of four bits.
2. Split the integer part 010100100001 into nibbles: 0101, 0010, 0001.
3. Translate each: $0101 = 5, 0010 = 2, 0001 = 1$, giving the integer digits 5, 2, 1.
4. The fractional part is the single nibble $0110 = 6$.
5. Assembling the digits gives 521.6.
6. Option (c) is correct.

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13. 2020 · Q59 Number Systems & Binary Arithmetic · Conceptual

QUESTION

Which is *not* a valid hexadecimal number?

OPTIONS

(a) 12345

(b) $1AB9F$

(c) $15F6G$

(d) $1A3BC$

ANSWER (AI-derived — no official key exists for this paper)

(c) $15F6G$

EXAM SHORTCUT (AI-derived explanation)

Hexadecimal digits are 0-9 and A-F. The string 15F6G contains G, which is out of range, so it is not a valid hexadecimal number.

TIPS & TRICKS (AI-derived explanation)

- Hex uses sixteen symbols: 0-9 then $A(10), B(11), C(12), D(13), \mathrm{E}(14), F(15)$. Anything beyond F is invalid.
- Scan each option for letters past F - the check takes seconds and needs no conversion.
- Option (a) 12345 uses only decimal digits, which are perfectly valid hex digits; a number can be simultaneously valid in several bases.
- The same reasoning applies elsewhere: 8 and 9 are invalid in octal, and 2 is invalid in binary.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The hexadecimal system is base 16 and uses exactly sixteen digit symbols: 0, 1, 2, ..., 9, A, B, C, D, E, F.
2. Option (a) 12345 uses only the digits 1 to 5, all valid hexadecimal digits.
3. Option (b) 1AB9F uses 1, A, B, 9 and F, all valid.
4. Option (d) 1A3BC uses 1, A, 3, B and C, all valid.
5. Option (c) 15F6G contains the letter G, which lies beyond F and is not a hexadecimal digit.
6. Hence 15F6G is not a valid hexadecimal number, option (c).

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14. 2021 · Q58 Number Systems & Binary Arithmetic · Numerical

QUESTION

Which binary number gives output 100101 when AND-ed with 101101?

OPTIONS

(a) 101101

(b) 100101

(c) 111010

(d) 110011

ANSWER (AI-derived — no official key exists for this paper)

(b) 100101

EXAM SHORTCUT (AI-derived explanation)

AND can only clear bits, never set them, so the answer must be a subset of 101101 that keeps the 1s of 100101. Testing 100101 AND 101101 gives 100101 exactly.

TIPS & TRICKS (AI-derived explanation)

- Bitwise AND yields 1 only where BOTH operands are 1, so the result is always a 'subset' of each operand's 1-bits - the result 100101 must therefore be attainable from 101101 by clearing bits.
- Compare 101101 with the target 100101: only the third bit from the left changes from 1 to 0, so the unknown must have 0 there and 1 wherever the target has 1.
- The target itself always works when it is a subset of the given operand - check that shortcut first, and here it succeeds immediately.
- Options (c) and (d) have 0 in the fourth position where the result needs a 1, so they fail before any full computation.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write the given operand and the required result bit by bit: 101101 AND $x = 100101$.
2. Bitwise AND produces 1 only where both bits are 1, so wherever the result has a 1 the unknown x must also have a 1: positions 1, 4 and 6 (from the left).
3. Wherever the given operand has a 1 but the result has a 0 - position 3 - the unknown must have a 0.
4. Positions 2 and 5 are 0 in the given operand, so x is unconstrained there.
5. Testing option (b): 101101 AND $100101 = 100101$, exactly the required result.
6. Option (a) gives 101101, and options (c) and (d) have 0 in position 4 so they cannot produce the required 1. Hence option (b).

---

15. 2021 · Q76 Number Systems & Binary Arithmetic · Numerical

QUESTION

Binary equivalent of Gray-coded number 10101110 is

OPTIONS

(a) 10101110

(b) 11011000

(c) 11001011

(d) 01010001

ANSWER (AI-derived — no official key exists for this paper)

(c) 11001011

EXAM SHORTCUT (AI-derived explanation)

Gray to binary: copy the leading bit, then XOR each previous BINARY bit with the next Gray bit. 10101110 gives $1, 1, 0, 0, 1, 0, 1, 1 = 11001011$.

TIPS & TRICKS (AI-derived explanation)

- Conversion rule: $b_1 = g_1$ and $b_i = b_{i-1}$ XOR g_i. Note that the running value uses the BINARY bit just produced, not the Gray bit - this is the step candidates reverse.
- The reverse direction (binary to Gray) is $g_1 = b_1$ and $g_i = b_{i-1}$ XOR b_i, which uses only binary bits. Keep the two rules distinct.
- Work left to right and write the growing binary string underneath the Gray string so each XOR is visually aligned.
- Check: converting 11001011 back to Gray gives 1, 1 XOR $1 = 0, 1$ XOR $0 = 1, 0$ XOR $0 = 0, 0$ XOR $1 = 1, 1$ XOR $0 = 1, 0$ XOR $1 = 1, 1$ XOR $1 = 0$, i.e. 10101110 - the original.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For Gray-to-binary conversion, the most significant binary bit equals the most significant Gray bit, and thereafter $b_i = b_{i-1}$ XOR g_i.
2. Gray code: $g = 1, 0, 1, 0, 1, 1, 1, 0$.
3. $b_1 = g_1 = 1$. $b_2 = b_1$ XOR $g_2 = 1$ XOR $0 = 1$. $b_3 = b_2$ XOR $g_3 = 1$ XOR $1 = 0$. $b_4 = b_3$ XOR $g_4 = 0$ XOR $0 = 0$.
4. $b_5 = b_4$ XOR $g_5 = 0$ XOR $1 = 1$. $b_6 = b_5$ XOR $g_6 = 1$ XOR $1 = 0$. $b_7 = b_6$ XOR $g_7 = 0$ XOR $1 = 1$. $b_8 = b_7$ XOR $g_8 = 1$ XOR $0 = 1$.
5. So the binary equivalent is 11001011.
6. Reconverting to Gray reproduces 10101110, confirming the result. Option (c) is correct.

---

16. 2021 · Q78 Number Systems & Binary Arithmetic · Numerical

QUESTION

The hexadecimal equivalent of $(6251)_8$ is

OPTIONS

(a) CA8

(b) BA9

(c) CA9

(d) CA0

ANSWER (AI-derived — no official key exists for this paper)

(c) CA9

EXAM SHORTCUT (AI-derived explanation)

Octal to binary in triples: $6251 = 110010101001$. Regroup in fours from the right: 1100 1010 1001 = CA9.

TIPS & TRICKS (AI-derived explanation)

- Octal to hexadecimal always goes through BINARY - there is no direct digit correspondence because 8 and 16 are different powers of 2.
- Three bits per octal digit, four bits per hex digit. Convert to binary, then regroup from the RIGHT.
- Check by decimal: $(6251)_8 = 6(512) + 2(64) + 5(8) + 1 = 3072 + 128 + 40 + 1 = 3241$, and $(CA9)_{16} = 12(256) + 10(16) + 9 = 3072 + 160 + 9 = 3241$. They agree.
- The same conversion appears again as 2023 Q79 - it is a repeated item, so the technique is worth drilling.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Convert each octal digit to three binary bits: $6 = 110, 2 = 010, 5 = 101, 1 = 001$.
2. Concatenating gives the binary number 110010101001.
3. Regroup into fours starting from the right: $1100|1010|1001$.
4. Convert each group to a hexadecimal digit: $1100 = C, 1010 = A, 1001 = 9$.
5. So $(6251)_8 = (CA9)_{16}$.
6. Check in decimal: $6(512) + 2(64) + 5(8) + 1 = 3241$, and $C(256) + A(16) + 9 = 3072 + 160 + 9 = 3241$. Option (c) is correct.

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17. 2022 · Q32 Number Systems & Binary Arithmetic · Conceptual

QUESTION

The highest digit possible in any number system is:

OPTIONS

(a) equal to base of the number system

(b) equal to one less than the base of the number system

(c) equal to one more than the radix of number system

(d) equal to two less than the base of the number system

ANSWER (AI-derived — no official key exists for this paper)

(b) equal to one less than the base of the number system

EXAM SHORTCUT (AI-derived explanation)

A base-b system uses the digits 0 through b - 1, so the highest digit is one less than the base.

TIPS & TRICKS (AI-derived explanation)

- Anchor with examples: binary (base 2) tops out at 1; octal (base 8) at 7; decimal (base 10) at 9; hexadecimal (base 16) at $F = 15$. Every one is base minus one.

- The base itself can never be a single digit - it is written as '10' in its own system, which is exactly why the maximum digit is b - 1.
- 'Radix' is a synonym for base, so option (c) - one more than the radix - is even further from the truth than option (a).
- This is the fastest way to spot invalid numerals: an 8 or 9 in an octal number, or a 2 in a binary one, is immediately illegal.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A positional number system with base (radix) b represents values using exactly b distinct digit symbols.
2. These symbols denote the values 0, 1, 2, ..., b - 1.
3. The largest of them is therefore b - 1, one less than the base.
4. The base itself is not a digit; in its own system it is written as the two-symbol numeral '10'.
5. Examples confirm this: binary's largest digit is 1, octal's is 7, decimal's is 9 and hexadecimal's is $F(=15)$.
6. Hence the highest digit equals one less than the base, option (b).

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18. 2022 · Q39 Number Systems & Binary Arithmetic · Numerical

QUESTION

In octal number system, the hexadecimal number $(32FC.75)$ is equivalent to:

OPTIONS

(a) $(31375.452)_8$

(b) $(73256.732)_8$

(c) $(31374.352)_8$

(d) $(26753.175)_8$

ANSWER (AI-derived — no official key exists for this paper)

(c) $(31374.352)_8$

EXAM SHORTCUT (AI-derived explanation)

Convert through binary: 32FC.$75 = 0011001011111100$. 0111 0101. Regrouping in threes gives 31374.352 in octal.

TIPS & TRICKS (AI-derived explanation)

- Hex to octal must pass through BINARY - four bits per hex digit out, three bits per octal digit in. There is no direct digit mapping.
- Group the integer part in threes from the BINARY POINT leftwards and the fractional part from the point rightwards, padding each end with zeros as needed.
- Integer part: 11001011111100 pads to $011001011111100 = 3, 1, 3, 7, 4$. Fraction: 01110101 pads to $011101010 = 3, 5, 2$.

- Decimal check: $(32FC)_{16} = 13052$ and $(31374)_8 = 3(4096) + 1(512) + 3(64) + 7(8) + 4 = 13052$. The integer parts agree.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Convert each hexadecimal digit to four binary bits: $3 = 0011, 2 = 0010, F = 1111, C = 1100$, and after the point $7 = 0111, 5 = 0101$.
2. So the number in binary is 0011 0010 1111 1100 . 0111 0101, i.e. 11001011111100.01110101.
3. For octal, regroup the integer part in threes from the binary point leftwards, padding with leading zeros: 011 001 011 111 100.
4. These groups give the octal digits 3, 1, 3, 7, 4.
5. Regroup the fractional part in threes from the point rightwards, padding with a trailing zero: 011 101 010, giving 3, 5, 2.
6. Hence $(32FC.75)_{16} = (31374.352)_8$, option (c).

---

19. 2023 · Q67 Number Systems & Binary Arithmetic · Numerical

QUESTION

Consider the following statements: 1. In number system if $(110)_X = (1100)_2$, then the value of X is 3. 2. The gray coded equivalent of the binary number $(01010010)_2$ is (01111011). Which of the statements given above is/are correct?

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(c) Both 1 and 2

EXAM SHORTCUT (AI-derived explanation)

$(1100)_2 = 12$ and $(110)_X = X^2 + X$, so $X^2 + X - 12 = 0$ gives $X = 3$. And converting 01010010 to Gray by XOR-ing adjacent bits gives 01111011. Both statements hold.

TIPS & TRICKS (AI-derived explanation)

- Convert the known side first: $(1100)_2 = 8 + 4 = 12$, then solve the quadratic $X^2 + X = 12$ by factorising $(X + 4)(X - 3) = 0$ and discarding the negative root.
- Binary to Gray: keep the leading bit, then XOR each pair of ADJACENT BINARY bits. (The reverse direction uses the running binary result, which is easy to confuse.)
- Work the XORs across: $0|0^1 = 1|1^0 = 1|0^1 = 1|1^0 = 1|0^0 = 0|0^1 = 1 \mid 1^0 = 1$, giving 01111011 - matching the claim.
- A base must exceed its largest digit; $X = 3$ is consistent with the digits 1 and 0 appearing in $(110)_X$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: $(1100)_2 = 8+4+0+0 = 12$, and $(110)_X = 1(X^2)+1(X)+0 = X^2+X$.
2. Setting $X^2+X = 12$ gives $X^2+X-12 = 0$, i.e. $(X+4)(X-3) = 0$, so $X = 3$ (the negative root is inadmissible). TRUE.
3. Statement 2: to convert binary to Gray, the leading Gray bit equals the leading binary bit and thereafter $g_i = b_{i-1}$ XOR b_i.
4. With $b =$ 0,1,0,1,0,0,1,0: $g_1 = 0$; $g_2 = 0$ XOR $1 = 1$; $g_3 = 1$ XOR $0 = 1$; $g_4 = 0$ XOR $1 = 1$; $g_5 = 1$ XOR $0 = 1$; $g_6 = 0$ XOR $0 = 0$; $g_7 = 0$ XOR $1 = 1$; $g_8 = 1$ XOR $0 = 1$.
5. So the Gray code is 01111011, exactly as stated. TRUE.
6. Both statements are correct, option (c).

---

20. 2023 · Q79 Number Systems & Binary Arithmetic · Numerical

QUESTION

What is the equivalent hexadecimal number of the octal number 6251?

OPTIONS

(a) 9CA

(b) AC9

(c) DB9

(d) CA9

ANSWER (AI-derived — no official key exists for this paper)

(d) CA9

EXAM SHORTCUT (AI-derived explanation)

Octal to binary in triples: $6251 = 110010101001$. Regrouping in fours from the right gives 1100 1010 1001 = CA9.

TIPS & TRICKS (AI-derived explanation)

- Octal to hexadecimal must pass through BINARY: three bits per octal digit out, four bits per hex digit in.
- Always regroup from the RIGHT for the integer part, padding the leftmost group with zeros if necessary.
- Decimal check: $(6251)_8 = 6(512)+2(64)+5(8)+1 = 3241$, and $(CA9)_{16} = 12(256)+10(16)+9 = 3241$. They agree.
- This is a verbatim repeat of 2021 Q78, so the conversion is clearly worth drilling until it is automatic.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Convert each octal digit to three binary bits: $6 = 110, 2 = 010, 5 = 101, 1 = 001$.
2. Concatenating gives the binary number 110010101001.
3. Regroup into fours from the right: $1100|1010|1001$.
4. Convert each group: $1100 = C, 1010 = A, 1001 = 9$.

5. So $(6251)_8 = (CA9)_{16}$.
6. Decimal check: $6(512)+2(64)+5(8)+1=3241$ and $C(256)+A(16)+9=3241$. Option (d) is correct.

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21. 2024 · Q46 Number Systems & Binary Arithmetic · Numerical

QUESTION

What is the representation of -59 in the 2's complement system?

OPTIONS

(a) 11000101

(b) 11000100

(c) 10000101

(d) 11000001

ANSWER (AI-derived — no official key exists for this paper)

(a) 11000101

EXAM SHORTCUT (AI-derived explanation)

$59 = 00111011$; invert to get 11000100; add 1 to get 11000101.

TIPS & TRICKS (AI-derived explanation)

- Two's complement = one's complement PLUS ONE. Stopping at the inversion gives 11000100, which is planted as option (b).
- $59 = 32+16+8+2+1 = 00111011$ - build from powers of two rather than by repeated division.
- The sign bit must be 1 for a negative number, which eliminates any option beginning with 0.
- Verify by addition: $00111011 + 11000101 = 100000000$, and discarding the carry leaves zero - the definitive check.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write the magnitude in 8-bit binary: $59 = 32+16+8+2+1$, so $59 = 00111011$.
2. Take the one's complement by inverting every bit: 11000100.
3. Add 1: $11000100 + 1 = 11000101$.
4. The leading bit is 1, correctly marking a negative value.
5. Check: $00111011 + 11000101 = 100000000$; discarding the carry out of the 8-bit word leaves 00000000, confirming the two are additive inverses.
6. Hence -59 is 11000101, option (a).

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22. 2024 · Q54 Number Systems & Binary Arithmetic · Numerical

QUESTION

Consider the following statements: I. The hexadecimal equivalent of $(1101010101)_2$ is 355. II. The complement of $(47)_{10}$ is $(53)_{10}$. Which of the statements given above is/are correct?

OPTIONS

(a) I only

(b) II only

(c) Both I and II

(d) Neither I nor II

ANSWER (AI-derived — no official key exists for this paper)

(c) Both I and II

EXAM SHORTCUT (AI-derived explanation)

Grouping 1101010101 in fours from the right gives $001101010101 = 355$ in hex, so statement I holds. And the radix $(10's)$ complement of 47 is $100 - 47 = 53$, so statement II holds too.

TIPS & TRICKS (AI-derived explanation)

- Binary to hex: pad on the LEFT to a multiple of four bits, then translate each nibble. Here 1101010101 becomes $0011 \mid 0101 \mid 0101 = 3, 5, 5$.
- Decimal check for statement I: $(1101010101)_2 = 512 + 256 + 64 + 16 + 4 + 1 = 853$, and $(355)_{16} = 3(256) + 5(16) + 5 = 853$. They agree.
- For a two-digit decimal number the $r's(10's)$ complement is 100 - n, so $100 - 47 = 53$; the $(r-1)'s(9's)$ complement would be $99 - 47 = 52$.
- When 'complement' is stated without qualification for base r it conventionally means the $r's$ complement - which is what makes statement II true as printed.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement I: group the binary digits in fours from the right, padding on the left: 1101010101 becomes $0011|0101|0101$.
2. Translating each group: $0011 = 3, 0101 = 5, 0101 = 5$, so the hexadecimal equivalent is 355.
3. Decimal verification: $(1101010101)_2 = 512 + 256 + 64 + 16 + 4 + 1 = 853$, and $(355)_{16} = 3(256) + 5(16) + 5 = 768 + 80 + 5 = 853$. TRUE.
4. Statement II: for a decimal number of n digits, the radix $(10's)$ complement is 10^n minus the number. With $n = 2$ and the number 47, this is $100 - 47 = 53$.
5. (The diminished-radix or $9's$ complement would be $99 - 47 = 52$; 'the complement' without qualification conventionally denotes the radix complement.) TRUE.
6. Both statements are correct, option (c).

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23. 2025 · Q35 Number Systems & Binary Arithmetic · Numerical

QUESTION

Consider the statements: I. The hexadecimal equivalent of octal number 366 is 0F6. II. The decimal equivalent of hexadecimal number A53 is 2643. Which is/are correct?

OPTIONS

(a) I only

(b) II only

(c) Both I and II

(d) Neither I nor II

ANSWER (AI-derived — no official key exists for this paper)

(c) Both I and II

EXAM SHORTCUT (AI-derived explanation)

Octal 366 is binary 011110110, which regroups in fours as $000011110110 = 0F6$. And hex $A_{53} = 10(256) + 5(16) + 3 = 2643$. Both statements hold.

TIPS & TRICKS (AI-derived explanation)

- Octal to hex must pass through BINARY: three bits per octal digit out, four bits per hex digit in. There is no direct digit correspondence.
- 366 (octal) becomes 011 110 110, and regrouping from the right gives 1111 0110 with a leading nibble of zeros - hence the printed leading 0 in 0F6.
- For statement II use place values 256, 16, 1: $10(256) = 2560, 5(16) = 80$, plus 3 gives 2643.
- Cross-check statement I in decimal: $(366)_8 = 3(64) + 6(8) + 6 = 246$, and $(F_6)_{16} = 15(16) + 6 = 246$. They agree.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement I: convert each octal digit to three bits: $3 = 011, 6 = 110, 6 = 110$, giving the binary 011110110.
2. Regroup in fours from the right, padding on the left: $0000|1111|0110$, which translates to 0, F, 6 - that is 0F6.
3. Decimal check: $(366)_8 = 3(64) + 6(8) + 6 = 246$, and $(F_6)_{16} = 15(16) + 6 = 246$. TRUE.
4. Statement II: $(A_{53})_{16} = A(16^2) + 5(16) + 3$ with $A = 10 = 10(256) + 80 + 3 = 2560 + 83 = 2643$. TRUE.
5. Both statements are correct, option (c).

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24. 2026 · Q63 Number Systems & Binary Arithmetic · Numerical

QUESTION

Consider: I. The equivalent octal number of decimal 45796 is 131344. II. The equivalent hexadecimal number of octal 365 is 0F5. Which is/are correct?

OPTIONS

(a) I only

(b) II only

(c) Both I and II

(d) Neither I nor II

ANSWER **(AI-derived — no official key exists for this paper)**

(c) Both I and II

EXAM SHORTCUT **(AI-derived explanation)**

Repeated division of 45796 by 8 gives remainders 4, 4, 3, 1, 3, 1, i.e. 131344 - statement I checks out. Octal 365 is 011 110 101 in binary; regrouping in fours gives $11110101 = F_5$, so statement II checks out too. Answer (c).

TIPS & TRICKS **(AI-derived explanation)**

- Decimal to octal: divide repeatedly by 8 and read the remainders bottom-up.
- Octal to hexadecimal always goes via binary: expand each octal digit to 3 bits, then regroup the bit string into blocks of 4 from the right.
- Octal $365 = 3(64) + 6(8) + 5 = 245$ in decimal, and $245 = 15(16) + 5 = F_5$ in hex - a fast independent check.
- A leading zero $(0F5)$ does not change the value; treat it as cosmetic, not as an error in the statement.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Statement I: convert 45796 to octal by repeated division by 8.
2. $45796 = 8(5724) + 4$; $5724 = 8(715) + 4$; $715 = 8(89) + 3$; $89 = 8(11) + 1$; $11 = 8(1) + 3$; $1 = 8(0) + 1$.
3. Reading the remainders from last to first gives $131344 = 131344$ (octal). Statement I is correct.
4. Statement II: expand octal 365 into binary, three bits per digit: $3 = 011, 6 = 110, 5 = 101$, giving 011110101.
5. Regroup from the right in blocks of four: 0 1111 0101, i.e. $11110101 = F_5$ in hexadecimal (written 0F5 with a leading zero).
6. Check in decimal: octal $365 = 3(64) + 6(8) + 5 = 192 + 48 + 5 = 245$, and $245 = 15(16) + 5 = F_5$. Statement II is correct.
7. Both statements hold.
8. Hence option (c) is correct.

C5 · COMPUTER APPLICATION AND DATA PROCESSING

Software, Operating Systems, Packages & Utilities

24 questions · 2018: 2 2019: 4 2020: 0 2021: 3 2022: 5 2023: 3 2024: 2 2025: 2 2026: 3

Weight. 24 questions (13.3% of Computer Application and Data Processing) · appeared in 8 of 9 papers · peak year 2022 (5)

Examiner's pattern. Operating-system concepts, broadly. Recurring: OS functions and goals, real-time vs embedded vs batch vs time-sharing, the kernel as the core of UNIX (asked in 2018 and again in 2023), the four deadlock conditions, process states, and page faults.

Must-know. Deadlock needs mutual exclusion, hold-and-wait, no pre-emption and circular wait — all four. Process states: new, ready, running, waiting, terminated. Firmware lives in ROM. Linux is open source; Windows CE is embedded.

1. 2018 · Q64 Operating Systems, Packages & Utilities · Factual

QUESTION

The core component of UNIX is

OPTIONS

(a) Command shell

(b) Kernel

(c) Directories and programs

(d) None of the above

ANSWER **(AI-derived — no official key exists for this paper)**

(b) Kernel

EXAM SHORTCUT **(AI-derived explanation)**

The kernel is the core of UNIX - it manages processes, memory, files and devices. The shell is only the outer command interpreter.

TIPS & TRICKS **(AI-derived explanation)**

- UNIX has an onion structure: hardware, then kernel, then shell, then applications/utilities. 'Core' always means the kernel.
- The shell (sh, bash, csh, ksh) is a user-level program that can be replaced without touching the operating system - hence it cannot be the core.
- Kernel duties to remember: process scheduling, memory management, file-system control, device drivers and system calls.
- This exact question is repeated in the 2023 paper (Q_{74}), so the recall is worth locking in.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. UNIX is layered: the hardware at the centre, surrounded by the kernel, then the shell, then user applications and utilities.

2. The kernel is the resident part of the operating system that manages processes, memory, the file system and device I/O, and it services all system calls.
3. The command shell is merely a command interpreter running as an ordinary user process; several different shells can be used interchangeably.
4. Directories and programs are file-system objects, not the operating core.
5. Hence the core component of UNIX is the kernel, option (b).

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2. 2018 · Q77 Low & High Level Languages, Compiler, Assembler · Definition

QUESTION

A software program stored in ROM that cannot easily be changed is known as

OPTIONS

(a) Hardware

(b) Firmware

(c) Linker

(d) Editor

ANSWER (AI-derived — no official key exists for this paper)

(b) Firmware

EXAM SHORTCUT (AI-derived explanation)

Software permanently held in ROM is firmware - the standard 'software living in hardware' hybrid.

TIPS & TRICKS (AI-derived explanation)

- Definition to lock in: firmware = software written into ROM/PROM/EPROM/flash, e.g. BIOS, device controllers, embedded system code.
- The name itself is the mnemonic: it is 'firm' because it sits between soft (easily changed) and hard (fixed) - not as changeable as software, not as immutable as hardware.
- A linker (option c) combines object modules into an executable and an editor (option d) creates source files; both are ordinary utility software stored on disk.
- Modern firmware in flash memory CAN be updated ('flashing the BIOS'), but it is still classified as firmware - the defining feature is where it resides, not absolute immutability.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Programs stored permanently in read-only memory are not loaded from disk each time; they are built into the machine.
2. Such programs are called FIRMWARE - the classic examples being the BIOS, bootstrap loader and the control programs of embedded devices.
3. Hardware refers to the physical components themselves, not to a stored program, so option (a) is wrong.
4. A linker and an editor are ordinary application/system utilities stored on secondary storage, so options (c) and (d) are wrong.

5. Hence the answer is firmware, option (b).

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3. 2019 · Q49 Operating Systems, Packages & Utilities · Conceptual

QUESTION

If someone wants to reduce the cost of developing and maintaining a computer software, which operating system(s) should be installed?

OPTIONS

(a) Windows only

(b) Linux only

(c) Mac

(d) Both Windows and Linux

ANSWER (AI-derived — no official key exists for this paper)

(b) Linux only

EXAM SHORTCUT (AI-derived explanation)

Linux is free and open source, with no licence fees and freely modifiable source, so it minimises both development and maintenance cost.

TIPS & TRICKS (AI-derived explanation)

- Cost drivers for an OS: licence fees, tooling costs, and the freedom to modify. Linux scores best on all three because it is open source under the GPL.
- Installing BOTH Windows and Linux (option d) would raise cost, not reduce it - the extra licence and the burden of maintaining two platforms.
- Linux also ships with free development tool chains (GCC, make, debuggers, editors), whereas equivalent proprietary tooling is normally paid for.
- Mac requires Apple hardware in addition to the OS, making it the most expensive of the listed options.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The question asks which operating system minimises the cost of DEVELOPING and MAINTAINING software.
2. Windows is proprietary and requires paid licences for the operating system and often for its development tools.
3. Mac OS is proprietary and additionally tied to Apple hardware, raising the total cost further.
4. Linux is free and open source: there are no licence fees, the source code may be studied and modified, and a complete free development tool chain (compilers, debuggers, build tools) is included.
5. Installing both Windows and Linux would incur the Windows costs as well as the overhead of supporting two platforms, so it cannot be the cheapest option.

6. Hence Linux only, option (b).

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4. 2019 · Q63 Operating Systems, Packages & Utilities · Conceptual

QUESTION

Which one of the following systems does **not** share common memory and common bus?

OPTIONS

(a) Tightly coupled system

(b) Distributed system

(c) Batch system

(d) Multiprogramming system

ANSWER (AI-derived — no official key exists for this paper)

(b) Distributed system

EXAM SHORTCUT (AI-derived explanation)

A distributed system is loosely coupled: its processors have their own local memory and communicate over a network, so no common memory or common bus exists.

TIPS & TRICKS (AI-derived explanation)

- Coupling is the key idea: tightly coupled (multiprocessor) systems share memory and a bus; loosely coupled (distributed) systems do not and communicate by message passing.
- Batch and multiprogramming are SCHEDULING strategies on a single machine - both still use one common memory and bus, so options (c) and (d) are the wrong category.
- Distributed systems are characterised by resource sharing, scalability and fault tolerance, but also by communication delay - the direct consequence of having no shared bus.
- Symmetric multiprocessing (SMP) is the standard example of a tightly coupled system and is the direct contrast to a distributed one.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A tightly coupled system (a multiprocessor) has several processors sharing a common main memory and a common system bus, communicating through that shared memory.
2. Batch processing and multiprogramming describe how jobs are scheduled on a single computer; the hardware still has one memory and one bus.
3. A distributed system consists of autonomous computers, each with its own local memory and its own bus, connected by a communication network.
4. Processes in a distributed system communicate by message passing rather than through shared memory, so there is no common memory and no common bus.
5. Hence the distributed system is the one that does not share common memory and a common bus, option (b).

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5. 2019 · Q68 Operating Systems, Packages & Utilities · Conceptual

QUESTION

A technique used by an OS for efficient management of memory space of a computer system is

OPTIONS

(a) paging

(b) swapping

(c) caching

(d) debugging

ANSWER (AI-derived — no official key exists for this paper)

(a) paging

EXAM SHORTCUT (AI-derived explanation)

Paging divides memory into fixed-size frames and pages, allowing non-contiguous allocation - the standard OS technique for efficient management of memory SPACE.

TIPS & TRICKS (AI-derived explanation)

- Paging's specific benefit is the elimination of EXTERNAL fragmentation: any free frame can hold any page, so memory space is used efficiently.
- Swapping moves whole processes between memory and disk; it extends capacity but is a coarser mechanism than paging and is usually described as a supplement to it.
- Caching (option c) speeds up ACCESS rather than managing space, and debugging (option d) is a program-development activity, not memory management.
- Related terms to keep straight: paging, segmentation, virtual memory, page fault, page replacement (FIFO, LRU, optimal) and thrashing.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Memory management is the operating system function that allocates and tracks main memory among competing processes.
2. Paging divides physical memory into fixed-size frames and each process's logical address space into pages of the same size, then maps pages to any available frames via a page table.
3. Because a process need not occupy contiguous physical memory, paging eliminates external fragmentation and lets the OS use every free frame - the efficient management of memory space the question describes.
4. Swapping transfers entire processes between main memory and backing store; caching speeds up access by holding frequently used data in fast memory; debugging is unrelated to memory management.
5. Hence the technique is paging, option (a).

6. 2019 · Q69 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Factual

QUESTION

Which of the following are functions of an operating system? 1. Memory management 2. Device management 3. Security management

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

Memory management, device management and security management are all core operating-system functions, so all three are correct.

TIPS & TRICKS (AI-derived explanation)

- The standard list of OS functions: process management, memory management, file management, device (I/O) management, security and protection, and user interface. Memorise the six.
- When every listed item is drawn from that standard list, 'all of the above' is almost always the key - verify the least familiar one and move on.
- Device management is exercised through DEVICE DRIVERS and I/O scheduling; security management covers authentication, access control and protection between processes.
- Utilities such as antivirus supplement OS security but do not replace it - the OS itself enforces user accounts, permissions and memory protection.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. An operating system acts as the resource manager and interface between the user and the hardware.
2. Memory management: allocating and deallocating main memory, keeping track of which parts are in use, and implementing paging or segmentation and virtual memory. A core OS function.
3. Device management: controlling peripheral devices through device drivers, buffering, spooling and I/O scheduling. A core OS function.
4. Security management: user authentication, access control, file permissions and protection of one process's memory from another. A core OS function.
5. All three listed items are genuine operating-system functions.
6. Hence option (d) is correct.

7. 2021 · Q55 Operating Systems, Packages & Utilities · Factual

QUESTION

Which is *not* a real-time operating system?

OPTIONS

(a) MTOS

(b) LINUX

(c) Lynx

(d) RTX

ANSWER (AI-derived — no official key exists for this paper)

(b) LINUX

EXAM SHORTCUT (AI-derived explanation)

MTOS, LynxOS and RTX are real-time operating systems; standard LINUX is a general-purpose OS.

TIPS & TRICKS (AI-derived explanation)

- A real-time OS guarantees BOUNDED response times; a general-purpose OS optimises average throughput and fairness instead. That distinction is the whole question.
- Common RTOS names worth recognising: VxWorks, QNX, RTLinux, LynxOS, RTX, MTOS, FreeRTOS, pSOS.
- Standard Linux is not an RTOS, although real-time VARIANTS exist (RTLinux, PREEMPT_RT) - the question refers to plain LINUX.
- Hard real time means a missed deadline is a system failure (avionics, pacemakers); soft real time merely degrades quality (video streaming).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A real-time operating system is designed to guarantee that tasks complete within strict, predictable time limits, using deterministic scheduling and bounded interrupt latency.
2. MTOS is a classic multitasking real-time executive.
3. LynxOS (Lynx) is a well-known hard real-time, POSIX-compatible RTOS used in embedded and avionics systems.
4. RTX is a real-time kernel widely used for embedded microcontroller applications.
5. LINUX in its standard form is a general-purpose, time-sharing operating system optimised for throughput and fairness rather than for guaranteed deadlines, although real-time variants exist.
6. Hence LINUX is not a real-time operating system, option (b).

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8. 2021 · Q73 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Conceptual

QUESTION

A major security problem for operating systems is

OPTIONS

(a) Authentication problem

(b) Physical problem

(c) Human problem

(d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(a) Authentication problem

EXAM SHORTCUT (AI-derived explanation)

The classical statement is that operating-system security rests on three problems - physical, human and authentication - of which AUTHENTICATION is the major one the OS itself must solve.

TIPS & TRICKS (AI-derived explanation)

- The standard taxonomy (Silberschatz) lists physical security, human security and operating-system security, with the AUTHENTICATION problem as the OS's own principal concern.
- Physical and human failings are external to the software; the operating system can only enforce identity checks, access control and protection domains.
- Authentication mechanisms to know: passwords, one-time passwords, biometrics, two-factor authentication - all aimed at binding a user to a subject identity.
- Once authentication fails, every downstream protection mechanism (access-control lists, capabilities, memory protection) is defeated, which is why it is called the major problem.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Operating-system security depends on correctly identifying the user or process on whose behalf every request is made.
2. That identification is the authentication problem: the system must verify that a user is who they claim to be, typically through passwords, tokens or biometrics.
3. If authentication is defeated, all the OS's other protection mechanisms - access control lists, protection domains, memory protection - are bypassed, because the attacker operates with a legitimate identity.
4. Physical security (protecting the machine room) and human factors (social engineering, careless staff) are genuine risks but lie outside the operating system's control.
5. Hence the major security problem for the operating system itself is the authentication problem, option (a).

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9. **2021 · Q79** Operating Systems, Packages & Utilities · Factual

QUESTION

Which is an example of an Embedded operating system?

OPTIONS

(a) Linux

(b) Windows 2000

(c) Windows CE

(d) MTOS

ANSWER **(AI-derived — no official key exists for this paper)**

(c) Windows CE

EXAM SHORTCUT **(AI-derived explanation)**

Windows CE was Microsoft's operating system built specifically for embedded and handheld devices.

TIPS & TRICKS **(AI-derived explanation)**

- CE stands for 'Compact Edition' - a stripped-down, modular Windows designed for devices with limited memory and no disk.
- Embedded operating systems are characterised by a small footprint, a fixed device-specific configuration, and often real-time capability.
- Windows 2000 is a desktop and server OS, and standard Linux is general purpose (though EMBEDDED Linux variants exist, the plain entry here is the general-purpose one).
- MTOS is a real-time executive rather than the standard textbook example of an embedded OS; Windows CE, VxWorks and Embedded Linux are the usual named examples.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. An embedded operating system is a compact OS designed to run on a device dedicated to a specific function, with limited memory and often no secondary storage.
2. Windows CE (Compact Edition) was developed by Microsoft precisely for such devices - handheld computers, PDAs, industrial controllers and point-of-sale terminals - with a modular kernel that could be trimmed to the hardware.
3. Windows 2000 is a full desktop and server operating system.
4. Linux in its standard distribution is a general-purpose operating system, although embedded variants of it exist.
5. MTOS is a real-time executive rather than the standard example of an embedded OS.
6. Hence Windows CE is the embedded operating system, option (c).

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10. **2022 · Q33** Operating Systems, Packages & Utilities · Conceptual

QUESTION

Consider the conditions: 1. Mutual exclusion, 2. Hold and wait, 3. No pre-emption. Under which of these may deadlock arise?

OPTIONS

(a) 1 and 2 only

(b) 1 and 3 only

(c) 2 and 3 only

(d) 1, 2 and 3

ANSWER **(AI-derived — no official key exists for this paper)**

(d) 1, 2 and 3

EXAM SHORTCUT **(AI-derived explanation)**

Mutual exclusion, hold and wait, and no pre-emption are three of Coffman's four necessary conditions for deadlock (the fourth being circular wait), so all three are involved.

TIPS & TRICKS **(AI-derived explanation)**

- Coffman's four conditions, all of which must hold simultaneously: mutual exclusion, hold and wait, no pre-emption, circular wait. Memorise them as a set of four.
- Deadlock PREVENTION works by breaking any one of the four - which is why each condition is genuinely necessary and none can be dropped.
- The question lists three of the four, so all of them are conditions under which deadlock may arise; 'all of the above' is the natural key.
- Circular wait is the one omitted here; it is the condition most often attacked in practice, by imposing a total ordering on resource requests.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Deadlock in an operating system arises only when four conditions hold simultaneously - Coffman's conditions.
2. Mutual exclusion: at least one resource is held in a non-sharable mode, so only one process may use it at a time. NECESSARY.
3. Hold and wait: a process holding at least one resource is waiting to acquire additional resources held by others. NECESSARY.
4. No pre-emption: resources cannot be forcibly taken from a process; they are released only voluntarily. NECESSARY.
5. The fourth condition, circular wait, is not listed here but completes the set.
6. All three listed conditions are conditions under which deadlock may arise, option (d).

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11. 2022 · Q40 Operating Systems, Packages & Utilities · Conceptual

QUESTION

Consider the statements about Batch processing OS: 1. All jobs are processed as per their priority. 2. Debugging of a program at execution time is not possible. 3. After execution of one job, the next job is fetched automatically. Which are correct?

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

Batch systems queue jobs and dispatch them by priority, run them without user interaction (so no interactive debugging), and load the next job automatically. All three are standard characteristics.

TIPS & TRICKS (AI-derived explanation)

- Defining feature of a batch OS: jobs are collected, queued and executed without user interaction, with the monitor loading the next job automatically as soon as one finishes.
- Statement 2 is the classic DISADVANTAGE quoted in every textbook - because the user is absent during execution, interactive debugging is impossible and errors surface only in the printed output.
- Statement 3 captures the whole point of the resident monitor: eliminating manual set-up time between jobs was the historical motivation for batch processing.
- Statement 1 reflects the standard scheduling description - jobs are ordered in the batch queue, commonly by priority (or by FCFS); the priority ordering is the feature usually listed.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A batch operating system collects jobs with similar requirements into batches and executes them one after another under a resident monitor, without any interaction with the user.
2. Statement 1: jobs waiting in the batch queue are ordered for execution, and priority-based ordering is the standard scheduling description given for such systems. CORRECT.
3. Statement 2: since the user is not present while the job runs, no interactive debugging is possible; errors can be diagnosed only from the printed output after the run. CORRECT.
4. Statement 3: the resident monitor automatically loads and starts the next job as soon as the current one terminates, which is precisely the automation that batch processing introduced. CORRECT.
5. All three describe genuine characteristics of batch processing.
6. Hence option (d) is correct.

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12. 2022 · Q72 Operating Systems, Packages & Utilities · Conceptual

QUESTION

What is the correct sequence of a cycle of a process in execution?

OPTIONS

(a) Ready → Waiting → Executing → Terminated → Blocked

(b) Ready → Waiting → Executing → Blocked

(c) Waiting → Ready → Executing → Blocked

(d) Waiting → Ready → Executing → Terminated

ANSWER (AI-derived — no official key exists for this paper)

(d) Waiting → Ready → Executing → Terminated

EXAM SHORTCUT (AI-derived explanation)

A process is admitted from the waiting/new queue, becomes Ready, is dispatched to Executing, and finally Terminates - the only option that ends in a terminal state.

TIPS & TRICKS (AI-derived explanation)

- Process state model: New -> Ready -> Running -> Terminated, with excursions to Blocked/Waiting and back to Ready whenever I/O is needed.
- A complete life cycle must END in Terminated; options ending in 'Blocked' describe an intermediate excursion rather than a completed cycle.
- A process never goes directly from Ready to Waiting - it must be dispatched to Running first and then block on a request. That rules out options (a) and (b).
- Only the scheduler moves a process from Ready to Running (dispatch), and only an I/O request moves it from Running to Blocked - keep the transition triggers in mind.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The standard process state diagram is: a new process is admitted to the READY queue, the scheduler dispatches it to the RUNNING (executing) state, and on completion it moves to TERMINATED.
2. While running, a process may issue an I/O request and move to the BLOCKED/WAITING state, returning to READY when the I/O completes.
3. Option (a) places Blocked after Terminated, which is impossible since termination is a final state.
4. Options (b) and (c) end at Blocked, which is an intermediate state rather than the end of a life cycle; option (b) additionally moves from Ready to Waiting directly, a transition that does not exist.
5. Option (d), Waiting -> Ready -> Executing -> Terminated, follows the admission, dispatch and completion path and ends in the terminal state.
6. Hence option (d) is correct.

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13. 2022 · Q78 Memory - RAM, ROM & Units of Computer Memory · Conceptual

QUESTION

Consider: 1. Shared region, 2. Cache memory, 3. Message passing, 4. Shared memory. Which are related to interprocess communication?

OPTIONS

(a) 1 and 2 only

(b) 1, 2 and 3

(c) 2 and 4 only

(d) 3 and 4 only

ANSWER (AI-derived — no official key exists for this paper)

(d) 3 and 4 only

EXAM SHORTCUT (AI-derived explanation)

The two standard models of interprocess communication are SHARED MEMORY and MESSAGE PASSING - items 3 and 4.

TIPS & TRICKS (AI-derived explanation)

- Memorise the pair: IPC = shared memory model + message passing model. Everything else in an options list is a distractor.
- Cache memory is a hardware performance device managed by the processor, entirely invisible to processes - it is never an IPC mechanism.
- Shared memory is fast (processes read and write a common region directly) but needs explicit synchronisation such as semaphores; message passing is slower but needs no shared address space and works across machines.
- 'Shared region' in item 1 is loose phrasing for the same idea as shared memory, but the option set pairs the two canonical mechanisms 3 and 4.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Interprocess communication (IPC) is the set of mechanisms by which cooperating processes exchange data and synchronise their actions.
2. Item 4, shared memory: the operating system maps a common region of memory into the address spaces of two or more processes, which then read and write it directly (with semaphores or mutexes for synchronisation). A standard IPC model.
3. Item 3, message passing: processes exchange data through send and receive primitives supplied by the kernel, requiring no shared address space and extending naturally across a network. The other standard IPC model.
4. Item 2, cache memory: a hardware buffer between the CPU and main memory, managed transparently by the processor and unrelated to communication between processes.
5. Hence the mechanisms related to interprocess communication are message passing and shared memory, option (d).

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14. 2022 · Q79 Operating Systems, Packages & Utilities · Conceptual

QUESTION

Which of the following are goals of an operating system? 1. Efficient use of CPU, 2. Efficient use of memory, 3. Efficient use of I/O devices

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

The operating system is the resource manager for the whole machine, so efficient use of the CPU, of memory and of I/O devices are all among its goals.

TIPS & TRICKS (AI-derived explanation)

- The OS is defined as a RESOURCE MANAGER; the three principal resources are the processor, memory and I/O devices, so all three appear in its goals.
- Concrete mechanisms: CPU scheduling for the processor, paging and virtual memory for memory, buffering and spooling for I/O.
- The complementary goal is CONVENIENCE - providing an easy interface for the user - which sits alongside efficiency in every textbook list.
- When all listed items are standard entries from the same textbook list, 'all of them' is almost always the key.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. An operating system acts as the manager of all the computer's resources and as the interface between the user and the hardware.
2. Goal 1, efficient use of the CPU: achieved through multiprogramming and CPU scheduling, keeping the processor busy while other jobs wait for I/O. A genuine goal.
3. Goal 2, efficient use of memory: achieved through allocation strategies, paging, segmentation and virtual memory, so that as many processes as possible are resident. A genuine goal.
4. Goal 3, efficient use of I/O devices: achieved through buffering, caching, spooling and device scheduling, so that slow peripherals do not stall the system. A genuine goal.
5. All three are standard goals, alongside the complementary aim of user convenience.
6. Hence option (d) is correct.

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15. 2023 · Q66 Operating Systems, Packages & Utilities · Definition

QUESTION

Which one of the following Operating Systems responds to an event within a predetermined time?

OPTIONS

(a) Time Sharing Operating System

(b) Embedded Operating System

(c) Real-time Operating System

(d) Batch Operating System

ANSWER (AI-derived — no official key exists for this paper)

(c) Real-time Operating System

EXAM SHORTCUT (AI-derived explanation)

Guaranteeing a response within a predetermined time is the defining property of a Real-time

Operating System.

TIPS & TRICKS (AI-derived explanation)

- The key words are 'within a predetermined time' - that is the definition of a DEADLINE, and deadlines are what real-time systems guarantee.
- Time-sharing systems optimise fair average response, not guaranteed worst-case response - a crucial distinction from real-time.
- Hard real time means a missed deadline is a system failure (avionics, pacemakers); soft real time means quality degrades (video streaming).
- Embedded systems OFTEN use an RTOS, but 'embedded' describes where the system runs, whereas 'real-time' describes its timing guarantee - the question asks about timing.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A Real-time Operating System is designed so that processing is completed within a specified, bounded time after an event occurs.
2. It achieves this through deterministic scheduling, priority-based pre-emption and bounded interrupt latency.
3. A time-sharing operating system divides processor time among many users to give good AVERAGE response, but offers no guarantee for any individual event.
4. A batch operating system executes jobs one after another with no interactive response requirement at all.
5. An embedded operating system is defined by the dedicated device it runs on rather than by any timing guarantee, though it often is a real-time system.
6. Hence the OS that responds within a predetermined time is the real-time operating system, option (c).

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16. 2023 · Q69 Operating Systems, Packages & Utilities · Factual

QUESTION

Which of the following Operating Systems is an open source?

OPTIONS

(a) Windows 10

(b) iOS 16

(c) Linux

(d) MS-DOS

ANSWER (AI-derived — no official key exists for this paper)

(c) Linux

EXAM SHORTCUT (AI-derived explanation)

Linux is released under the GNU General Public License with freely available source code; Windows, iOS and MS-DOS are all proprietary.

TIPS & TRICKS (AI-derived explanation)

- Open source means the source code is published and may be studied, modified and redistributed - Linux under the GPL is the standard example.
- Windows, iOS and MS-DOS are all closed, commercially licensed products from Microsoft and Apple.
- Other open-source systems worth naming: FreeBSD, OpenBSD, Android (AOSP core), and the GNU tool chain that accompanies Linux.
- Being free of charge and being open source are different things - the defining test is access to and freedom to modify the source.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. An open-source operating system publishes its source code under a licence permitting study, modification and redistribution.
2. Linux, created by Linus Torvalds and released under the GNU General Public License, is the archetypal open-source operating system; its kernel source is publicly available and maintained by a worldwide community.
3. Windows 10 is proprietary software licensed by Microsoft, with its source code closed.
4. iOS 16 is Apple's proprietary mobile operating system, restricted to Apple hardware.
5. MS-DOS was Microsoft's proprietary disk operating system of the 1980s.
6. Hence the open-source operating system is Linux, option (c).

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17. 2023 · Q74 Operating Systems, Packages & Utilities · Factual

QUESTION

Which one of the following is the core component of UNIX?

OPTIONS

(a) Shell

(b) Kernel

(c) Process

(d) Directories

ANSWER (AI-derived — no official key exists for this paper)

(b) Kernel

EXAM SHORTCUT (AI-derived explanation)

The kernel is the core of UNIX; the shell is only the outer command interpreter.

TIPS & TRICKS (AI-derived explanation)

- UNIX layers from the inside out: hardware, kernel, shell, applications. 'Core' always means the kernel.
- Kernel duties: process scheduling, memory management, file-system control, device drivers, system calls.
- The shell can be replaced (sh, bash, csh, ksh, zsh) without touching the operating system, which proves it is not the core.

- This item repeats 2018 Q64 word for word - a clear signal that the layered UNIX structure is worth memorising.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. UNIX has a layered architecture: the hardware at the centre, surrounded by the kernel, then the shell, then user applications and utilities.
2. The kernel is the resident core that manages processes, memory, the file system and device I/O, and services every system call.
3. The shell is a command interpreter running as an ordinary user process; several different shells may be used interchangeably.
4. A process is a program in execution - an entity MANAGED by the kernel, not a component of the operating system itself.
5. Directories are file-system objects, again managed by the kernel.
6. Hence the core component of UNIX is the kernel, option (b).

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18. 2024 · Q42 Operating Systems, Packages & Utilities · Conceptual

QUESTION

Consider the following statements in respect of Operating System: I. It is the principal component of system software. II. It is responsible for overall management of the computer resources. III. It provides an interface between the computer and user and helps in implementing the application programs. Which of the statements given above are correct?

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(d) I, II and III

EXAM SHORTCUT (AI-derived explanation)

The operating system is the principal system software, the manager of all machine resources, and the interface between user and hardware - all three statements are standard definitions.

TIPS & TRICKS (AI-derived explanation)

- Three standard descriptions of an OS: principal system software, resource manager, and user-hardware interface. Each statement matches one.
- Resource management covers the processor, memory, files and I/O devices - the four resources every textbook lists.
- The OS also provides the environment in which application programs are loaded and run, which is the second half of statement III.

- When all three statements paraphrase the same textbook definition from different angles, 'all of them' is almost always the key.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement I: system software comprises the operating system, device drivers, language translators and utilities, and the operating system is its principal component - without it no other software can run. TRUE.
2. Statement II: the operating system allocates and manages the processor, main memory, files and I/O devices among competing processes. TRUE.
3. Statement III: it provides the interface - command line or graphical - through which the user directs the machine, and it loads and runs application programs, providing them with system services. TRUE.
4. All three are standard descriptions of an operating system.
5. Hence option (d) is correct.

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19. 2024 · Q53 Memory - RAM, ROM & Units of Computer Memory · Conceptual

QUESTION

Which one of the following methods is used for interprocess communication in distributed systems?

OPTIONS

(a) Shared cache memory

(b) Message passing

(c) Shared region

(d) Cache memory

ANSWER (AI-derived — no official key exists for this paper)

(b) Message passing

EXAM SHORTCUT (AI-derived explanation)

Distributed systems have no shared physical memory, so processes must communicate by MESSAGE PASSING.

TIPS & TRICKS (AI-derived explanation)

- The two IPC models are shared memory and message passing; in a DISTRIBUTED system the machines have separate memories, so only message passing is available.
- Every option except (b) presupposes some shared memory or cache, which by definition does not exist across independent machines.
- Message passing is realised in practice by sockets, remote procedure calls (RPC) and message queues - worth naming as examples.
- The trade-off: message passing is slower than shared memory but requires no common address space and scales across a network.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A distributed system consists of autonomous computers, each with its own local memory, connected only by a communication network.
2. Because there is no physical memory common to the machines, shared-memory mechanisms - shared regions, shared caches - cannot be used across nodes.
3. The remaining interprocess communication model is message passing, in which processes exchange data using send and receive primitives supplied by the operating system.
4. In practice this is implemented through sockets, remote procedure calls or message queues.
5. Options (a), (c) and (d) all presuppose memory shared between the communicating processes, which distributed systems lack.
6. Hence the method is message passing, option (b).

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20. 2025 · Q33 Operating Systems, Packages & Utilities · Factual

QUESTION

In which type of scheduling is the process allocated to the CPU for a specific time slice?

OPTIONS

(a) FCFS

(b) Shortest Job First

(c) Priority

(d) Round Robin

ANSWER (AI-derived — no official key exists for this paper)

(d) Round Robin

EXAM SHORTCUT (AI-derived explanation)

Allocating the CPU to each process for a fixed time slice (quantum) in rotation is Round Robin scheduling.

TIPS & TRICKS (AI-derived explanation)

- 'Time slice' or 'quantum' is the signature of Round Robin - no other classical algorithm uses one.
- Round Robin is PRE-EMPTIVE: when the quantum expires the process is moved to the back of the ready queue whether or not it has finished.
- FCFS and (non-pre-emptive) SJF run each process to completion, and priority scheduling selects by priority rather than by a clock - none involves a time slice.
- Quantum size matters: too large and Round Robin degenerates to FCFS; too small and context-switching overhead dominates.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Round Robin scheduling maintains a circular ready queue and allocates the CPU to each process for a fixed time slice, or quantum.

2. If a process does not complete within its quantum it is pre-empted and placed at the end of the queue, and the next process runs.
3. First-Come First-Served runs each process to completion in arrival order, with no time limit.
4. Shortest Job First selects the process with the smallest burst time; in its non-pre-emptive form it also runs to completion.
5. Priority scheduling selects the highest-priority process, again without any fixed slice.
6. Hence the time-slice algorithm is Round Robin, option (d).

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21. 2025 · Q36 Operating Systems, Packages & Utilities · Factual

QUESTION

Consider: I. Multi-user II. Multitasking III. Multiprocessor. Which are correct in respect of the Linux operating system?

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(d) I, II and III

EXAM SHORTCUT (AI-derived explanation)

Linux supports many simultaneous users, many concurrent tasks and symmetric multiprocessing - all three descriptions apply.

TIPS & TRICKS (AI-derived explanation)

- Linux inherits the UNIX design: multi-user with separate accounts and permissions, multitasking with pre-emptive scheduling, and SMP support for multiple processors.
- Multi-user means several people can be logged in at once with protected private files - not merely that several accounts exist.
- Multitasking here is PRE-EMPTIVE: the kernel interrupts a running process at the end of its quantum rather than waiting for it to yield.
- When every item in a list is a standard textbook property of the same system, 'all of them' is almost always the key.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Linux is a UNIX-like operating system and inherits the full set of UNIX capabilities.
2. Item I, multi-user: several users can be logged in simultaneously, each with a separate account, home directory and permission set, and the kernel isolates their processes and files. TRUE.
3. Item II, multitasking: the kernel schedules many processes concurrently using pre-emptive time slicing, so several programs progress at once. TRUE.

4. Item III, multiprocessor: the Linux kernel supports symmetric multiprocessing, distributing processes across multiple CPUs or cores. TRUE.
5. All three descriptions apply to Linux.
6. Hence option (d) is correct.

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22. 2026 · Q64 Operating Systems, Packages & Utilities · Conceptual

QUESTION

Which of the following is *not* a process state?

OPTIONS

(a) Ready

(b) Executing

(c) Deadlock

(d) Terminated

ANSWER (AI-derived — no official key exists for this paper)

(c) Deadlock

EXAM SHORTCUT (AI-derived explanation)

The process states are New, Ready, Running (executing), Waiting/Blocked and Terminated. Deadlock is a condition affecting a set of processes, not a state in the diagram. Answer (c).

TIPS & TRICKS (AI-derived explanation)

- Memorise the five-state model: New -> Ready -> Running -> (Waiting -> Ready) -> Terminated.
- Deadlocked processes are technically in the WAITING/blocked state; deadlock describes why they are stuck, not a distinct state label.
- The four Coffman conditions for deadlock - mutual exclusion, hold and wait, no preemption, circular wait - are about resource allocation, again not a process state.
- 'Executing' is simply another name for the Running state, so option (b) is a genuine state under a different label.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. In the standard process state model an operating system tracks: New (being created), Ready (waiting for the CPU), Running or Executing (holding the CPU), Waiting or Blocked (awaiting an event or I/O), and Terminated (finished).
2. Options (a) Ready, (b) Executing and (d) Terminated are all genuine states in this model.
3. Deadlock is not a state of an individual process. It is a system condition in which a set of processes are permanently blocked because each holds a resource the next one needs (circular wait).
4. Processes caught in a deadlock are recorded by the operating system as being in the Waiting/Blocked state.
5. Hence option (c) is the one that is not a process state, and option (c) is correct.

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23. 2026 · Q65 Operating Systems, Packages & Utilities · Definition

QUESTION

Paging with swapping is also known as:

OPTIONS

(a) demand paging

(b) segmentation

(c) swapping

(d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(a) demand paging

EXAM SHORTCUT (AI-derived explanation)

Bringing pages into memory only when they are referenced - paging combined with swapping - is called demand paging. Answer (a).

TIPS & TRICKS (AI-derived explanation)

- Demand paging = paging + swapping: a page is loaded only on demand, i.e. when a page fault occurs.
- Segmentation (option b) divides a program into logical variable-sized units; it is a different memory-management scheme, not a name for paging.
- Plain swapping (option c) moves an ENTIRE process between memory and backing store, whereas demand paging moves individual pages - the granularity is the whole distinction.
- Demand paging is what makes virtual memory practical, since a process can run with only part of its address space resident.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Paging divides the logical address space into fixed-size pages and physical memory into frames of the same size.
2. When paging is combined with swapping, pages are transferred between main memory and the backing store as required rather than all at once.
3. A page is brought in only when it is actually referenced and found to be absent (a page fault). This lazy, on-demand transfer scheme is called demand paging.
4. Segmentation (option b) is an alternative scheme using variable-sized logical segments; swapping alone (option c) relocates an entire process, not individual pages.
5. Hence option (a) is correct.

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24. 2026 · Q77 Operating Systems, Packages & Utilities · Definition

QUESTION

A page fault in an operating system:

OPTIONS

(a) refers to the page used in the previous page reference

(b) occurs when a page program occurs in a page memory

(c) refers to an access to a page not currently in memory

(d) refers to an error specific page

ANSWER (AI-derived — no official key exists for this paper)

(c) refers to an access to a page not currently in memory

EXAM SHORTCUT (AI-derived explanation)

A page fault is the trap raised when a program references a page that is not currently resident in main memory. Answer (c).

TIPS & TRICKS (AI-derived explanation)

- 'Fault' here does not mean an error: page faults are a normal, expected part of demand paging.
- Sequence on a fault: trap to the OS -> locate the page on disk -> find or free a frame (page replacement) -> read the page in -> update the page table -> restart the faulting instruction.
- Effective access time = $(1 - p)$ *memory access time + p* page fault service time, where p is the fault rate - the formula behind most numerical questions on this topic.
- Options (a), (b) and (d) are deliberately garbled non-definitions; read them carefully and they do not describe any coherent mechanism.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Under demand paging, only some of a process's pages are resident in physical memory at any time; the page table marks each page as valid (resident) or invalid (not resident).
2. When the process references an address on a page whose valid bit is not set, the memory management unit raises a trap called a page fault.
3. The operating system then locates the page in secondary storage, selects a frame (evicting a victim page if necessary), reads the page in, updates the page table, and restarts the interrupted instruction.
4. So a page fault refers precisely to an access to a page that is not currently in memory.
5. The remaining options do not describe this mechanism.
6. Hence option (c) is correct.

C6 · COMPUTER APPLICATION AND DATA PROCESSING

Languages, Compiler, Assembler, Interpreter; Low & High Level

16 questions · 2018: 4 2019: 1 2020: 1 2021: 3 2022: 0 2023: 0 2024: 1 2025: 2 2026: 4

Weight. 16 questions (8.9% of Computer Application and Data Processing) · appeared in 7 of 9 papers · peak year 2018 (4)

Examiner's pattern. Translators and language levels. The compiler-versus-interpreter distinction is asked in some form nearly every year: a compiler translates the whole program then runs it, an interpreter translates and executes line by line. Also: linker vs loader, the lexical analyser's output, and which errors are detectable.

Must-know. Compilers and interpreters detect syntax and semantic errors but NOT logical errors — a program with a logical error still runs. The lexical analyser outputs a stream of tokens. Assembly and machine language are low-level.

1. 2018 · Q70 Low & High Level Languages, Compiler, Assembler · Conceptual

QUESTION

Which statement is correct?

OPTIONS

(a) A compiler translates and executes instructions simultaneously.

(b) An interpreter translates and executes instructions simultaneously.

(c) An interpreter can only translate machine-language programs.

(d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(b) An interpreter translates and executes instructions simultaneously.

EXAM SHORTCUT (AI-derived explanation)

An interpreter translates and executes one statement at a time, so translation and execution happen together. A compiler translates the whole program first and executes nothing.

TIPS & TRICKS (AI-derived explanation)

- One-line contrast: compiler = translate ALL, then run; interpreter = translate ONE line, run it, move on. Everything else follows from this.
- A compiler produces an object/executable file; an interpreter produces no separate object file, which is why interpreted programs are slower but easier to debug.
- Option (c) reverses the machine: an interpreter translates HIGH-level source, it does not translate machine language (machine language needs no translation).
- Java is the standard hybrid to remember - compiled to bytecode, then interpreted (or JIT-compiled) by the JVM.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A compiler reads the entire source program, translates it into object code, and reports all errors before any execution takes place; execution is a separate later step.
2. An interpreter takes the source program one statement at a time, translates that statement and immediately executes it before moving to the next.
3. Therefore translation and execution occur simultaneously in an interpreter, which makes option (b) correct and option (a) incorrect.
4. Option (c) is wrong because an interpreter translates high-level source code; machine language requires no translation at all.
5. Since (b) is a correct statement, option (d) cannot be right.
6. Hence option (b) is correct.

---

2. 2018 · Q76 Low & High Level Languages, Compiler, Assembler · Conceptual

QUESTION

A compiler can diagnose

OPTIONS

(a) Grammatical errors only

(b) Logical errors only

(c) Both grammatical and logical errors

(d) Neither grammatical nor logical errors

ANSWER (AI-derived — no official key exists for this paper)

(a) Grammatical errors only

EXAM SHORTCUT (AI-derived explanation)

A compiler checks the program against the grammar of the language, so it catches syntax (grammatical) errors. Logical errors are semantically valid code and are invisible to it.

TIPS & TRICKS (AI-derived explanation)

- Error taxonomy: syntax/grammatical errors are caught at COMPILE time; run-time errors surface during execution; logical errors are never reported at all - the program simply gives wrong answers.
- A logical error such as writing a + b when a - b was intended is perfectly grammatical, so no compiler could possibly flag it.
- Modern compilers do emit warnings for suspicious constructs, but a warning is not a diagnosis of a logical error - keep to the textbook distinction in this paper.
- This pairs directly with Q78 of the same paper: a program with a logical error compiles AND runs successfully, just incorrectly.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A compiler performs lexical, syntactic and (limited) semantic analysis of the source program against the formal grammar of the language.

2. Violations of that grammar - missing semicolons, unbalanced brackets, undeclared identifiers, type mismatches - are grammatical (syntax) errors, and the compiler reports them.
3. A logical error is a mistake in the algorithm: the code is grammatically valid but computes the wrong thing. The compiler has no knowledge of the programmer's intention, so it cannot detect this.
4. Therefore a compiler diagnoses grammatical errors only.
5. Hence option (a) is correct.

---

3. 2018 · Q78 Low & High Level Languages, Compiler, Assembler · Conceptual

QUESTION

A computer program can run successfully even if there is a

OPTIONS

(a) Syntax error

(b) Logical error

(c) Run-time error

(d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(b) Logical error

EXAM SHORTCUT (AI-derived explanation)

Syntax errors stop compilation and run-time errors abort execution. A logical error breaks neither, so the program runs to completion - just with wrong output.

TIPS & TRICKS (AI-derived explanation)

- Ask 'does this error prevent the program from finishing?'. Syntax errors: yes, at compile time. Run-time errors: yes, during execution. Logical errors: no.
- This is the mirror image of Q76 in the same paper - the two items together define the syntax/logic boundary, so answer them as a pair.
- Classic logical errors: an off-by-one loop bound, using $<$ instead of $<=$, or the wrong formula. The program executes happily and reports a wrong answer.
- 'Run successfully' in this question means 'runs to completion', not 'produces the correct result' - read the phrase precisely.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A syntax error violates the grammar of the language, so the compiler refuses to produce an executable and the program cannot run at all.
2. A run-time error - division by zero, an out-of-range array index, a failed memory allocation - causes abnormal termination during execution.
3. A logical error is a flaw in the algorithm. The code is syntactically valid and performs no illegal operation, so it compiles and executes to completion.
4. The only symptom is that the output is wrong; the run itself is successful.

5. Hence a program can run successfully in the presence of a logical error, option (b).

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4. 2018 · Q80 Low & High Level Languages, Compiler, Assembler · Factual

QUESTION

Which is a low-level programming language?

OPTIONS

(a) FORTRAN

(b) Machine language

(c) COBOL

(d) C

ANSWER (AI-derived — no official key exists for this paper)

(b) Machine language

EXAM SHORTCUT (AI-derived explanation)

Low-level means machine-dependent and close to the hardware: machine language and assembly language. FORTRAN, COBOL and C are high-level languages.

TIPS & TRICKS (AI-derived explanation)

- Only two languages are low-level: machine language (pure binary, level 1) and assembly language (mnemonics, level 2). Everything with English-like statements is high-level.
- C is often called a 'middle-level' language because it allows pointer and bit manipulation, but in UPSC classification it is a HIGH-level language - do not be tempted by option (d).
- Machine language is the only one that needs no translator at all; assembly needs an assembler, and high-level languages need a compiler or interpreter.
- FORTRAN (scientific) and COBOL (business) are the two classic first-generation high-level languages and appear as distractors throughout these papers.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Low-level languages are machine-dependent and correspond closely to the processor's instruction set: machine language (binary opcodes) and assembly language (symbolic mnemonics).
2. High-level languages use English-like statements, are machine-independent, and require a compiler or interpreter for translation.
3. FORTRAN and COBOL are classic high-level languages, and C, although it permits low-level operations, is also classified as a high-level language.
4. Machine language consists of binary instructions executed directly by the CPU and needs no translation.
5. Hence machine language is the low-level language here, option (b).

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5. 2019 · Q44 Low & High Level Languages, Compiler, Assembler · Definition

QUESTION

A software development module that arranges the object code of all modules generated by language translators into a single program, is

OPTIONS

(a) editor

(b) debugger

(c) linker

(d) loader

ANSWER (AI-derived — no official key exists for this paper)

(c) linker

EXAM SHORTCUT (AI-derived explanation)

Combining the separately compiled object modules into one executable program is precisely the linker's job.

TIPS & TRICKS (AI-derived explanation)

- Tool chain in order: editor writes source, compiler/assembler produces object code, LINKER combines object modules and libraries into an executable, loader places it in memory, debugger helps find errors.
- The linker works on OBJECT code; the loader works on the finished EXECUTABLE. That distinction settles the choice between options (c) and (d).
- A linker also resolves external references - a call in one module to a function defined in another - which is exactly the 'single program' aspect the question describes.
- Static linking embeds library code at build time; dynamic linking defers it to load time, which is why shared libraries (DLLs, .so files) exist.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Language translators (compilers and assemblers) convert each source module separately into object code, leaving unresolved references between modules.
2. The LINKER takes all these object modules, together with any required library routines, resolves the cross-references between them, and produces a single executable program.
3. An editor is used only to create and modify source text.
4. A debugger assists in locating and correcting run-time and logical errors after linking.
5. A loader loads the already-linked executable into main memory for execution; it does not combine object modules.
6. Hence the answer is the linker, option (c).

6. 2020 · Q47 Low & High Level Languages, Compiler, Assembler · Factual

QUESTION

Which language was designed to work with Microsoft's .NET platform?

OPTIONS

(a) Python

(b) Visual Basic

(c) Java

(d) C#

ANSWER (AI-derived — no official key exists for this paper)

(d) C#

EXAM SHORTCUT (AI-derived explanation)

C# was created by Microsoft specifically as the flagship language of the .NET platform.

TIPS & TRICKS (AI-derived explanation)

- C# (2000, designed by Anders Hejlsberg at Microsoft) was built from scratch for the Common Language Runtime of .NET - the only language in the list designed FOR .NET.
- Visual Basic predates .NET (VB6 and earlier); VB.NET is a later port, so VB was not 'designed to work with' .NET.
- Java was designed by Sun for the Java Virtual Machine, and .NET was in large part Microsoft's answer to it - a useful contrast to remember.
- Python predates .NET by a decade; IronPython is merely a later .NET implementation of the language.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The .NET platform, released by Microsoft in the early 2000s, runs managed code on the Common Language Runtime.
2. C# was designed at Microsoft expressly for this platform, with its type system and features mapped directly onto the CLR.
3. Visual Basic existed long before .NET as VB6 and earlier; VB.NET is a subsequent adaptation, not an original .NET design.
4. Java was designed by Sun Microsystems for the Java Virtual Machine, an entirely separate runtime.
5. Python was created in 1991, independent of Microsoft; IronPython is only a later .NET implementation.
6. Hence the language designed for .NET is C#, option (d).

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7. 2021 · Q53 Low & High Level Languages, Compiler, Assembler · Factual

QUESTION

Which is *not* correct about Python?

OPTIONS

(a) It is an interpreted language.

(b) It is a platform-dependent language.

(c) Its syntax is clear.

(d) It is free and open source.

ANSWER (AI-derived — no official key exists for this paper)

(b) It is a platform-dependent language.

EXAM SHORTCUT (AI-derived explanation)

Python is platform-INDEPENDENT: the same source runs wherever the interpreter is available. The other three statements are all true of Python.

TIPS & TRICKS (AI-derived explanation)

- Interpreted languages are portable by construction - the interpreter absorbs the platform differences, so 'interpreted' and 'platform-dependent' sit awkwardly together.
- Python's headline properties for exams: interpreted, high-level, dynamically typed, platform-independent, free and open source (PSF licence), with clean indentation-based syntax.
- The question asks which is NOT correct, so scan for the one claim that contradicts the others rather than for the true statements.
- Contrast with C, where an executable compiled for one platform will not run on another - that is genuine platform dependence.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Option (a): Python source code is executed by an interpreter rather than compiled to native machine code ahead of time. TRUE of Python.
2. Option (c): Python is well known for readable, indentation-based syntax that is close to pseudocode. TRUE.
3. Option (d): Python is distributed free under an open-source licence maintained by the Python Software Foundation. TRUE.
4. Option (b): because Python programs are run by an interpreter that is itself available on Windows, Linux, macOS and other systems, the same source code runs unchanged across platforms. Python is therefore platform-INDEPENDENT.
5. The statement that Python is platform-dependent is the incorrect one.
6. Hence option (b) is the answer.

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8. 2021 · Q57 Operating Systems, Packages & Utilities · Conceptual

QUESTION

Which is *not* system software?

OPTIONS

(a) Linker

(b) Operating system

(c) Word processor

(d) Assembler

ANSWER **(AI-derived — no official key exists for this paper)**

(c) Word processor

EXAM SHORTCUT **(AI-derived explanation)**

Linkers, operating systems and assemblers all serve the machine; a word processor serves the USER and is therefore application software.

TIPS & TRICKS **(AI-derived explanation)**

- Test by asking 'who is this for?'. System software manages the computer; application software performs a task for the user.
- System software includes the OS, device drivers, and all translators and build tools - compilers, assemblers, linkers, loaders.
- Application software includes word processors, spreadsheets, browsers, media players and database applications.
- Utility software (antivirus, defragmenter) is a maintenance sub-category of system software - useful to keep as a third bucket.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. System software manages and controls computer hardware and provides the platform on which application programs run.
2. A linker combines object modules into an executable - part of the program-development tool chain and hence system software.
3. An operating system manages processes, memory, files and devices - the central item of system software.
4. An assembler translates assembly language into machine code - again a translator and hence system software.
5. A word processor is used to create and edit documents for the user's own purposes, making it application software.
6. Hence the word processor is not system software, option (c).

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9. **2021 · Q80** Low & High Level Languages, Compiler, Assembler · Factual

QUESTION

What is the output of a lexical analyzer?

OPTIONS

(a) Parse tree

(b) List of tokens

(c) Intermediate code

(d) Machine code

ANSWER (AI-derived — no official key exists for this paper)

(b) List of tokens

EXAM SHORTCUT (AI-derived explanation)

The lexical analyser is the first phase of a compiler and its output is a stream of TOKENS handed to the parser.

TIPS & TRICKS (AI-derived explanation)

- Compiler phases in order: lexical analysis (tokens) -> syntax analysis (parse tree) -> semantic analysis -> intermediate code -> optimisation -> code generation (machine code). Each option here is the output of a DIFFERENT phase.
- The lexical analyser also strips whitespace and comments and builds symbol-table entries - useful supporting detail.
- A token is a pair (type, value): identifier, keyword, constant, operator, punctuation. Recognising them is a regular-language task, which is why lexers are built from finite automata.
- The parse tree (option a) is the output of the SYNTAX analyser, which consumes the token list - a one-step-later confusion the examiner is testing.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The lexical analyser (scanner) is the first phase of a compiler.
2. It reads the source program character by character, groups the characters into meaningful units, and removes whitespace and comments.
3. Each unit is emitted as a TOKEN - an identifier, keyword, constant, operator or punctuation symbol - typically together with an entry in the symbol table.
4. This stream of tokens is passed to the syntax analyser (parser), which builds the parse tree from it.
5. Intermediate code and machine code are produced by still later phases.
6. Hence the output of a lexical analyser is a list of tokens, option (b).

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10. 2024 · Q47 Low & High Level Languages, Compiler, Assembler · Conceptual

QUESTION

Consider the following statements: I. Both compiler and interpreter can translate the program written in high-level language into machine-level instructions. II. The interpreter translates the whole program into machine language program before executing any of the instructions. Which of the statements given above is/are correct?

OPTIONS

(a) I only

(b) II only

(c) Both I and II

(d) Neither I nor II

ANSWER (AI-derived — no official key exists for this paper)

(a) I only

EXAM SHORTCUT (AI-derived explanation)

Statement I is true - both translators convert high-level source to machine instructions. Statement II describes the COMPILER, not the interpreter, so it is false.

TIPS & TRICKS (AI-derived explanation)

- One-line contrast: compiler translates the WHOLE program before any execution; interpreter translates and executes ONE statement at a time.
- Statement II simply swaps the two definitions - the classic falsification device in this topic.
- Both are translators, which is why statement I is safe; the difference lies in WHEN translation happens, not in what is produced.
- Consequences: a compiler produces a separate object/executable file and reports all errors up front, whereas an interpreter produces no object file and stops at the first error encountered.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement I: both compilers and interpreters are language translators that convert a program written in a high-level language into machine-level instructions the processor can execute. TRUE.
2. Statement II: it is the COMPILER that reads and translates the entire program before any execution takes place, producing an object program.
3. An interpreter takes the source one statement at a time, translates that statement and immediately executes it before moving on, so it never translates the whole program in advance.
4. The statement therefore attributes the compiler's behaviour to the interpreter. FALSE.
5. Only statement I is correct.
6. Hence option (a) is correct.

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11. 2025 · Q38 Low & High Level Languages, Compiler, Assembler · Conceptual

QUESTION

In high-level programming languages, what role does the runtime environment play?

OPTIONS

(a) It provides libraries and frameworks for application development.

(b) It executes the compiled machine code directly on the CPU.

(c) It manages memory allocation, garbage collection, and other runtime tasks.

(d) It translates source code into machine code during compilation.

ANSWER (AI-derived — no official key exists for this paper)

(c) It manages memory allocation, garbage collection, and other runtime tasks.

EXAM SHORTCUT (AI-derived explanation)

The runtime environment supports the program WHILE it executes - allocating and reclaiming memory, running garbage collection and handling exceptions.

TIPS & TRICKS (AI-derived explanation)

- The clue is in the name: RUNTIME environment means services provided during execution, not before it.
- Translation to machine code (option d) happens at COMPILE time, so it is by definition not a runtime activity.
- Libraries and frameworks (option a) are development-time resources; they may be loaded at runtime but providing them is not the runtime system's defining role.
- Executing machine code directly (option b) is the CPU's job; the runtime sits above it, managing resources on the program's behalf.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A runtime environment is the software layer that supports a program while it is executing.
2. Its responsibilities include allocating memory for objects and activation records, reclaiming unreachable memory through garbage collection, managing the call stack, handling exceptions, and mediating access to operating-system services.
3. Translating source code into machine code is the compiler's task and happens before execution, so option (d) is wrong.
4. Executing machine instructions is performed by the CPU itself, not by the runtime layer, so option (b) is wrong.
5. Libraries and frameworks are resources used during development; supplying them is not what defines a runtime environment.
6. Hence the runtime environment manages memory allocation, garbage collection and other runtime tasks, option (c).

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12. 2025 · Q72 Operating Systems, Packages & Utilities · Definition

QUESTION

What is the main purpose of the linker in the compilation process?

OPTIONS

(a) To perform optimization of compiled code

(b) To translate source code into machine code

(c) To check syntax of the program

(d) To combine object files and libraries into an executable program

ANSWER (AI-derived — no official key exists for this paper)

(d) To combine object files and libraries into an executable program

EXAM SHORTCUT (AI-derived explanation)

The linker's job is in its name - it links. It combines separately compiled object modules with library routines to produce a single executable. Answer (d).

TIPS & TRICKS (AI-derived explanation)

- Pipeline order: preprocessor -> compiler -> assembler -> linker -> loader. Each option here names a different stage, so match the stage to the description.
- Translation to machine code is the compiler/assembler's job (option b); syntax checking is the compiler front-end's job (option c); optimisation is the compiler's middle/back end (option a).
- Static linking bakes library code into the executable; dynamic linking resolves it at load or run time via shared libraries (.so / .dll).
- Undefined reference errors come from the linker, not the compiler - a useful diagnostic memory aid.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. After compilation each source file becomes an object file containing machine code with unresolved external references (calls to functions defined elsewhere or in libraries).
2. The linker takes these object files together with the required library modules, resolves all external symbol references, assigns final relative addresses, and merges everything into one executable file.
3. Option (a) describes the optimiser inside the compiler; option (b) describes the compiler/assembler; option (c) describes the compiler's syntax analysis phase.
4. Only option (d) states the linker's actual function.
5. Hence option (d) is correct.

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13. 2026 · Q61 Operating Systems, Packages & Utilities · Definition

QUESTION

Which is the central part of the UNIX OS that manages and controls communication between hardware and software components?

OPTIONS

(a) Shell

(b) Kernel

(c) Compiler

(d) Device driver

ANSWER (AI-derived — no official key exists for this paper)

(b) Kernel

EXAM SHORTCUT (AI-derived explanation)

The core of any UNIX-like operating system, sitting between hardware and software, is the kernel. The shell is only the command interpreter around it. Answer (b).

TIPS & TRICKS (AI-derived explanation)

- UNIX architecture in layers, from inside out: hardware -> kernel -> shell -> application programs.
- The kernel manages processes, memory, file systems, device drivers and system calls - it is the resident core loaded at boot.
- The shell (sh, bash, ksh, csh) is a user-level command interpreter; several shells can run over one kernel, which is why it cannot be the 'central part'.
- A device driver (option d) controls one particular device and typically runs inside the kernel - a component of it, not the whole.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. UNIX is structured in layers: the hardware at the centre, surrounded by the kernel, then the shell, then user applications and utilities.
2. The kernel is the resident core of the operating system. It manages the CPU, memory and devices, schedules processes, implements the file system, and provides the system-call interface through which all software reaches the hardware.
3. This is exactly the description of managing and controlling communication between hardware and software components.
4. The shell (option a) is a command interpreter that runs as an ordinary user program on top of the kernel; the compiler (option c) is a translation utility, not part of the OS core; a device driver (option d) handles a single class of device and is a kernel module, not the kernel itself.
5. Hence option (b) is correct.

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14. 2026 · Q67 Low & High Level Languages, Compiler, Assembler · Conceptual

QUESTION

In respect of Python: I. It can be used to create web applications. II. It is an interpreted language. III. It is platform-independent. Which are correct?

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(d) I, II and III

EXAM SHORTCUT (AI-derived explanation)

Python is interpreted, runs on Windows, Linux and macOS alike, and powers web frameworks such as Django and Flask. All three statements are true. Answer (d).

TIPS & TRICKS (AI-derived explanation)

- Python's core characteristics: high-level, interpreted (compiled to bytecode then executed by the PVM), dynamically typed, platform-independent, object-oriented, with a very large standard library.
- Web frameworks Django, Flask and FastAPI make statement I obviously true.
- Platform independence follows from the bytecode-plus-virtual-machine model, the same mechanism that makes Java portable.
- In UPSC computer-awareness items, when all listed statements are standard textbook properties with no factual error, 'all of the above' is the expected answer.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement I: Python is widely used for web development through frameworks such as Django, Flask and FastAPI. Correct.
2. Statement II: Python source is compiled to bytecode which is then executed by the Python Virtual Machine, so it is classified as an interpreted (rather than natively compiled) language. Correct.
3. Statement III: because execution takes place on the virtual machine, the same Python program runs unchanged on Windows, Linux, macOS and other platforms with an interpreter available. Correct.
4. All three statements are correct.
5. Hence option (d) is correct.

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15. 2026 · Q70 Operating Systems, Packages & Utilities · Factual

QUESTION

The UNIX operating system is written in which language?

OPTIONS

(a) C

(b) COBOL

(c) JAVA

(d) FORTRAN

ANSWER (AI-derived — no official key exists for this paper)

(a) C

EXAM SHORTCUT (AI-derived explanation)

UNIX was rewritten in C by Dennis Ritchie and Ken Thompson at Bell Labs in 1973 - the first portable operating system. Answer (a).

TIPS & TRICKS (AI-derived explanation)

- The C language was created by Dennis Ritchie specifically to rewrite UNIX, so the two are historically inseparable.
- Rewriting UNIX in a high-level language is what made it portable across machine architectures - previously operating systems were written in assembly.
- Chronology check: COBOL (1959) and FORTRAN (1957) are business and scientific languages, and Java (1995) postdates UNIX (1969) entirely.
- Linux, the BSDs, macOS's Darwin core and Android's kernel are all descendants or reimplementations written predominantly in C.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. UNIX was created at Bell Laboratories by Ken Thompson and Dennis Ritchie beginning in 1969, initially in assembly language for the PDP-7 and PDP-11.
2. In 1973 the kernel was rewritten in C, a language Ritchie had developed for the purpose.
3. This made UNIX the first widely used operating system written in a high-level language, and therefore portable to different hardware by recompilation.
4. COBOL is a business data-processing language, FORTRAN a scientific computing language, and Java was released in 1995, long after UNIX.
5. Hence option (a) is correct.

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16. 2026 · Q73 Low & High Level Languages, Compiler, Assembler · Conceptual

QUESTION

In an absolute loading scheme, which function is performed by the assembler instead of the loader?

OPTIONS

(a) Combining object modules

(b) Allocation

(c) Linking

(d) Loading

ANSWER (AI-derived — no official key exists for this paper)

(b) Allocation

Source note. Textbook presentations of the absolute loading scheme split the four loader functions differently. In Dhamdhere's table, allocation and linking are done by the PROGRAMMER, relocation by the translator (assembler) and loading by the loader; on that reading the intended word would be 'relocation', which is not among the options. Among the choices offered, allocation is the function the assembler takes over from the loader, since in an absolute scheme the assembler fixes the actual load addresses at assembly time. Note also that options (a) 'combining object modules' and (c) 'linking' name the same function, so neither can be the intended key.

EXAM SHORTCUT (AI-derived explanation)

In absolute loading the assembler already fixes the actual memory addresses, so the loader has nothing to allocate - the assembler has done it. Options (a) and (c) both mean linking, and (d) is the one thing the loader still does, so (b) is the answer.

TIPS & TRICKS (AI-derived explanation)

- The four classical loader functions are allocation, linking, relocation and loading; each loading scheme distributes them differently between programmer, assembler and loader.
- Absolute loader: addresses are fixed at assembly time, so the loader merely copies the code to the stated addresses - it is the simplest and fastest, but the program cannot be moved.
- Relocating and linking loaders, by contrast, leave address binding to load time, which is what makes relocatable object modules possible.
- Spot equivalent options: 'combining object modules' and 'linking' describe one and the same function, so an examiner cannot intend either as the unique key.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A loader classically performs four functions: allocation (reserving memory space), linking (resolving external references between modules), relocation (adjusting address-dependent fields), and loading (placing the code in memory).
2. In an absolute loading scheme the assembler translates the source into an absolute object program in which every address is already the final memory address at which the code will run.
3. Consequently the memory space is committed at assembly time: the assembler has performed the allocation function, and the loader is left with nothing to decide - it simply copies the image to the fixed addresses.
4. Option (d) loading is precisely what the loader still does, so it cannot be the answer; options (a) and (c) both describe linking, which in this scheme is handled outside the assembler.
5. Hence option (b) is correct among the choices offered (see the note above on the textbook variant that names relocation).

C7 · COMPUTER APPLICATION AND DATA PROCESSING

Networks: LAN, WAN, Internet, Intranet

29 questions · 2018: 0 2019: 6 2020: 3 2021: 3 2022: 3 2023: 3 2024: 5 2025: 4 2026: 2

Weight. 29 questions (16.1% of Computer Application and Data Processing) · appeared in 8 of 9 papers · peak year 2019 (6)

Examiner's pattern. The biggest computer topic (29 items). Layer-mapping questions dominate: ICMP and ARP at the network layer, SMTP/FTP/HTTP at the application layer. ARP resolving an IP to a MAC address was asked in both 2021 and 2026. Also: topology failure behaviour, DNS, port numbers, transmission media, IEEE 802.11 and connecting devices.

Must-know. ARP: IP -> MAC. DNS: name -> IP. Ports: FTP 21, SMTP 25, HTTP 80, POP3 110, IMAP 143. In a star topology only the hub is a single point of failure; a ring breaks on any single node failure. Wi-Fi = IEEE 802.11.

1. **2019 · Q46** Network - LAN, WAN, Internet, Intranet · Conceptual

QUESTION

Which of the statements regarding Network Protocol are **not** correct? 1. Identification of type of physical connection cannot be done by Network Protocol. 2. Error detection and correction are done by Network Protocol. 3. Message formatting cannot be done by Network Protocol. 4. Network Protocol does initiation and termination.

OPTIONS

(a) 1 and 2

(b) 3 only

(c) 2 and 3

(d) 1 and 3

ANSWER **(AI-derived — no official key exists for this paper)**

(d) 1 and 3

EXAM SHORTCUT **(AI-derived explanation)**

Network protocols DO identify the physical connection and DO format messages, so statements 1 and 3 - both phrased as 'cannot' - are the incorrect ones.

TIPS & TRICKS **(AI-derived explanation)**

- Read the question direction: it asks which statements are NOT correct. The two negatively phrased items are the natural suspects.
- The standard list of protocol functions: physical connection identification, message formatting (syntax and semantics), error detection and correction, sequencing, flow control, and connection initiation and termination.
- Statements 2 and 4 assert genuine protocol functions and are therefore correct, so they cannot be part of the answer - that eliminates options (a) and (c) immediately.

- A protocol is defined as a set of rules governing syntax, semantics and timing of communication; message formatting is literally the 'syntax' component.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The recognised functions of a network protocol include identifying the type of physical connection, formatting messages, detecting and correcting errors, sequencing, flow control, and initiating and terminating a connection.
2. Statement 1 claims that identification of the type of physical connection CANNOT be done by a network protocol. This contradicts a genuine protocol function, so statement 1 is NOT correct.
3. Statement 2 says error detection and correction are done by network protocols - this is a genuine function, so statement 2 IS correct.
4. Statement 3 claims message formatting CANNOT be done by a network protocol. Message formatting is the syntax function of a protocol, so statement 3 is NOT correct.
5. Statement 4 says network protocols handle initiation and termination - a genuine function, so statement 4 IS correct.
6. The statements that are not correct are 1 and 3, option (d).

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2. 2019 · Q50

Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which one of the following protocols is used to send email messages?

OPTIONS

(a) TCP

(b) SMTP

(c) POP

(d) UDP

ANSWER (AI-derived — no official key exists for this paper)

(b) SMTP

EXAM SHORTCUT (AI-derived explanation)

SMTP - Simple Mail Transfer Protocol - is the protocol used to SEND email. POP is for retrieving it.

TIPS & TRICKS (AI-derived explanation)

- Direction is the discriminator: SMTP pushes mail OUT, while POP3 and IMAP pull mail IN. The question says 'send', so SMTP.
- TCP and UDP are transport-layer protocols that carry many kinds of traffic; they are not email protocols, so options (a) and (d) are out of category.
- Port numbers worth remembering: SMTP 25, POP3 110, IMAP 143, HTTP 80, HTTPS 443, FTP 21, Telnet 23, DNS 53.
- IMAP differs from POP3 by leaving messages on the server and synchronising across devices; both are retrieval protocols.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Email uses different protocols for sending and for retrieving messages.
2. SMTP (Simple Mail Transfer Protocol) is the application-layer protocol used to transfer outgoing mail from a client to a mail server and between mail servers.
3. POP (Post Office Protocol) is used by a client to RETRIEVE messages from a mail server, not to send them.
4. TCP and UDP are transport-layer protocols that provide connection-oriented and connectionless delivery respectively; they carry email traffic but are not email protocols.
5. Hence the protocol used to send email messages is SMTP, option (b).

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3. 2019 · Q61 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which one of the following translates IP address to a readable manner such as website URLs?

OPTIONS

(a) DNS

(b) SMTP

(c) DHCP

(d) POP

ANSWER (AI-derived — no official key exists for this paper)

(a) DNS

EXAM SHORTCUT (AI-derived explanation)

DNS - the Domain Name System - is the internet's directory service, mapping between human-readable names and numeric IP addresses.

TIPS & TRICKS (AI-derived explanation)

- Mnemonic: DNS = 'Directory Name Service' - it is the phone book of the internet, translating names to numbers and back.
- DHCP (option c) does something different: it ASSIGNS IP addresses to hosts dynamically. Name-to-number translation versus address allocation is the distinction being tested.
- SMTP and POP are email protocols and have nothing to do with address resolution, so options (b) and (d) are out of category.
- Reverse DNS performs the number-to-name direction described in the stem; both directions are handled by the same Domain Name System.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Computers route traffic using numeric IP addresses, but users work with readable domain names such as www.example.com.
2. The Domain Name System (DNS) is the distributed directory service that maps between the two forms - names to addresses, and addresses back to names by reverse lookup.

3. SMTP is the protocol for sending email, and POP is the protocol for retrieving it; neither performs address translation.
4. DHCP dynamically allocates IP addresses and configuration parameters to hosts; it does not translate addresses into names.
5. Hence the answer is DNS, option (a).

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4. **2019 · Q64** Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which one of the following network cables is capable of faster data transmission?

OPTIONS

(a) Thin Ethernet cable

(b) Thick Ethernet cable

(c) Fiber optic cable

(d) UTP cable

ANSWER (AI-derived — no official key exists for this paper)

(c) Fiber optic cable

EXAM SHORTCUT (AI-derived explanation)

Fibre optic carries light rather than electrical signals, giving by far the highest bandwidth of the four - gigabits and beyond over long distances.

TIPS & TRICKS (AI-derived explanation)

- Speed ranking of common media: fibre optic > UTP (twisted pair) > coaxial (thick then thin Ethernet). Fibre wins every 'fastest medium' question.
- Fibre's other advantages worth quoting: immunity to electromagnetic interference, very low attenuation over long runs, and high security since it is hard to tap.
- Thin Ethernet (10Base2) and thick Ethernet (10Base5) are legacy coaxial standards capped around 10 Mbps - historical distractors.
- The identical concept appears again as Q67 in this very paper, so a single fact answers two marks.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Thin Ethernet (10Base2) and thick Ethernet (10Base5) are coaxial cable standards limited to about 10 Mbps.
2. Unshielded twisted pair (UTP) supports higher rates - 100 Mbps to 10 Gbps over short distances in modern categories - but is still limited by electrical attenuation and interference.
3. Fibre optic cable transmits data as pulses of light through a glass or plastic core, achieving very high bandwidth with extremely low attenuation over long distances.
4. It is also immune to electromagnetic interference, which allows the high rates to be sustained in practice.

5. Hence fibre optic cable permits the fastest data transmission, option (c).

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5. 2019 · Q66 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

The name of the network developed by the Government of India in public interest for various government activities is

OPTIONS

(a) ARPANET

(b) NICNET

(c) Quicknet

(d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(b) NICNET

EXAM SHORTCUT (AI-derived explanation)

NICNET is the nationwide government network built by the National Informatics Centre for Indian government activities.

TIPS & TRICKS (AI-derived explanation)

- NICNET = National Informatics Centre NETwork, launched in the 1980s to link central ministries, states and districts for e-governance.
- ARPANET (option a) is the American research network that became the ancestor of the internet - a foreign, historical distractor.
- Related Indian networks worth knowing: ERNET (education and research), INDONET, NKN (National Knowledge Network) and SWAN (State Wide Area Network).
- 'Government of India, public interest, government activities' in the stem is the signature of NIC and hence of NICNET.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. NICNET is the nationwide computer communication network established by the National Informatics Centre (NIC) under the Government of India.
2. It connects central government ministries and departments with state governments and district administrations, supporting e-governance and public information services.
3. ARPANET was the US Defense Department research network of the late 1960s from which the internet evolved, and has no connection with the Government of India.
4. 'Quicknet' is not an Indian government network.
5. Hence the network developed by the Government of India is NICNET, option (b).

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6. 2019 · Q67 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which one of the following communication media supports higher Internet bandwidth?

OPTIONS

(a) UTP cable

(b) Coaxial cable

(c) Fiber optic cable

(d) Thin Ethernet

ANSWER (AI-derived — no official key exists for this paper)

(c) Fiber optic cable

EXAM SHORTCUT (AI-derived explanation)

Fibre optic supports by far the highest bandwidth of the listed media.

TIPS & TRICKS (AI-derived explanation)

- This restates Q64 of the same paper - the examiner is testing the same ranking twice, so the medium comparison is clearly worth memorising.
- Bandwidth ranking: fibre optic > coaxial > UTP > thin Ethernet in classical textbook terms, with fibre always at the top.
- Fibre carries light rather than electrical current, so its bandwidth is not limited by the resistance, capacitance and interference that cap copper media.
- Modern backbone links and undersea cables are all fibre for exactly this reason - a real-world anchor for the fact.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The question asks which medium supports the highest internet bandwidth.
2. UTP cable and thin Ethernet are copper media whose bandwidth is limited by attenuation and electromagnetic interference.
3. Coaxial cable offers more bandwidth than twisted pair but is still an electrical medium subject to the same physical limits.
4. Fibre optic cable transmits information as light through a glass core, offering bandwidth orders of magnitude greater than any copper medium and negligible interference.
5. Hence fibre optic cable supports the highest internet bandwidth, option (c).

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7. 2020 · Q44 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which is *not* an application-layer protocol?

OPTIONS

(a) ARP

(b) FTP

(c) DNS

(d) SNMP

ANSWER **(AI-derived — no official key exists for this paper)**

(a) ARP

EXAM SHORTCUT **(AI-derived explanation)**

FTP, DNS and SNMP are all application-layer protocols. ARP maps IP addresses to MAC addresses and operates at the network/data-link boundary, not the application layer.

TIPS & TRICKS **(AI-derived explanation)**

- Application-layer roll call: HTTP, FTP, SMTP, POP3, IMAP, DNS, SNMP, Telnet, DHCP. If a protocol is not on that list, suspect a lower layer.
- ARP's job - resolving an IP address to a hardware (MAC) address - is meaningful only on a local link, which is the giveaway that it sits far below the application layer.
- Layer anchors: application (HTTP, FTP), transport (TCP, UDP), network (IP, ICMP), data link (ARP, Ethernet).
- Do not confuse ARP with RARP (MAC to IP) or with DNS (name to IP); all three are 'resolution' protocols but at different layers.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. FTP (File Transfer Protocol) provides file transfer services directly to users and applications - application layer.
2. DNS (Domain Name System) resolves domain names to IP addresses for applications - application layer.
3. SNMP (Simple Network Management Protocol) manages network devices from management applications - application layer.
4. ARP (Address Resolution Protocol) maps an IP address to the corresponding physical MAC address on a local network segment.
5. ARP therefore operates at the network/data-link boundary of the TCP/IP model, not the application layer.
6. Hence ARP is not an application-layer protocol, option (a).

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8. **2020 · Q53** Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which is *not* a web browser?

OPTIONS

(a) Google Chrome

(b) Microsoft Internet Explorer

(c) Apple iOS

(d) Mozilla Firefox

ANSWER (AI-derived — no official key exists for this paper)

(c) Apple iOS

EXAM SHORTCUT (AI-derived explanation)

Chrome, Internet Explorer and Firefox are browsers. Apple iOS is a mobile OPERATING SYSTEM, not a browser.

TIPS & TRICKS (AI-derived explanation)

- Sort by category before comparing details: three items are applications and one is an operating system - the odd one out is immediate.
- Apple's browser is SAFARI; iOS is the operating system that runs it. Confusing the vendor's OS with its browser is the intended trap.
- Other browsers worth recognising: Safari, Opera, Microsoft Edge, Brave, UC Browser.
- A browser is application software that requests and renders web pages over HTTP/HTTPS; an operating system manages hardware and runs such applications.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A web browser is application software that retrieves web pages over HTTP/HTTPS and renders them for the user.
2. Google Chrome is a browser developed by Google.
3. Microsoft Internet Explorer was Microsoft's browser, later succeeded by Edge.
4. Mozilla Firefox is the open-source browser from the Mozilla Foundation.
5. Apple iOS is the mobile operating system that runs on iPhones and iPads; the browser supplied with it is Safari.
6. Hence Apple iOS is not a web browser, option (c).

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9. **2020 · Q57** Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which is the wireless technology standard?

OPTIONS

(a) IEEE 803.11b

(b) IEEE 803.11e

(c) IEEE 801.11c

(d) IEEE 802.11b

ANSWER (AI-derived — no official key exists for this paper)

(d) IEEE 802.11b

EXAM SHORTCUT (AI-derived explanation)

The wireless LAN family is IEEE 802.11. Only option (d) has the correct 802 prefix; the others print 803 or 801.

TIPS & TRICKS (AI-derived explanation)

- IEEE 802 is the family for local and metropolitan area networks: 802.3 is Ethernet, 802.11 is Wi-Fi, 802.15 is Bluetooth, 802.16 is WiMAX.
- This item is really a proof-reading test - scan the digits before the first dot and reject anything that is not 802.
- 802.11 letter suffixes and their nominal rates: $b = 11$ Mbps, $a = 54$ Mbps, $g = 54$ Mbps, $n = 600$ Mbps, ac = gigabit class.
- Wi-Fi is the marketing name for certified 802.11 equipment - the two terms are used interchangeably in these papers.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Wireless local area networking is standardised by the IEEE 802.11 family of standards.
2. Option (a) IEEE 803.11b and option (b) IEEE 803.11e use the non-existent prefix 803.
3. Option (c) IEEE 801.11c uses the non-existent prefix 801.
4. Option (d) IEEE 802.11b is a genuine standard, the 1999 amendment providing up to 11 Mbps in the 2.4 GHz band.
5. Hence the wireless technology standard is IEEE 802.11b, option (d).

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10. 2021 · Q52 Network - LAN, WAN, Internet, Intranet · Conceptual

QUESTION

Which is/are advantage(s) of star topology? 1. If the hub fails, the network is unaffected. 2. It requires more cable to connect nodes. 3. It allows easy error detection and correction.

OPTIONS

(a) 1 only

(b) 2 only

(c) 3 only

(d) 2 and 3 only

ANSWER (AI-derived — no official key exists for this paper)

(c) 3 only

EXAM SHORTCUT (AI-derived explanation)

Statement 1 is false - the hub is a single point of failure. Statement 2 describes a DISADVANTAGE. Only statement 3, easy fault isolation, is a genuine advantage.

TIPS & TRICKS (AI-derived explanation)

- Star topology: every node connects to a central hub or switch. Advantages - easy fault isolation, simple to add or remove nodes, one failed link affects only that node. Disadvantages - hub is a single point of failure, more cable needed.
- Read the question direction: it asks for ADVANTAGES, so a true statement that describes a drawback (statement 2) still does not qualify.
- Statement 1 inverts the best-known weakness of the star: if the hub fails, the ENTIRE network goes down.
- Contrast with bus topology (less cable, harder fault isolation) and ring topology (each node's failure can break the loop unless dual rings are used).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. In a star topology every node is connected by its own cable to a central hub or switch.
2. Statement 1: the hub is the single point through which all traffic passes, so if the hub fails the whole network fails. The claim that the network is unaffected is FALSE.
3. Statement 2: a star does require more cable than a bus, since each node needs its own dedicated link. This is a genuine property, but it is a DISADVANTAGE, not an advantage, so it does not answer the question.
4. Statement 3: because each node has a separate link to the hub, a fault can be traced to a single cable or port, and a failed link affects only that one node. Easy error detection and isolation is the principal ADVANTAGE of the star.
5. Only statement 3 is an advantage.
6. Hence option (c) is correct.

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11. 2021 · Q54 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which protocol is commonly used to retrieve e-mail from a mail server?

OPTIONS

(a) FTP

(b) IMAP

(c) HTML

(d) TELNET

ANSWER (AI-derived — no official key exists for this paper)

(b) IMAP

EXAM SHORTCUT (AI-derived explanation)

IMAP (Internet Message Access Protocol) is the standard protocol for retrieving mail from a server; SMTP sends it.

TIPS & TRICKS (AI-derived explanation)

- Direction again: SMTP pushes mail OUT, POP3 and IMAP pull it IN. The word 'retrieve' selects the incoming pair.
- IMAP versus POP3: IMAP leaves messages on the server and synchronises folders across devices; POP3 typically downloads and deletes. IMAP is the modern default.
- FTP transfers files, HTML is a markup language (not a protocol at all), and TELNET provides remote terminal access - all out of category here.
- Ports: IMAP 143 (993 with TLS), POP3 110 (995 with TLS), SMTP 25 (587 for submission).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Retrieving email from a mail server to a client is done by a mail access protocol.
2. IMAP (Internet Message Access Protocol) is the standard protocol for this purpose; it allows the client to read, organise and search messages that remain stored on the server.
3. FTP is a file transfer protocol and is not used for mailbox access.
4. HTML is a markup language for describing web pages, not a network protocol.
5. TELNET provides a remote terminal login session.
6. Hence the protocol used to retrieve email is IMAP, option (b).

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12. 2021 · Q71 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which protocol determines the MAC address?

OPTIONS

(a) ICMP

(b) IP

(c) ARP

(d) FTP

ANSWER (AI-derived — no official key exists for this paper)

(c) ARP

EXAM SHORTCUT (AI-derived explanation)

ARP - Address Resolution Protocol - is precisely the protocol that finds the MAC address corresponding to a known IP address.

TIPS & TRICKS (AI-derived explanation)

- The name is the answer: Address Resolution Protocol resolves a network-layer (IP) address into a link-layer (MAC) address.

- The reverse direction, MAC to IP, was handled by RARP (now largely replaced by DHCP/BOOTP) - keep the pair straight.
- ICMP carries error and diagnostic messages such as ping and 'destination unreachable'; IP handles routing and addressing; FTP transfers files. None resolves hardware addresses.
- ARP works by broadcasting 'who has this IP?' on the local segment and caching the reply in the ARP table - the cache is why repeated lookups are fast.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. To deliver a packet on a local network, a host needs the destination's physical (MAC) address, but applications and routing use IP addresses.
2. ARP (Address Resolution Protocol) broadcasts a request asking which host owns a given IP address; the owner replies with its MAC address, which is then cached in the ARP table.
3. ICMP is used for error reporting and diagnostics (for example ping), not address resolution.
4. IP provides logical addressing and routing but does not itself determine hardware addresses.
5. FTP is an application-layer file transfer protocol.
6. Hence ARP determines the MAC address, option (c).

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13. 2022 · Q38 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which one of the following is a dedicated phone line that connects a computer to the internet?

OPTIONS

(a) Dial-up access

(b) Digital subscriber line

(c) Integrated Services Digital Network

(d) Leased line

ANSWER (AI-derived — no official key exists for this paper)

(d) Leased line

EXAM SHORTCUT (AI-derived explanation)

A leased line is a permanent, dedicated circuit rented from a carrier and reserved solely for one subscriber.

TIPS & TRICKS (AI-derived explanation)

- 'Dedicated' is the keyword: a leased line is reserved full-time for one customer, with no dialling and no sharing.
- Dial-up (option a) establishes a temporary connection over the ordinary switched telephone network - the opposite of dedicated.
- DSL and ISDN run over the existing local loop and are shared, switched services; they are always-on in the DSL case but are not dedicated point-to-point circuits.

- Leased lines offer guaranteed symmetric bandwidth and an SLA, which is why organisations pay a premium for them.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A leased line is a permanent telecommunications circuit rented from a carrier and reserved exclusively for one subscriber, connecting two fixed points around the clock.
2. Because the circuit is not shared or switched, it provides guaranteed, symmetric bandwidth with no dialling or contention.
3. Dial-up access sets up a temporary connection through the public switched telephone network and occupies the line only during the call.
4. DSL provides an always-on service over the ordinary telephone local loop but is a shared, contended service rather than a dedicated circuit.
5. ISDN is a switched digital service in which calls are set up and torn down as required.
6. Hence the dedicated phone line is the leased line, option (d).

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14. 2022 · Q75 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Conceptual

QUESTION

Which one of the following provides application-independent security and privacy over the internet?

OPTIONS

(a) IP security protocol

(b) Secure socket layer

(c) Internet protocol

(d) Transmission control protocol

ANSWER (AI-derived — no official key exists for this paper)

(a) IP security protocol

EXAM SHORTCUT (AI-derived explanation)

IPsec operates at the NETWORK layer, so it protects all traffic transparently regardless of the application; SSL sits above the transport layer and each application must be written to use it.

TIPS & TRICKS (AI-derived explanation)

- Layer determines transparency: IPsec at layer 3 secures every packet automatically, while SSL/TLS at the transport-session boundary must be invoked by the application (HTTPS, FTPS and so on).
- 'Application-independent' is the key phrase - it points to the layer BELOW the applications, which is where IPsec lives.
- IPsec provides authentication (AH) and confidentiality (ESP) and is the basis of most VPNs, which is why it can protect legacy applications unchanged.
- Plain IP and TCP (options c and d) provide no security at all - they are the protocols IPsec was designed to protect.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The question asks for a mechanism that secures internet communication without the applications needing to be aware of it.
2. IPsec operates at the network (IP) layer, encrypting and authenticating IP packets themselves. Because it sits below the transport and application layers, every application's traffic is protected transparently, with no modification to the applications.
3. SSL (and its successor TLS) operates above the transport layer; an application must be explicitly written to open an SSL connection, which is why HTTPS, FTPS and SMTPS exist as separate application-level variants.
4. IP and TCP provide addressing, routing and reliable delivery but offer no security services of their own.
5. Hence the application-independent mechanism is the IP security protocol, option (a).

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15. 2022 · Q77 Network - LAN, WAN, Internet, Intranet · Conceptual

QUESTION

Consider network topologies: 1. Bus, 2. Star, 3. Ring, 4. Mesh. In which does the failure of one computer not affect the others?

OPTIONS

(a) 1, 2 and 4

(b) 1, 3 and 4

(c) 3 and 4 only

(d) 1 and 2 only

ANSWER (AI-derived — no official key exists for this paper)

(a) 1, 2 and 4

EXAM SHORTCUT (AI-derived explanation)

In bus, star and mesh topologies a failed node simply drops off the network. In a RING the signal must pass through every node, so one failure breaks the loop for everyone.

TIPS & TRICKS (AI-derived explanation)

- The discriminator is whether traffic must PASS THROUGH the failed node. Only the ring routes every message through each station in turn.
- In a bus, nodes tap a shared cable passively; in a star, each node has its own link to the hub; in a mesh, redundant paths bypass any failure. None depends on a particular node.
- Note the caveat for the star: a HUB failure disables the network, but the question asks about the failure of a computer, not of the hub.
- Ring networks mitigate this with dual counter-rotating rings (as in FDDI) precisely because the single-ring weakness is so severe.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The question is whether the failure of one computer disrupts the rest of the network.
2. Item 1, bus: all nodes tap a single shared backbone passively, so a failed node simply stops transmitting and the others continue. UNAFFECTED. (A break in the backbone cable itself is a different matter.)
3. Item 2, star: each node has its own dedicated link to the central hub, so a failed node affects only its own link. UNAFFECTED by another computer's failure.
4. Item 3, ring: data circulate from node to node around the loop, and every station must forward the signal. A single failed node breaks the ring and halts communication for all. AFFECTED.
5. Item 4, mesh: multiple redundant paths connect the nodes, so traffic is rerouted around any failure. UNAFFECTED.
6. Hence the failure of one computer does not affect the others in topologies 1, 2 and 4, option (a).

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16. 2023 · Q64 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

In TCP/IP reference model, consider the following protocols belonging to various layers: 1. SMTP 2. FTP 3. DHCP 4. IP 5. TCP Which of the above are the Application Layer Protocols?

OPTIONS

(a) 1, 2 and 3

(b) 1, 3 and 4

(c) 1, 3 and 5

(d) 3, 4 and 5

ANSWER (AI-derived — no official key exists for this paper)

(a) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

SMTP, FTP and DHCP are application-layer protocols; IP is network layer and TCP is transport layer.

TIPS & TRICKS (AI-derived explanation)

- TCP/IP layer map: application (HTTP, FTP, SMTP, DNS, DHCP, SNMP, Telnet), transport (TCP, UDP), internet (IP, ICMP, ARP), network access (Ethernet).
- The two easiest to place are IP and TCP - they name the model itself and sit at the internet and transport layers respectively, so any option containing them is wrong.
- DHCP surprises candidates because it configures addresses, but it runs over UDP as an ordinary application service - hence application layer.
- Once items 4 and 5 are excluded, only option (a) remains without further checking.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The TCP/IP reference model has four layers: application, transport, internet and network access.
2. Item 1, SMTP: transfers email between clients and servers - an application service. APPLICATION.

3. Item 2, FTP: transfers files on behalf of users - APPLICATION.
4. Item 3, DHCP: dynamically assigns IP addresses and configuration to hosts, running as a client-server service over UDP - APPLICATION.
5. Item 4, IP provides logical addressing and routing at the INTERNET layer, and item 5, TCP provides reliable end-to-end delivery at the TRANSPORT layer.
6. Hence the application-layer protocols are 1, 2 and 3, option (a).

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17. 2023 · Q70 Network - LAN, WAN, Internet, Intranet · Conceptual

QUESTION

Consider the following statements: 1. TCP/IP is the communication protocol used in Internet. 2. A Network Interface Card (NIC) is used to connect a computer to Internet. 3. IP address is also known as hardware address that is assigned uniquely to every computer connected to the Internet. Which of the statements given above are correct?

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(a) 1 and 2 only

EXAM SHORTCUT (AI-derived explanation)

Statements 1 and 2 are correct. Statement 3 confuses the IP address with the MAC address - the hardware address is the MAC, not the IP.

TIPS & TRICKS (AI-derived explanation)

- Address distinction: the MAC address is the permanent HARDWARE address burned into the NIC; the IP address is a LOGICAL address assigned by the network and changeable.
- ARP exists precisely to translate between the two, which is proof that they are different things.
- TCP/IP is the protocol suite of the internet - statement 1 is definitional and safe.
- The NIC provides the physical interface to the network, so statement 2 is also straightforward; the false statement is the one that conflates two address types.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: the internet runs on the TCP/IP protocol suite, with IP providing addressing and routing and TCP reliable delivery. TRUE.
2. Statement 2: a Network Interface Card provides the physical and data-link connection between a computer and the network, and hence to the internet. TRUE.
3. Statement 3: the HARDWARE address is the MAC address, a 48-bit identifier permanently assigned to the NIC by its manufacturer.

4. The IP address is a LOGICAL address assigned by the network administrator or by DHCP, and it changes when the machine moves to a different network.
5. Since ARP is needed to translate an IP address into a MAC address, the two are plainly distinct. Statement 3 is FALSE.
6. Only statements 1 and 2 are correct, option (a).

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18. 2023 · Q73 Network - LAN, WAN, Internet, Intranet · Definition

QUESTION

A device that converts digital signal into analog signal and vice-versa is known as:

OPTIONS

(a) Network switch

(b) Modem

(c) Network hub

(d) Network router

ANSWER (AI-derived — no official key exists for this paper)

(b) Modem

EXAM SHORTCUT (AI-derived explanation)

A modem MODulates digital data onto an analog carrier and DEModulates it back - the name is the definition.

TIPS & TRICKS (AI-derived explanation)

- The acronym is the answer: MODulator-DEModulator. Digital to analog and back is exactly what it does.
- Switches, hubs and routers all handle DIGITAL signals only - none performs signal-form conversion, so they are out of category.
- Layer placement: a hub works at layer 1, a switch at layer 2, a router at layer 3; a modem operates at the physical layer converting signal form.
- Modems were needed because telephone lines carried analog signals; DSL and cable modems apply the same principle at much higher frequencies.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A modem (MOdulator-DEModulator) converts the digital signals produced by a computer into analog signals suitable for transmission over an analog medium such as a telephone line.
2. At the receiving end it performs the reverse conversion, demodulating the analog carrier back into digital data.
3. A network switch forwards digital frames between ports using MAC addresses (layer 2).
4. A network hub simply repeats incoming digital signals to all its ports (layer 1).
5. A router forwards digital packets between networks using IP addresses (layer 3). None of these three converts between digital and analog form.

6. Hence the device is the modem, option (b).

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19. 2024 · Q45 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

At which layer of TCP/IP do ICMP and ARP function?

OPTIONS

(a) Network layer

(b) Application layer

(c) Transport layer

(d) Physical layer

ANSWER (AI-derived — no official key exists for this paper)

(a) Network layer

EXAM SHORTCUT (AI-derived explanation)

ICMP and ARP are both support protocols for IP, so in the TCP/IP model they are placed at the network (internet) layer.

TIPS & TRICKS (AI-derived explanation)

- ICMP carries error and control messages for IP itself (ping, destination unreachable, time exceeded), so it sits alongside IP at the network layer.
- ARP resolves IP addresses to MAC addresses; it straddles the network-data link boundary but is conventionally grouped with the network layer in the TCP/IP model.
- Neither carries user data end to end, so they cannot be transport or application protocols - options (b) and (c) are out of category.
- Companion facts: ICMP messages are encapsulated inside IP datagrams, and ARP requests are broadcast on the local link only.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The TCP/IP model has four layers: application, transport, network (internet) and network access.
2. ICMP (Internet Control Message Protocol) carries error reports and diagnostic messages about IP delivery, such as 'destination unreachable' and the echo request and reply used by ping. Its messages are encapsulated in IP datagrams, so it belongs with IP at the network layer.
3. ARP (Address Resolution Protocol) maps an IP address to the corresponding MAC address so that a datagram can be delivered on the local link. It supports IP addressing and is conventionally placed at the network layer of the TCP/IP model.
4. Neither protocol provides end-to-end data transfer (transport layer) nor user services (application layer).
5. Hence both function at the network layer, option (a).

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20. 2024 · Q50 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

In an e-mail service, the protocols used are SMTP, POP3 and IMAP. What are the port numbers in sequence?

OPTIONS

(a) 110, 143, 25

(b) 25, 143, 110

(c) 143, 110, 25

(d) 25, 110, 143

ANSWER (AI-derived — no official key exists for this paper)

(d) 25, 110, 143

EXAM SHORTCUT (AI-derived explanation)

SMTP uses port 25, POP3 uses 110 and IMAP uses 143 - in the order asked, 25, 110, 143.

TIPS & TRICKS (AI-derived explanation)

- Memorise the email trio in ascending order: 25 (SMTP), 110 (POP3), 143 (IMAP). They happen to run low to high in the same order the question lists them.
- Other well-known ports: HTTP 80, HTTPS 443, FTP 21 (control) and 20 (data), Telnet 23, DNS 53, SSH 22.
- Secure variants: SMTPS $465/587$, POP3S 995, IMAPS 993 - useful if a question asks about encrypted mail.
- Read the requested SEQUENCE carefully; options (a), (b) and (c) contain the same three numbers in different orders, so the ordering is the entire question.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. SMTP (Simple Mail Transfer Protocol) transfers outgoing mail and uses well-known port 25.
2. POP3 (Post Office Protocol version 3) retrieves mail from a server, downloading and typically deleting it, and uses port 110.
3. IMAP (Internet Message Access Protocol) also retrieves mail but leaves it on the server for synchronisation across devices, and uses port 143.
4. The question asks for the ports in the sequence SMTP, POP3, IMAP.
5. That sequence is 25, 110, 143.
6. Option (d) is correct.

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21. 2024 · Q52 Network - LAN, WAN, Internet, Intranet · Definition

QUESTION

Sending messages using IPSEC protocol on the internet between two IPSEC compliant router servers is also termed as:

OPTIONS

(a) tunneling

(b) leased line

(c) packet filtering

(d) message switching

ANSWER (AI-derived — no official key exists for this paper)

(a) tunneling

EXAM SHORTCUT (AI-derived explanation)

Encapsulating IPSec-protected packets inside ordinary IP packets between two gateway routers is called TUNNELLING - the basis of a VPN.

TIPS & TRICKS (AI-derived explanation)

- IPSec has two modes: TRANSPORT mode (host to host, only the payload protected) and TUNNEL mode (gateway to gateway, the whole original packet encapsulated). Router-to-router means tunnel mode.
- The tunnel is what creates a Virtual Private Network across the public internet - a useful real-world anchor.
- A leased line is a physical dedicated circuit, not a software encapsulation, so option (b) is a different category entirely.
- Packet filtering is a firewall function and message switching is a store-and-forward transmission technique - neither describes IPSec encapsulation.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. IPSec can operate in transport mode, protecting only the payload between two hosts, or in tunnel mode, in which the entire original IP packet is encrypted and encapsulated inside a new IP packet.
2. When two IPSec-compliant routers exchange traffic across the public internet, they use tunnel mode: each router encapsulates the packets of its internal network and the peer router decapsulates them.
3. This encapsulation of one protocol's packets inside another's is called TUNNELLING, and it is the mechanism underlying a Virtual Private Network.
4. A leased line is a physical dedicated circuit rented from a carrier, not an encapsulation technique.
5. Packet filtering is a firewall function, and message switching is a store-and-forward transmission method.
6. Hence the term is tunnelling, option (a).

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22. 2024 · Q55 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which of the following are network layer protocols? I. HTTP II. FTP III. NMP IV. SMTP Select the correct answer using the code given below:

OPTIONS

(a) I, II and III

(b) I, II and IV

(c) I, III and IV

(d) II, III and IV

ANSWER (AI-derived — no official key exists for this paper)

(b) I, II and IV

Source note. SOURCE NOTE: HTTP, FTP and SMTP are strictly APPLICATION-layer protocols of the TCP/IP model, not network-layer protocols (the network layer carries IP, ICMP and ARP). 'NMP' is not a standard TCP/IP protocol name - the recognised network-management protocol is SNMP, itself an application-layer protocol. Under the literal reading of the stem none of the four items is a network-layer protocol; option (b) is selected because it names the three genuine, universally recognised protocols in the list, and the discrepancy is flagged rather than the stem altered.

EXAM SHORTCUT (AI-derived explanation)

HTTP, FTP and SMTP are the three genuine, standard protocols listed; 'NMP' is not a recognised TCP/IP protocol name (the network-management protocol is SNMP).

TIPS & TRICKS (AI-derived explanation)

- Correct layer map: application - HTTP, FTP, SMTP, DNS, SNMP, Telnet; transport - TCP, UDP; network - IP, ICMP, ARP, RARP.
- If a question about the 'network layer' lists only application-layer names, work by elimination on which items are genuine protocols at all.
- SNMP (Simple Network Management Protocol) is the real protocol; 'NMP' with the S omitted is not a standard designation.
- Keep the true network-layer list short and firm - IP, ICMP, ARP - so that mislabelled options are easy to spot.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. In the TCP/IP reference model the network (internet) layer carries IP, ICMP, ARP and RARP.
2. Item I, HTTP: transfers web pages between browser and server - an APPLICATION-layer protocol.
3. Item II, FTP: transfers files on behalf of users - an APPLICATION-layer protocol.
4. Item IV, SMTP: transfers outgoing email - an APPLICATION-layer protocol.
5. Item III, 'NMP': not a standard TCP/IP protocol name; the recognised network-management protocol is SNMP, which is itself an application-layer protocol.
6. None of the four is genuinely a network-layer protocol, so the item is flagged; among the printed options only (b) names the three universally recognised protocols HTTP, FTP and SMTP, and it is selected on that basis.

23. 2024 · Q56 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which of the following are network connecting devices? I. Repeater II. Bridge III. Hub IV. Bluetooth Select the correct answer using the code given below:

OPTIONS

(a) I, II and III

(b) I, II and IV

(c) I, III and IV

(d) II, III and IV

ANSWER (AI-derived — no official key exists for this paper)

(a) I, II and III

EXAM SHORTCUT (AI-derived explanation)

Repeater, bridge and hub are all network connecting devices. Bluetooth is a short-range wireless communication STANDARD, not a device.

TIPS & TRICKS (AI-derived explanation)

- Connecting devices by layer: repeater and hub at layer 1 (physical), bridge and switch at layer 2 (data link), router at layer 3 (network), gateway above that.
- Bluetooth is a technology/standard (IEEE 802.15) like Wi-Fi, not a piece of connecting hardware - a category difference, not a technical detail.
- A repeater regenerates a weakened signal; a hub repeats it to all ports; a bridge forwards frames selectively between segments using MAC addresses.
- Sorting the list by category before comparing details identifies the odd one out immediately.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Network connecting devices are hardware components used to link segments or nodes of a network.
2. Item I, repeater: regenerates and retransmits a weakened signal to extend the reach of a segment - a physical-layer connecting device.
3. Item II, bridge: connects two network segments and forwards frames selectively using MAC addresses - a data-link-layer connecting device.
4. Item III, hub: a multiport repeater that connects several nodes into one segment - a physical-layer connecting device.
5. Item IV, Bluetooth: a short-range wireless communication STANDARD (IEEE 802.15), not a connecting device in its own right.
6. Hence items I, II and III are the connecting devices, option (a).

24. 2025 · Q31 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

In which topology are all nodes connected to a single backbone cable?

OPTIONS

(a) Mesh

(b) Star

(c) Hybrid

(d) Bus

ANSWER (AI-derived — no official key exists for this paper)

(d) Bus

EXAM SHORTCUT (AI-derived explanation)

A single shared backbone cable with all nodes tapping into it is the BUS topology.

TIPS & TRICKS (AI-derived explanation)

- Topology shapes: bus = one shared backbone; star = all nodes to a central hub; ring = closed loop; mesh = every node to every other; hybrid = a combination.
- The phrase 'single backbone cable' is the definition of a bus - no other topology has one shared medium.
- Bus needs terminators at both ends to prevent signal reflection, and a break in the backbone brings the whole network down.
- Bus uses the least cable of any topology but offers the worst fault isolation - the classic trade-off to quote.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. In a bus topology every node is attached, by a short drop cable, to one common backbone cable that runs the length of the network.
2. All transmissions travel along that shared medium and are seen by every node, with terminators at each end absorbing the signal.
3. A star topology connects each node by its own cable to a central hub or switch, so there is no shared backbone.
4. A mesh topology provides direct links between pairs of nodes, and a hybrid topology combines two or more basic types.
5. Hence the topology with a single backbone cable is the bus, option (d).

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25. 2025 · Q32 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which device connects computers in a star topology?

OPTIONS

(a) Router

(b) Hub

(c) Repeater

(d) Bridge

ANSWER (AI-derived — no official key exists for this paper)

(b) Hub

EXAM SHORTCUT (AI-derived explanation)

In a star topology every node connects to a central HUB (or switch), which forms the hub of the star.

TIPS & TRICKS (AI-derived explanation)

- The star's defining component is the central device - classically a hub, in modern networks a switch.
- A repeater merely regenerates a signal along one link and a bridge joins two segments; neither serves as the centre of a star.
- A router connects different NETWORKS at layer 3, not the nodes of a single LAN - a category difference.
- Consequence worth remembering: the hub is the star's single point of failure, which is why hub failure disables the whole network.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. In a star topology each node is connected by its own dedicated cable to a central device, which relays traffic between them.
2. That central device is a hub (or, in modern installations, a switch), giving the layout its star shape.
3. A repeater simply regenerates a weakened signal along a single link and connects only two segments.
4. A bridge joins two network segments and filters frames between them; it does not act as a central connection point for many nodes.
5. A router forwards packets between distinct networks at the network layer.
6. Hence the device at the centre of a star topology is the hub, option (b).

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26. 2025 · Q75 Network - LAN, WAN, Internet, Intranet · Conceptual

QUESTION

In respect of the working of a router: I. It interconnects computer networks. II. It determines on which outgoing link an IP packet is forwarded. III. The destination address of the IP packet is examined and it is routed to the next router. Which are correct?

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(d) I, II and III

EXAM SHORTCUT (AI-derived explanation)

All three describe exactly what a router does at the network layer: interconnect networks, choose the outgoing link, and forward on the destination IP address. Answer (d).

TIPS & TRICKS (AI-derived explanation)

- Layer map: hub = physical (layer 1), switch/bridge = data link (layer 2), router = network (layer 3), gateway = up to application layer.
- A router forwards on the destination IP address using its routing table (longest-prefix match); a switch forwards on the destination MAC address.
- Routers separate broadcast domains, which is why they are the device that interconnects distinct networks - statement I.
- Each hop re-examines the destination address and passes the packet to the next router - this hop-by-hop behaviour is statement III.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement I: a router operates at the network layer and its primary purpose is to interconnect two or more distinct computer networks. True.
2. Statement II: on receiving a packet, the router consults its routing table and determines the outgoing interface (link) on which the packet should be forwarded. True.
3. Statement III: the forwarding decision is made by examining the destination IP address in the packet header, and the packet is passed to the next-hop router towards the destination. True.
4. All three statements are correct.
5. Hence option (d) is correct.

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27. 2025 · Q79 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Wi-Fi is often used as a synonym for which IEEE technology?

OPTIONS

(a) 802.1

(b) 802.5

(c) 802.9

(d) 802.11

ANSWER (AI-derived — no official key exists for this paper)

(d) 802.11

EXAM SHORTCUT (AI-derived explanation)

Wi-Fi = IEEE 802.11. Direct recall, option (d).

TIPS & TRICKS (AI-derived explanation)

- Memorise the IEEE 802 family: 802.3 Ethernet, 802.5 Token Ring, 802.11 Wireless LAN $(Wi - Fi), 802.15$ Bluetooth/WPAN, 802.16 WiMAX.
- 802.1 covers bridging and network management, 802.9 was integrated voice/data (isoEthernet, now withdrawn) - both are pure distractors here.
- Wi-Fi generations: $802.11b/a/g/n(Wi - Fi4)/ac(Wi - Fi5)/ax(Wi - Fi6)$/be $(Wi - Fi7)$.
- 'Wi-Fi' is a trademark of the Wi-Fi Alliance certifying interoperable 802.11 equipment; it is not itself a standard number.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The IEEE 802 committee standardises local and metropolitan area networks, with each working group numbered 802.x.
2. Working group 802.11 defines wireless local area networking - the media access control and physical layer specifications for WLANs.
3. Products certified as interoperable with 802.11 are branded Wi-Fi by the Wi-Fi Alliance, so the two terms are used synonymously.
4. 802.1 is bridging/network management, 802.5 is Token Ring, 802.9 was integrated services (isoEthernet) - none relate to Wi-Fi.
5. Hence option (d) is correct.

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28. 2026 · Q62 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which IEEE wireless standard has the maximum data rate (Mbps)?

OPTIONS

(a) 802.11a

(b) 802.11ad

(c) 802.11g

(d) 802.11b

ANSWER (AI-derived — no official key exists for this paper)

(b) 802.11ad

EXAM SHORTCUT (AI-derived explanation)

802.11ad (WiGig) operates in the 60 GHz band with rates up to about 6.7 Gbps, far above $802.11a/g$ (54 Mbps) and 802.11b (11 Mbps). Answer (b).

TIPS & TRICKS (AI-derived explanation)

- Data-rate ladder to memorise: $802.11b = 11$ Mbps, 802.11a and $802.11g = 54$ Mbps, $802.11n = 600$ Mbps, 802.11ac = several Gbps, 802.11ad = about 6.7 Gbps.
- The two-letter suffixes (ac, ad, ax, be) are always the newer, faster generations; single-letter ones (a, b, g, n) are older.
- 802.11ad uses the 60 GHz millimetre-wave band, which gives huge bandwidth but very short range - essentially in-room use.
- 802.11a and 802.11g share the same 54 Mbps ceiling but differ in band (5 GHz versus 2.4 GHz), so neither can be 'the maximum'.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Compare the nominal maximum physical-layer data rates of the listed IEEE 802.11 amendments.
2. 802.11b (2.4 GHz, DSSS): 11 Mbps.
3. 802.11a (5 GHz, OFDM): 54 Mbps.
4. 802.11g (2.4 GHz, OFDM): 54 Mbps.
5. 802.11ad (60 GHz, WiGig): up to approximately 6757 Mbps (about 6.7 Gbps).
6. 802.11ad is therefore by far the fastest of the four.
7. Hence option (b) is correct.

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29. 2026 · Q69

Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which protocol determines the MAC address corresponding to an IP address?

OPTIONS

(a) TCP

(b) FTP

(c) ARP

(d) SMTP

ANSWER (AI-derived — no official key exists for this paper)

(c) ARP

EXAM SHORTCUT (AI-derived explanation)

Address Resolution Protocol maps an IP address to the corresponding MAC address. Answer (c).

TIPS & TRICKS (AI-derived explanation)

- Direction matters: ARP goes IP -> MAC (logical to physical); RARP goes MAC -> IP (physical to logical).

- ARP works by broadcasting 'who has this IP?' on the local link; the owner replies with its MAC address, which is then cached in the ARP table.
- TCP (option a) is a transport-layer protocol, FTP (b) and SMTP (d) are application-layer protocols - none of them resolves addresses.
- ARP sits between the network and data-link layers, which is exactly why it is the protocol that bridges IP addressing and MAC addressing.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Communication on a local network requires the physical (MAC) address of the destination interface, but applications address hosts by IP address.
2. The Address Resolution Protocol (ARP) performs this mapping: the sender broadcasts an ARP request asking which host owns a given IP address, and the owner replies with its MAC address, which is then cached in the ARP table.
3. TCP is a transport-layer protocol providing reliable byte streams; FTP transfers files; SMTP transfers mail - none of these performs address resolution.
4. (The reverse mapping, MAC to IP, was historically done by RARP.)
5. Hence option (c) is correct.

C8 · COMPUTER APPLICATION AND DATA PROCESSING

Computer Security: Virus, Antivirus, Firewall, Spyware, Malware

15 questions · 2018: 0 2019: 3 2020: 2 2021: 0 2022: 1 2023: 2 2024: 2 2025: 1 2026: 4

Weight. 15 questions (8.3% of Computer Application and Data Processing) · appeared in 7 of 9 papers · peak year 2026 (4)

Examiner's pattern. Threat taxonomy plus firewall concepts. Know precisely what separates a virus, a worm, a trojan and spyware — the discriminator is usually whether user action is needed to spread (a worm needs none). Phishing appears as a short scenario. Firewall terminology and packet filtering recur.

Must-know. A worm self-propagates with no user interaction; a virus needs a host and user action. A polymorphic virus rewrites its own code to evade detection. Non-repudiation stops a sender denying transmission. ISO 27001 is the information-security management standard.

1. 2019 · Q45 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Conceptual

QUESTION

Antivirus is a software that comes in the category of 1. system software 2. application software 3. utility software 4. debugger

OPTIONS

(a) 2

(b) 3 and 4 only

(c) 3 only

(d) 1, 3 and 4

ANSWER (AI-derived — no official key exists for this paper)

(c) 3 only

EXAM SHORTCUT (AI-derived explanation)

Antivirus programs maintain and protect the system rather than doing end-user work, so they are classed as UTILITY software - item 3 only.

TIPS & TRICKS (AI-derived explanation)

- Three-way split: system software runs the machine (OS, device drivers), utility software maintains and protects it (antivirus, disk defragmenter, backup, compression), application software does the user's work (word processor, spreadsheet).
- A debugger (item 4) is a programming/development tool, not a security product - it is never a classification for antivirus.
- Utilities are often described as a SUBSET of system software, which is why option (d) looks tempting; the standard single-best classification asked for here is utility software.
- Anchor examples of utilities: antivirus, disk cleanup, defragmenter, file compression, backup - all housekeeping tools.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Software is conventionally classified as system software, utility software, or application software.
2. System software (the operating system, device drivers, language translators) manages the hardware and provides the platform on which other programs run.
3. Utility software performs maintenance and protection tasks for the computer system - examples are antivirus programs, disk defragmenters, backup tools and file compressors.
4. Application software performs tasks directly for the end user, such as word processing or spreadsheets.
5. An antivirus program protects and maintains the system rather than performing end-user work, so it is utility software; a debugger is a program-development tool and is unrelated.
6. Hence only item 3 applies, option (c).

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2. 2019 · Q47 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Definition

QUESTION

A software that gathers information about a person and organization without their knowledge is called

OPTIONS

(a) firmware

(b) adware

(c) virus

(d) spyware

ANSWER (AI-derived — no official key exists for this paper)

(d) spyware

EXAM SHORTCUT (AI-derived explanation)

Software that secretly gathers information about a person or organisation is spyware, by definition.

TIPS & TRICKS (AI-derived explanation)

- Malware taxonomy: virus attaches to a host file and replicates; worm spreads by itself over a network; trojan disguises itself as useful software; SPYWARE secretly collects information; adware displays unwanted advertisements; ransomware encrypts and demands payment.
- 'Without their knowledge' is the signature phrase of spyware - the covert data collection is the defining property.
- Adware (option b) may also track behaviour, but its defining purpose is displaying advertisements, so it is not the best answer.
- Firmware (option a) is not malware at all - it is legitimate software stored in ROM, included purely as a distractor.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Spyware is malicious software installed on a computer that covertly monitors activity and collects information - keystrokes, browsing habits, credentials, personal or organisational data - and transmits it to a third party without the user's knowledge or consent.
2. Firmware is legitimate software permanently stored in ROM, such as the BIOS, and has nothing to do with covert data collection.
3. Adware displays unsolicited advertisements; although some variants track behaviour, its defining purpose is advertising, not covert information gathering.
4. A virus is self-replicating code that attaches itself to other programs and is defined by its replication, not by data theft.
5. Hence the correct answer is spyware, option (d).

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3. 2019 · Q65 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Definition

QUESTION

Which technique does a hacker use for listening to private voice or data transmission, often using a wiretap?

OPTIONS

(a) Spoofing

(b) Snooping

(c) Eavesdropping

(d) Piggybacking

ANSWER (AI-derived — no official key exists for this paper)

(c) Eavesdropping

EXAM SHORTCUT (AI-derived explanation)

Intercepting a private voice or data transmission in transit - classically by a wiretap - is eavesdropping.

TIPS & TRICKS (AI-derived explanation)

- Attack vocabulary: eavesdropping = intercepting data IN TRANSIT; snooping = unauthorised browsing of STORED data; spoofing = impersonating a legitimate identity; piggybacking = riding on someone else's authorised access.
- The words 'wiretap' and 'transmission' point unambiguously to interception of data in motion.
- Eavesdropping is a PASSIVE attack - it breaches confidentiality without altering the data - whereas spoofing is active and breaches authenticity.
- The standard countermeasure is encryption, which leaves the interception possible but the content unreadable.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Eavesdropping is the unauthorised, real-time interception of a private communication while it is in transit, classically by physically tapping a telephone or network line.
2. Snooping refers to unauthorised access to or browsing of another's stored data or files rather than to live interception.
3. Spoofing is the falsification of an identity - an IP address, email sender or MAC address - to impersonate a trusted party.
4. Piggybacking is gaining access by riding on a legitimate user's authorised session or entry.
5. The description of listening to private voice or data transmission using a wiretap matches eavesdropping, option (c).

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4. 2020 · Q52 Memory - RAM, ROM & Units of Computer Memory · Definition

QUESTION

What technology is used to hide information inside a picture?

OPTIONS

(a) Root kits

(b) Bit mapping

(c) Steganography

(d) Image rendering

ANSWER (AI-derived — no official key exists for this paper)

(c) Steganography

EXAM SHORTCUT (AI-derived explanation)

Concealing a message inside an image (or audio or video) file is steganography - literally 'covered writing'.

TIPS & TRICKS (AI-derived explanation)

- Distinguish the two: cryptography makes a message UNREADABLE, steganography makes its very existence UNNOTICED. They are complementary, not alternatives.
- The usual technique is least-significant-bit substitution: altering the last bit of each pixel's colour value changes the image imperceptibly.
- A rootkit (option a) hides malicious software from the operating system, not information inside a picture - a different kind of concealment.
- Bit mapping and image rendering are ordinary graphics terms describing how images are stored and drawn, with no concealment aspect.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Steganography is the practice of concealing a message, file or image inside another file so that the presence of the hidden data is not apparent.

2. In digital images this is commonly done by least-significant-bit substitution, replacing the lowest-order bit of each pixel value with bits of the secret message; the visual change is imperceptible.
3. This differs from cryptography, which scrambles a message but does not hide the fact that a message exists.
4. Root kits conceal malicious software from the operating system and its users, not data within pictures.
5. Bit mapping and image rendering are standard graphics techniques for representing and displaying images.
6. Hence the technology is steganography, option (c).

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5. 2020 · Q56 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Factual

QUESTION

ISO 27001 is a specification/standard for

OPTIONS

(a) Quality Management

(b) Information Security Management

(c) Environmental Management

(d) Health Management

ANSWER (AI-derived — no official key exists for this paper)

(b) Information Security Management

EXAM SHORTCUT (AI-derived explanation)

ISO 27001 is the international standard for Information Security Management Systems.

TIPS & TRICKS (AI-derived explanation)

- Learn the ISO family by number: 9001 Quality, 14001 Environmental, 27001 Information Security, 45001 Occupational Health and Safety, 22000 Food Safety.
- The 27000 series as a whole is devoted to information security, so any 27xxx number points to that domain.
- ISO 27001 specifies requirements for establishing, implementing, maintaining and continually improving an ISMS, and organisations can be certified against it.
- Its companion ISO 27002 gives the code of practice - the detailed controls - while 27001 gives the certifiable requirements.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. ISO standards are grouped by subject area, with each family covering a distinct management discipline.
2. ISO 9001 covers Quality Management Systems.
3. ISO 14001 covers Environmental Management Systems.

4. ISO 27001 is the standard of the ISO/IEC 27000 series that specifies the requirements for an Information Security Management System (ISMS), covering risk assessment, controls and continual improvement.
5. Health and safety management is covered by ISO 45001, not 27001.
6. Hence ISO 27001 is for Information Security Management, option (b).

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6. 2022 · Q37 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Factual

QUESTION

Data Encryption Standard (DES) is:

OPTIONS

(a) Asymmetric-key cryptographic algorithm

(b) Symmetric-key cryptographic algorithm

(c) High speed data transmission algorithm

(d) Low speed data transmission algorithm

ANSWER (AI-derived — no official key exists for this paper)

(b) Symmetric-key cryptographic algorithm

EXAM SHORTCUT (AI-derived explanation)

DES uses the SAME 56-bit key to encrypt and decrypt, which makes it a symmetric-key algorithm.

TIPS & TRICKS (AI-derived explanation)

- Symmetric = one shared secret key (DES, 3DES, AES, Blowfish, RC4). Asymmetric = a public/private key pair (RSA, Diffie-Hellman, ECC, DSA).
- DES specifics worth knowing: 64-bit block, 56-bit effective key, 16 Feistel rounds, superseded by AES because the key length became brute-forcible.
- Symmetric algorithms are far faster than asymmetric ones, which is why real systems use RSA to exchange a key and AES to encrypt the data.
- Options (c) and (d) are out of category entirely - DES is a cipher, not a transmission scheme.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The Data Encryption Standard is a block cipher that encrypts 64-bit blocks using a 56-bit key through 16 Feistel rounds.
2. The same key is used both to encrypt and to decrypt the data, and it must be shared secretly between sender and receiver.
3. Algorithms with this property are called symmetric-key (or secret-key) algorithms.
4. Asymmetric algorithms such as RSA use a public key for encryption and a mathematically related private key for decryption; DES has no such key pair.
5. DES is a cryptographic algorithm, not a data transmission scheme, so options (c) and (d) are irrelevant.

6. Hence DES is a symmetric-key cryptographic algorithm, option (b).

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7. 2023 · Q72 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Conceptual

QUESTION

Which of the following statements is/are correct? 1. Encryption process converts cipher text into plain text. 2. Digital signature employs two keys – private key and public key. 3. Secret Key Cryptography uses a single key for both, encryption and decryption. Select the correct answer using the code given below:

OPTIONS

(a) 2 only

(b) 2 and 3 only

(c) 3 only

(d) 1, 2 and 3

ANSWER **(AI-derived — no official key exists for this paper)**

(b) 2 and 3 only

EXAM SHORTCUT **(AI-derived explanation)**

Statement 1 reverses the definition - encryption turns plain text into cipher text. Statements 2 and 3 correctly describe digital signatures and secret-key cryptography.

TIPS & TRICKS **(AI-derived explanation)**

- Direction check: ENcryption goes plain -> cipher; DEcryption goes cipher -> plain. Reversal is the standard falsification device.
- Digital signatures use asymmetric keys: the sender signs with the PRIVATE key and anyone verifies with the matching PUBLIC key - two keys, as statement 2 says.
- Secret-key (symmetric) cryptography uses one shared key for both operations - DES and AES are the standard examples.
- Symmetric ciphers are fast but require secure key distribution; asymmetric ones solve distribution but are slow, which is why real systems combine them.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Statement 1: encryption transforms readable PLAIN text into unintelligible CIPHER text; it is decryption that performs the reverse. The statement inverts the definition. FALSE.
2. Statement 2: a digital signature is created with the sender's PRIVATE key and verified by anyone using the corresponding PUBLIC key, so two mathematically related keys are involved. TRUE.
3. Statement 3: secret-key (symmetric) cryptography uses a single shared key for both encryption and decryption - DES and AES are standard examples. TRUE.
4. Only statements 2 and 3 are correct.
5. Hence option (b) is correct.

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8. 2023 · Q80 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Factual

QUESTION

Which of the following are Firewall Related Terminology? 1. Gateway 2. Proxy server 3. Screening routers Select the correct answer using the code given below:

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

Gateways, proxy servers and screening routers are all standard firewall components, so all three terms belong to firewall terminology.

TIPS & TRICKS (AI-derived explanation)

- The three classical firewall implementations: packet-filtering (screening) routers, application-level gateways (proxies), and circuit-level gateways.
- A screening router filters packets by IP address, port and protocol at the network layer - the simplest firewall form.
- A proxy server terminates the connection and re-originates it, inspecting content at the application layer - deeper but slower filtering.
- 'Bastion host', 'DMZ' and 'stateful inspection' are other terms from this vocabulary worth recognising alongside these three.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A firewall controls traffic between a trusted internal network and an untrusted external one, and is built from several standard components.
2. Item 1, gateway: an application-level or circuit-level gateway relays traffic between the two networks while enforcing the security policy. FIREWALL TERM.
3. Item 2, proxy server: an application-level gateway that terminates the client connection, inspects the request and re-originates it on the client's behalf, hiding the internal network. FIREWALL TERM.
4. Item 3, screening router: a packet-filtering router that permits or denies packets on the basis of source and destination addresses, ports and protocols - the simplest firewall form. FIREWALL TERM.
5. All three belong to firewall terminology.
6. Hence option (d) is correct.

9. 2024 · Q43 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Conceptual

QUESTION

Consider the following tasks: I. Search and replace II. Encryption III. Virus scanner IV. Backup V. Data recovery How many of the above are tasks of Utility Programs?

OPTIONS

(a) Only two

(b) Only three

(c) Only four

(d) All five

ANSWER (AI-derived — no official key exists for this paper)

(c) Only four

EXAM SHORTCUT (AI-derived explanation)

Encryption, virus scanning, backup and data recovery are all system-maintenance tasks performed by utility programs. 'Search and replace' is a text-editing feature of application software, so four of the five qualify.

TIPS & TRICKS (AI-derived explanation)

- Utility programs MAINTAIN and PROTECT the system: antivirus, backup, data recovery, encryption, compression, disk defragmentation, file management.
- Search and replace is an editing operation inside a word processor or editor - application software performing work for the user, not housekeeping for the system.
- Ask of each item: 'does this look after the computer, or does it do the user's work?'. That test sorts the five instantly.
- Note the distinction from system software proper (the OS and translators); utilities are a maintenance sub-category sitting alongside them.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Utility programs are system software that maintain, analyse, configure and protect the computer system.
2. Item II, encryption: protecting stored or transmitted data is a standard security utility. UTILITY.
3. Item III, virus scanner: antivirus software is the archetypal utility program. UTILITY.
4. Item IV, backup: copying data for safekeeping is a standard maintenance utility. UTILITY. Item V, data recovery: restoring lost or damaged files is likewise a maintenance utility. UTILITY.
5. Item I, search and replace: this is an editing operation provided by a word processor or text editor, i.e. application software doing the user's work rather than maintaining the system. NOT a utility task.
6. Four of the five are tasks of utility programs, option (c).

10. 2024 · Q44 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Definition

QUESTION

Which of the following security services prevents either sender or receiver from denying a transmitted message?

OPTIONS

(a) Authentication

(b) Confidentiality

(c) Non-repudiation

(d) Integrity

ANSWER (AI-derived — no official key exists for this paper)

(c) Non-repudiation

EXAM SHORTCUT (AI-derived explanation)

Preventing a party from later denying that a message was sent or received is precisely the definition of non-repudiation.

TIPS & TRICKS (AI-derived explanation)

- The five classical security services: confidentiality (secrecy), integrity (no alteration), authentication (identity), non-repudiation (no denial), availability (access).
- The word 'denying' in the stem maps directly onto 'repudiation' - the vocabulary itself gives the answer.
- Digital signatures provide non-repudiation because only the holder of the private key could have produced the signature.
- Authentication proves who you ARE at the time; non-repudiation proves who acted LATER, in a way that stands up to subsequent denial - a subtle but examinable distinction.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Non-repudiation is the security service that prevents either party from later denying an action they performed.
2. Non-repudiation of origin prevents the sender from denying having sent the message; non-repudiation of delivery prevents the receiver from denying having received it.
3. It is typically implemented with digital signatures, since only the holder of the private key could have generated the signature.
4. Authentication verifies identity at the time of access, confidentiality prevents unauthorised disclosure, and integrity ensures the message has not been altered - none of these addresses subsequent denial.
5. Hence the required service is non-repudiation, option (c).

11. 2025 · Q78 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Conceptual

QUESTION

A packet filter firewall can:

OPTIONS

(a) filter specific users from accessing network services.

(b) block all IP packets from entering the networks.

(c) check all IP packets and, if valid, allow entry or exit.

(d) block any kind of viruses from entering the networks.

ANSWER (AI-derived — no official key exists for this paper)

(c) check all IP packets and, if valid, allow entry or exit.

EXAM SHORTCUT (AI-derived explanation)

A packet filter inspects each IP packet's header against rules and permits or denies it. Option (b) blocks everything (useless), (a) needs user identity which headers do not carry, and (d) needs payload/signature scanning. Only (c) survives.

TIPS & TRICKS (AI-derived explanation)

- Packet filters work on header fields only: source and destination IP, source and destination port, protocol and flags. Anything requiring user identity or payload content is outside their scope.
- Firewall hierarchy: packet filter (stateless, layer $3/4$) -> stateful inspection -> application/proxy gateway (layer 7) -> next-generation firewall with deep packet inspection.
- Virus detection needs signature or heuristic scanning of the payload - that is antivirus or an application-layer gateway, not a packet filter, so (d) is wrong.
- Beware absolutes: an option saying a device blocks 'all' or 'any kind of' something is usually a distractor.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A packet filtering firewall examines the header of every IP packet crossing the boundary and compares it against an ordered access control list.
2. Based on the match it either allows the packet through (in or out) or drops it - exactly what option (c) states.
3. Option (a) is wrong: filtering by individual user requires authentication information, which is not present in the IP or TCP header; that is the role of an application-level or identity-aware firewall.
4. Option (b) is wrong: blocking all IP packets would disconnect the network entirely and defeats the purpose of a firewall, which is selective control.
5. Option (d) is wrong: viruses live in packet payloads and require content scanning; a header-based packet filter cannot detect them.
6. Hence option (c) is correct.

12. 2026 · Q68 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Factual

QUESTION

Which is an International Standard for Information Security Management?

OPTIONS

(a) ISO/IEC 27001

(b) ISO 9001

(c) ISO 14001

(d) ISO 45001

ANSWER (AI-derived — no official key exists for this paper)

(a) ISO/IEC 27001

EXAM SHORTCUT (AI-derived explanation)

ISO/IEC 27001 is the information security management standard. The others cover quality (9001), environment (14001) and occupational health and safety (45001). Answer (a).

TIPS & TRICKS (AI-derived explanation)

- Memorise the four common management-system standards: 9001 quality, 14001 environment, 27001 information security, 45001 occupational health and safety.
- The 27000 series as a whole deals with information security; 27002 gives the control guidance and 27005 covers risk management.
- 'ISO/IEC' as a joint prefix is itself a clue - IEC involvement signals an information-technology standard.
- ISO/IEC 27001 specifies an Information Security Management System (ISMS) built on confidentiality, integrity and availability.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. ISO/IEC 27001 specifies the requirements for establishing, implementing, maintaining and continually improving an Information Security Management System (ISMS), covering confidentiality, integrity and availability of information.
2. ISO 9001 is the Quality Management System standard.
3. ISO 14001 is the Environmental Management System standard.
4. ISO 45001 is the Occupational Health and Safety Management System standard.
5. Only ISO/IEC 27001 addresses information security management.
6. Hence option (a) is correct.

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13. 2026 · Q74 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Definition

QUESTION

A virus that modifies its code each time it spreads, to evade antivirus detection, is known as:

OPTIONS

(a) stealth virus

(b) multipartite virus

(c) polymorphic virus

(d) boot sector virus

ANSWER (AI-derived — no official key exists for this paper)

(c) polymorphic virus

EXAM SHORTCUT (AI-derived explanation)

A virus that rewrites its own code on each infection to defeat signature scanning is by definition polymorphic. Answer (c).

TIPS & TRICKS (AI-derived explanation)

- Match name to behaviour: polymorphic = changes its code/encryption key each time; stealth = hides the changes it has made from the OS; multipartite = infects both boot sector and files; boot sector = infects the MBR.
- Polymorphic viruses typically encrypt the payload with a fresh key and mutate the decryptor, so no fixed byte signature exists.
- Metamorphic viruses go further, rewriting the entire body rather than just the decryptor - a common follow-up distinction.
- This is why modern antivirus software relies on heuristic and behavioural detection rather than signature matching alone.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A polymorphic virus alters its own code - typically by encrypting its body with a different key on each infection and mutating the accompanying decryption routine - so that no two copies share the same byte pattern.
2. Because signature-based antivirus scanners look for fixed byte sequences, this constant mutation defeats detection.
3. A stealth virus (option a) instead intercepts system calls to conceal the changes it has already made; a multipartite virus (option b) is classified by infecting both boot sectors and executable files; a boot sector virus (option d) is classified by its target location.
4. Only the polymorphic virus is defined by modifying its code on each spread.
5. Hence option (c) is correct.

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14. 2026 · Q76 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Conceptual

QUESTION

Rohit gets an e-mail saying his credit card will be cancelled unless he clicks a link and signs into his online banking account, which he does. A week later, ₹10,000 is missing from his account. What has Rohit fallen victim to?

OPTIONS

(a) Trojan Horse

(b) Online blackmail

(c) Phishing scam

(d) Virus

ANSWER (AI-derived — no official key exists for this paper)

(c) Phishing scam

EXAM SHORTCUT (AI-derived explanation)

A fraudulent email with an urgent threat and a link to a fake login page that harvests credentials is a textbook phishing scam. Answer (c).

TIPS & TRICKS (AI-derived explanation)

- Phishing signature: an unsolicited message plus manufactured urgency plus a link to a spoofed site that captures credentials - all three appear in this scenario.
- The theft happened because Rohit voluntarily entered his credentials on a fake site, not because software was installed - which rules out Trojan (a) and virus (d).
- Blackmail (option b) would require a demand backed by a threat to release information; here the attacker simply impersonated the bank.
- Related vocabulary: spear phishing targets a named individual, whaling targets executives, vishing uses voice calls and smishing uses SMS.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Rohit received an unsolicited email impersonating his bank, containing a manufactured urgency (the card will be cancelled) and a link to a page that asked for his online banking credentials.
2. By signing in, he supplied his username and password to an attacker-controlled site, which the attacker then used to withdraw money.
3. Deceiving a user into revealing credentials or sensitive data through a spoofed message and counterfeit website is precisely the definition of phishing.
4. A Trojan Horse (option a) or virus (option d) would require malicious software to be executed on his machine, which did not happen; online blackmail (option b) involves coercion by threatening to reveal information, which is also absent.
5. Hence option (c) is correct.

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15. 2026 · Q79 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Definition

QUESTION

A malware capable of transferring from one computer to another without any user interaction is known as:

OPTIONS

(a) Trojan

(b) worm

(c) boot sector virus

(d) spyware

ANSWER (AI-derived — no official key exists for this paper)

(b) worm

EXAM SHORTCUT (AI-derived explanation)

Self-replicating malware that spreads across a network on its own, needing no host file and no user action, is a worm. Answer (b).

TIPS & TRICKS (AI-derived explanation)

- Worm versus virus: a worm is standalone and self-propagating; a virus needs a host file and some user action to execute it.
- A Trojan (option a) is disguised as legitimate software and depends entirely on the user running it, so it cannot self-propagate.
- Spyware (option d) is classified by what it does - covertly gathering information - not by how it spreads.
- Famous worms - Morris, Code Red, SQL Slammer, Conficker, WannaCry - all spread by exploiting network service vulnerabilities without user interaction.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A computer worm is standalone malicious software that replicates itself and transmits copies to other computers over a network, typically by exploiting a vulnerability in a network service.
2. Because it does not attach itself to a host program and does not need to be opened or executed by a user, it propagates without any user interaction.
3. A virus, including a boot sector virus (option c), attaches to a host program or boot record and requires that host to be executed.
4. A Trojan (option a) masquerades as useful software and must be run deliberately by the victim; spyware (option d) is defined by its information-stealing payload rather than by autonomous spreading.
5. Hence option (b) is correct.

C9 · COMPUTER APPLICATION AND DATA PROCESSING

Programming Basics: Algorithm, Flowchart, Control Structures, Arrays, Functions

16 questions · 2018: 4 2019: 1 2020: 1 2021: 1 2022: 0 2023: 3 2024: 2 2025: 4 2026: 0

Weight. 16 questions (8.9% of Computer Application and Data Processing) · appeared in 7 of 9 papers · peak year 2018 (4)

Examiner's pattern. Programming fundamentals, tested conceptually. Recurring: debugging (breakpoints, the order of the debug steps), flowcharts, arrays and strings, subroutines and modularity, operator precedence and associativity, bitwise operators, stable sorting algorithms, and tracing a short loop.

Must-know. Stable sorts: bubble, insertion, merge. Quicksort and heapsort are not stable. Unary and assignment operators associate right-to-left. && and ! are logical, not bitwise; &, |, ^, >>, << are bitwise. Space complexity = memory required by an algorithm.

1. 2018 · Q68 Number Systems & Binary Arithmetic · Conceptual

QUESTION

Which of these are *not* bitwise operators? 1. `&&` 2. `&` 3. `>>` 4. `!` 5. `%`

OPTIONS

(a) 2 and 3 only

(b) 1, 3, 4 and 5

(c) 1, 2 and 5

(d) 1, 4 and 5

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 4 and 5

EXAM SHORTCUT (AI-derived explanation)

Bitwise operators act on individual bits: & and >> qualify. Logical && and !, and the arithmetic %, do not. So the non-bitwise ones are 1, 4 and 5.

TIPS & TRICKS (AI-derived explanation)

- The bitwise family in C is & | ^ ~ << >>. Anything outside that list - &&, ||, !, %, +, - - is not bitwise.
- Single symbol versus doubled symbol is the discriminator: & is bitwise AND while && is logical AND; | is bitwise OR while || is logical OR.
- ~ is the bitwise NOT and ! is the logical NOT - these two are the pair most often confused, and ! appears here as item 4.
- % is the modulus (remainder) operator, purely arithmetic. It is included precisely because its symbol looks unfamiliar enough to be mistaken for a bit operation.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Bitwise operators operate on the individual bits of their operands. In C these are & (AND), | (OR), ^ (XOR), ~ (NOT), << (left shift) and >> (right shift).
2. Item 1, &&, is the logical AND operator; it yields a truth value, not a bit pattern. NOT bitwise.
3. Item 2, &, is the bitwise AND. Bitwise.
4. Item 3, >>, is the right-shift operator. Bitwise.
5. Item 4, !, is the logical NOT (the bitwise complement is ~). NOT bitwise. Item 5, %, is the arithmetic modulus operator. NOT bitwise.
6. The operators that are NOT bitwise are 1, 4 and 5, so option (d) is correct.

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2. 2018 · Q71 Algorithm & Flowchart · Conceptual

QUESTION

Code: `` while (x != y) { if (x > y) x = x - y; else y = y - x; } `` What does this compute?

OPTIONS

(a) $x \bmod y$ via repeated subtraction

(b) $x \div y$ via repeated subtraction

(c) The GCD of x and y

(d) The LCM of x and y

ANSWER (AI-derived — no official key exists for this paper)

(c) The GCD of x and y

EXAM SHORTCUT (AI-derived explanation)

Repeatedly replacing the larger by the difference is the subtraction form of Euclid's algorithm. The loop stops when $x = y$, and that common value is $\gcd(x, y)$.

TIPS & TRICKS (AI-derived explanation)

- Trace one example: $x = 12, y = 18$ gives 12,6 then 6,6 - and $\gcd(12, 18) = 6$. A single trace beats any amount of staring at the code.
- The invariant is $\gcd(x, y)$ is unchanged by replacing the larger with the difference; the loop terminates because the sum x + y strictly decreases.
- The modulo form of Euclid uses $x = x$ % y and is much faster; this subtraction version is its slower ancestor and is a favourite exam disguise.
- LCM (option d) cannot be right because the loop only ever DECREASES the values, whereas the LCM is at least as large as the larger input.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The loop runs while x is not equal to y, each time subtracting the smaller value from the larger.
2. The key invariant is $\gcd(x, y) = \gcd(x - y, y)$ when $x > y$, so every iteration preserves the greatest common divisor of the pair.

3. Each iteration strictly decreases x + y while keeping both values positive, so the loop must terminate; it terminates only when $x = y$.
4. At termination the common value equals the preserved $\gcd$ of the original pair.
5. For example, starting with $x = 12, y = 18$: $(12, 18) \to (12, 6) \to (6, 6)$, and $\gcd(12, 18) = 6$.
6. This is the subtraction form of Euclid's algorithm, so the code computes the GCD, option (c).

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3. 2018 · Q72 Variables, Control Structures, Arrays, Functions, Loops · Conceptual

QUESTION

Consider the following statements: 1. An array is a collection of elements of the same type. 2. Strings are character arrays where each character uses one byte. 3. An array can store elements of different types. 4. Strings can be data arrays too.

OPTIONS

(a) 1 and 3 only

(b) 1 and 2 only

(c) 3 and 4 only

(d) 1, 2, 3 and 4

ANSWER (AI-derived — no official key exists for this paper)

(b) 1 and 2 only

EXAM SHORTCUT (AI-derived explanation)

Statement 3 is plainly false - an array is homogeneous - and every option except (b) contains statement 3. So the key must be 1 and 2 only.

TIPS & TRICKS (AI-derived explanation)

- Elimination by a single false statement is the fastest route in 'which are correct' items: identify the certainly-false one and strike every option containing it.
- An array is by definition a HOMOGENEOUS collection stored in contiguous memory; heterogeneous collections are structures/records, not arrays.
- Statement 2 is true in the C sense: a string is a null-terminated char array and each char occupies one byte.
- Statement 4 ('strings can be data arrays too') overlaps statement 2 and is not offered in any surviving combination; the option set forces (b), which is why the item is really a test of spotting the false statement 3.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: an array is a collection of elements of the SAME data type stored in contiguous memory locations. TRUE.
2. Statement 2: in C a string is stored as an array of characters terminated by the null character, and each char occupies one byte. TRUE.

3. Statement 3: an array cannot hold elements of different types - that is the defining restriction which distinguishes an array from a structure. FALSE.
4. Since statement 3 is false, every option containing it - (a), (c) and (d) - is eliminated.
5. The only surviving option is (b), 1 and 2 only.
6. Hence option (b) is correct.

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4. 2018 · Q75 Memory - RAM, ROM & Units of Computer Memory · Factual

QUESTION

Which operators follow right-to-left precedence? 1. Unary operator 2. Assignment operator 3. Logical operator 4. Bit-manipulation operator

OPTIONS

(a) 1 and 2 only

(b) 1, 2 and 4

(c) 3 and 4 only

(d) 1 only

ANSWER (AI-derived — no official key exists for this paper)

(a) 1 and 2 only

EXAM SHORTCUT (AI-derived explanation)

Right-to-left associativity belongs to the unary operators and the assignment operators (and the conditional ?:). Logical and bit-manipulation operators associate left to right.

TIPS & TRICKS (AI-derived explanation)

- Memorise the short right-to-left list: unary (+, -, ++, --, !, ~, *, &, sizeof, casts), assignment (=, +=, -=, ...), and the ternary ?:. Everything else in C is left-to-right.
- Assignment must be right-to-left so that $a = b = c$ works: c is assigned to b first, then b to a.
- Unary operators must be right-to-left so that - -x and !!x parse from the operand outwards.
- Logical && and || are strictly left-to-right - that is exactly what makes short-circuit evaluation well defined, so item 3 cannot be right-associative.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Associativity determines the grouping order when operators of equal precedence appear together.
2. Item 1, unary operators: these bind from the operand outwards, i.e. right to left. TRUE.
3. Item 2, assignment operators: $a = b = c$ is grouped as $a = (b = c)$, i.e. right to left. TRUE.
4. Item 3, logical operators (&& and ||): these associate left to right, which is what makes short-circuit evaluation deterministic. FALSE.
5. Item 4, bit-manipulation operators (&, |, ^, <<, >>): these also associate left to right. FALSE.
6. Only items 1 and 2 follow right-to-left precedence, so option (a) is correct.

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5. 2019 · Q48 Variables, Control Structures, Arrays, Functions, Loops · Conceptual

QUESTION

Consider the following statements in respect of subroutine: 1. It reduces the length of code. 2. It is a program stored somewhere else in memory and called to execute it. 3. It does not require the use of stack for its execution. *(answer options for Q48 were not captured in the source scan — page break falls right after statement 3; not fabricated here)*

OPTIONS

The four options for this question are absent from the source scan (a page break in the original booklet falls here). Nothing has been invented to fill the gap.

ANSWER (AI-derived — no official key exists for this paper)

Not keyed. No option in the printed paper is correct for this item; see the source note below.

Source note. SOURCE DEFECT - the four answer options for this item are absent from the source scan; the page break in the original booklet falls immediately after statement 3. The question stem is reproduced exactly as it appears in the source. No options have been invented, so this item is displayed for study but excluded from scoring.

EXAM SHORTCUT (AI-derived explanation)

AI-derived: statements 1 and 2 are true of subroutines, statement 3 is false - a subroutine call pushes the return address (and usually its local frame) onto the STACK, so stack use is required.

TIPS & TRICKS (AI-derived explanation)

- A subroutine is stored once and called from many places, which is exactly why it reduces the length of code - statement 1 is true.
- The call mechanism is: push the return address onto the stack, jump to the subroutine, execute, then pop the return address and return. Stack use is intrinsic, so statement 3 is false.
- Recursion is possible only because of that stack - each nested call gets its own activation record.
- Distinguish a subroutine (control transfers and returns) from a MACRO (code is textually expanded in place, so no stack and no call overhead).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: a subroutine is written once and invoked wherever needed, so repeated code is replaced by a single definition plus short call instructions. This reduces the length of the code. TRUE.
2. Statement 2: the subroutine body resides at a separate location in memory and control is transferred to it by a call instruction, returning afterwards to the caller. TRUE.
3. Statement 3: executing a subroutine call requires the return address (and typically the parameters and local variables) to be pushed onto the stack so that control can return correctly and so that nesting and recursion are possible. A subroutine therefore DOES require the stack. FALSE.
4. On this analysis the correct statements are 1 and 2 only.
5. However, the printed answer options are missing from the source, so no option letter can be selected and the item is excluded from scoring.

6. 2020 · Q42 Algorithm & Flowchart · Conceptual

QUESTION

Consider the following statements about an expression: 1. The operators + and - must be followed by an operand. 2. Two arithmetic operators must not appear side by side. 3. The number of left brackets must equal the number of right brackets.

OPTIONS

(a) 1 and 3 only

(b) 1 and 2 only

(c) 2 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

All three are standard syntax rules for a well-formed arithmetic expression: an operator needs a following operand, operators may not be adjacent, and brackets must balance.

TIPS & TRICKS (AI-derived explanation)

- These are the classic rules a compiler's parser enforces; each corresponds to a common syntax error message.
- Bracket balancing is checked with a STACK - push on '(' and pop on ')'. The stack must end empty and never underflow.
- Adjacent operators such as a + *b are rejected; the apparent exception a* -b is really a unary minus attached to its operand, not two binary operators.
- When every statement in a set is a textbook syntax rule, 'all of the above' is the natural key - verify the one you are least sure of and move on.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: an arithmetic operator such as + or - is binary or unary but in either case must be followed by an operand; an expression ending in an operator is malformed. TRUE.
2. Statement 2: two binary arithmetic operators cannot appear consecutively, as in a + *b, since the first operator would then have no operand. TRUE. (a* -b is not a counterexample: the minus there is a unary sign attached to b.)
3. Statement 3: parentheses must balance, so the count of '(' must equal the count of ')' and no closing bracket may appear before its opening partner. TRUE.
4. All three are conditions that a syntax analyser checks when validating an expression.
5. Hence option (d) is correct.

7. 2021 · Q75 Operating Systems, Packages & Utilities · Conceptual

QUESTION

Fragments: 1. Replication of a bug 2. Understanding the bug 3. Testing the bug 4. Fixing the bug. Which are parts of the debugging process?

OPTIONS

(a) 1, 2 and 4 only

(b) 1 and 2 only

(c) 1, 2, 3 and 4

(d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(c) 1, 2, 3 and 4

EXAM SHORTCUT (AI-derived explanation)

Debugging is a full cycle: reproduce the bug, understand its cause, fix it, then test that the fix works and nothing else broke. All four fragments belong.

TIPS & TRICKS (AI-derived explanation)

- Standard debugging cycle: reproduce, isolate/understand, correct, verify by testing (and add a regression test). Every listed item maps to one of these.
- Reproduction comes first because a bug that cannot be reproduced cannot be reliably diagnosed or confirmed as fixed.
- Testing after fixing is essential - it both confirms the repair and guards against regressions elsewhere, so it is part of debugging, not separate from it.
- When every option fragment names a recognised stage of the same process, 'all of them' is the natural key.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Debugging is the systematic process of locating and removing defects from a program.
2. Fragment 1, replication: the bug must first be reproduced reliably, since an intermittent or unreproducible fault cannot be diagnosed or verified as fixed. PART of debugging.
3. Fragment 2, understanding: the programmer must trace the execution and identify the root cause rather than the symptom. PART of debugging.
4. Fragment 4, fixing: the defective code is corrected. PART of debugging.
5. Fragment 3, testing: after the correction the program is re-run on the failing case and on the wider test suite to confirm the repair and check for regressions. PART of debugging.
6. All four fragments are stages of the debugging process, option (c).

8. **2023 · Q68** Low & High Level Languages, Compiler, Assembler · Conceptual

QUESTION

Division of a computer program in terms of functions and modules is one of the vital features of:

OPTIONS

(a) Assembly language

(b) Machine language

(c) Structural language

(d) Low level language

ANSWER (AI-derived — no official key exists for this paper)

(c) Structural language

EXAM SHORTCUT (AI-derived explanation)

Breaking a program into functions and modules is the defining feature of STRUCTURED (structural) programming languages.

TIPS & TRICKS (AI-derived explanation)

- Structured programming rests on three ideas: sequence, selection and iteration, plus decomposition into independent subprograms - which is exactly what the question describes.
- Assembly, machine and low-level languages work with individual instructions and jumps; they have no native concept of a function or module.
- Modularity brings reusability, easier testing and maintainability - the standard advantages to quote.
- Classic structured languages: C, Pascal, ALGOL. Structured programming was the historical reaction against unrestricted GOTO-based code.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Structured (structural) programming organises a program as a hierarchy of self-contained blocks, built from sequence, selection and iteration, and decomposed into functions, procedures and modules.
2. This decomposition allows each unit to be written, tested and maintained independently, and to be reused elsewhere.
3. Machine language consists of raw binary instructions with no notion of a function or module.
4. Assembly language works at the level of individual mnemonic instructions and branch targets; subroutines exist only as manually managed jumps.
5. Low-level languages generally are machine-oriented and lack the block structure the question describes.
6. Hence the feature belongs to structural languages, option (c).

9. 2023 · Q77 Algorithm & Flowchart · Factual

QUESTION

Which of the following algorithms are stable? 1. Bubble sort 2. Quick sort 3. Merge sort 4. Insertion sort Select the correct answer using the code given below:

OPTIONS

(a) 1, 2 and 3

(b) 1, 2 and 4

(c) 1, 3 and 4

(d) 2, 3 and 4

ANSWER (AI-derived — no official key exists for this paper)

(c) 1, 3 and 4

EXAM SHORTCUT (AI-derived explanation)

Bubble, merge and insertion sorts are stable; quick sort is not, because its partitioning swaps distant elements and can reorder equal keys.

TIPS & TRICKS (AI-derived explanation)

- Stable sorts: bubble, insertion, merge, counting, radix. Unstable: quick, heap, selection (in its usual form).
- Stability means records with EQUAL keys keep their original relative order - essential when sorting successively on several fields.
- Quick sort's partition step swaps elements across long distances, which can carry an equal key past another - hence instability.
- Merge sort is stable provided the merge step takes from the LEFT run when the two keys tie - a detail worth remembering.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A sorting algorithm is stable if records with equal keys retain their original relative order in the sorted output.
2. Item 1, bubble sort: it swaps only ADJACENT elements, and only when strictly out of order, so equal keys are never exchanged. STABLE.
3. Item 3, merge sort: in the merge step, ties are resolved in favour of the left run, preserving the original order. STABLE.
4. Item 4, insertion sort: each element is inserted after all elements with keys less than OR EQUAL to it, so equal keys keep their order. STABLE.
5. Item 2, quick sort: partitioning exchanges elements that may be far apart, which can move one equal key past another. NOT stable in its standard in-place form.
6. Hence the stable algorithms are 1, 3 and 4, option (c).

10. 2023 · Q78 Operating Systems, Packages & Utilities · Conceptual

QUESTION

Consider the following statements: 1. Steps for developing software program remain same irrespective of the Operating System. 2. If the program is using exp() function, then the object code of this function should be brought from the 'math' library of the system and linked to the main program. Which of the statements given above is/are correct?

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(c) Both 1 and 2

EXAM SHORTCUT (AI-derived explanation)

The development cycle - analyse, design, code, compile, test, debug - is the same on any operating system, and a call to $\exp()$ must indeed be resolved by linking the math library's object code. Both statements hold.

TIPS & TRICKS (AI-derived explanation)

- Statement 1 is about the METHODOLOGY, not the tools: the sequence of development steps is platform-independent even though compilers and file formats differ.
- Statement 2 describes the linker's job: the compiler emits a call with an unresolved reference, and the linker supplies the object code for $\exp()$ from the math library.
- In practice this is why C programs on UNIX must be compiled with the -lm flag - a concrete anchor for statement 2.
- Library functions are pre-compiled object code, not source, which is why LINKING rather than compiling is the step that resolves them.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: the software development process - problem analysis, algorithm design, coding, compilation, testing and debugging, then maintenance - is a methodology independent of the particular operating system.
2. Although the specific compilers, file formats and tools differ between platforms, the sequence of development steps does not. TRUE.
3. Statement 2: when a program calls a standard library function such as $\exp()$, the compiler emits a call with an unresolved external reference.
4. The linker then locates the pre-compiled object code for $\exp()$ in the mathematical library and binds it into the executable.
5. This is exactly why C programs using such functions must be linked against the math library. TRUE.

6. Both statements are correct, option (c).

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11. 2024 · Q41 Variables, Control Structures, Arrays, Functions, Loops · Conceptual

QUESTION

Consider the following statements in respect of debugger: I. Debugger is a program used to detect errors and bugs in a program. II. In order to debug the program, a debugger helps us perform step-by-step execution of a program. III. In order to debug the program, a debugger helps us perform stopping the execution of the program until the errors are corrected. Which of the statements given above are correct?

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(d) I, II and III

EXAM SHORTCUT (AI-derived explanation)

Detecting bugs, single-stepping through code and halting execution at breakpoints are all standard debugger facilities, so all three statements hold.

TIPS & TRICKS (AI-derived explanation)

- Core debugger facilities to remember: breakpoints, single-stepping, watching variables, examining the call stack, and modifying values at run time.
- Statement II describes SINGLE-STEPPING and statement III describes BREAKPOINTS - two distinct, genuine features, not duplicates.
- A debugger locates run-time and LOGICAL errors, the ones a compiler cannot report - which is exactly why it is a separate tool.
- When every statement in a set names a textbook feature of the same tool, 'all of them' is the natural key.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A debugger is a software tool that assists the programmer in locating and correcting faults in a program.
2. Statement I: it is used to detect errors and bugs - particularly run-time and logical errors that the compiler cannot report. TRUE.
3. Statement II: single-stepping lets the programmer execute one statement at a time and observe the effect on variables and control flow. TRUE.
4. Statement III: breakpoints let the programmer halt execution at chosen points, inspect the program state and resume once the problem is understood or corrected. TRUE.

5. All three describe standard debugger facilities.
6. Hence option (d) is correct.

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12. 2024 · Q59 Low & High Level Languages, Compiler, Assembler · Factual

QUESTION

Which of the following are looping statements in general of a computer programming language? I. While II. Break III. For IV. Continue V. Do while Select the correct answer using the code given below:

OPTIONS

(a) I and II

(b) II and IV only

(c) II, III and IV

(d) I, III and V

ANSWER (AI-derived — no official key exists for this paper)

(d) I, III and V

EXAM SHORTCUT (AI-derived explanation)

While, for and do-while are the three loop constructs; break and continue are jump statements that alter the flow WITHIN a loop but are not loops themselves.

TIPS & TRICKS (AI-derived explanation)

- Three loop constructs in C-family languages: while (entry-controlled), for (entry-controlled with initialisation and update), do-while (exit-controlled, body runs at least once).
- break and continue are JUMP statements, alongside goto and return - they modify loop behaviour but do not create iteration.
- The distinguishing test: a looping statement can repeat a block on its own; break and continue cannot repeat anything.
- do-while is the only exit-controlled loop, guaranteeing at least one execution of the body - a common follow-up question.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A looping (iteration) statement repeatedly executes a block of code while a condition holds.
2. Item I, while: tests the condition before each iteration - an entry-controlled loop. LOOP.
3. Item III, for: combines initialisation, condition and update in one header and is likewise entry-controlled. LOOP.
4. Item V, do while: executes the body first and tests the condition afterwards, so the body runs at least once - an exit-controlled loop. LOOP.
5. Item II, break: a jump statement that terminates the enclosing loop or switch immediately. Item IV, continue: a jump statement that skips to the next iteration. Neither creates iteration by itself.

6. Hence the looping statements are I, III and V, option (d).

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13. 2025 · Q37 Algorithm & Flowchart · Definition

QUESTION

Which term specifies the amount of memory required by an algorithm to perform its task?

OPTIONS

(a) Time complexity

(b) Space complexity

(c) Hybrid complexity

(d) Processing complexity

ANSWER (AI-derived — no official key exists for this paper)

(b) Space complexity

EXAM SHORTCUT (AI-derived explanation)

The memory an algorithm needs is its SPACE complexity; time complexity measures the number of operations instead.

TIPS & TRICKS (AI-derived explanation)

- Two complexities, two resources: TIME complexity counts operations, SPACE complexity counts memory. The question names memory, so space.
- Space complexity has two parts: the fixed part (code, constants) and the variable part (dynamic data structures, recursion stack).
- Both are expressed in big-O notation as functions of the input size n - e.g. merge sort is O($n \log n$) time but O(n) space.
- 'Hybrid complexity' and 'processing complexity' are not standard terms at all, so they can be dismissed immediately.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Algorithm analysis measures two resources: the time taken and the memory required.
2. Time complexity expresses the number of elementary operations as a function of the input size.
3. Space complexity expresses the amount of memory required as a function of the input size, comprising a fixed part (instructions and constants) and a variable part (dynamically allocated data and the recursion stack).
4. The question asks specifically about the amount of memory, which is space complexity.
5. 'Hybrid complexity' and 'processing complexity' are not standard terms in algorithm analysis.
6. Hence the answer is space complexity, option (b).

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14. 2025 · Q39 Variables, Control Structures, Arrays, Functions, Loops · Definition

QUESTION

What is the purpose of breakpoints in debugging?

OPTIONS

(a) To mark a point in the code where execution should pause for inspection.

(b) To measure the time elapsed to run a block of code.

(c) To skip a block of code during execution.

(d) To identify memory leaks in the program.

ANSWER (AI-derived — no official key exists for this paper)

(a) To mark a point in the code where execution should pause for inspection.

EXAM SHORTCUT (AI-derived explanation)

A breakpoint marks a line at which the debugger pauses execution so the programmer can inspect the program state.

TIPS & TRICKS (AI-derived explanation)

- Breakpoint = pause point. Once execution halts, the programmer can examine variables, the call stack and memory before resuming or single-stepping.
- Option (c) inverts the meaning: a breakpoint stops execution AT a point, it does not skip code.
- Timing a block (option b) is profiling, and detecting memory leaks (option d) is the work of a memory analyser - different tools entirely.
- Conditional breakpoints, which trigger only when a stated condition holds, are the natural extension worth mentioning.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A breakpoint is a marker placed by the programmer at a chosen line of source code within a debugger.
2. When execution reaches that line, the debugger suspends the program before executing it.
3. While paused, the programmer can inspect variable values, the call stack and memory contents, and may then resume, single-step or terminate the run.
4. Measuring elapsed time is the function of a profiler, and detecting memory leaks is the function of a memory analyser.
5. A breakpoint does not skip code - it halts execution at the marked point.
6. Hence a breakpoint marks a point where execution pauses for inspection, option (a).

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15. 2025 · Q71 Algorithm & Flowchart · Definition

QUESTION

Which statement best describes the use of flowcharts in problem solving?

OPTIONS

(a) Flowcharts are used to visualize data structure.

(b) Flowcharts are used to create graphical user interfaces.

(c) Flowcharts consist of short, readable, formally-styled English statements.

(d) Flowcharts are diagrammatic representations of the logic for solving a task.

ANSWER (AI-derived — no official key exists for this paper)

(d) Flowcharts are diagrammatic representations of the logic for solving a task.

EXAM SHORTCUT (AI-derived explanation)

A flowchart is a diagram of the logic of a solution. Option (c) describes pseudocode (English-like statements), not flowcharts; (a) and (b) are unrelated. Answer (d).

TIPS & TRICKS (AI-derived explanation)

- Distinguish the three: algorithm = ordered steps in words; flowchart = the same logic drawn with standard symbols; pseudocode = English-like statements in a program-ish style.
- Standard flowchart symbols: oval = start/stop, parallelogram = input/output, rectangle = process, diamond = decision, arrows = flow of control.
- Option (c) is the textbook definition of pseudocode - a classic swap-the-definition distractor.
- Flowcharts are language-independent, which is exactly why they are drawn before coding begins.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A flowchart is a graphical or diagrammatic representation of the sequence of steps and decisions needed to solve a problem, drawn with standardised symbols connected by directed arrows.
2. Option (a) is wrong: data structures are visualised with structure diagrams, trees or ER diagrams, not flowcharts.
3. Option (b) is wrong: graphical user interfaces are designed with GUI builders and wireframes; flowcharts describe logic, not screens.
4. Option (c) is wrong: 'short, readable, formally-styled English statements' is the standard definition of pseudocode.
5. Option (d) matches the definition precisely.
6. Hence option (d) is correct.

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16. 2025 · Q76 Variables, Control Structures, Arrays, Functions, Loops · Definition

QUESTION

In which construct is the same set of statements executed several times based on a specified condition?

OPTIONS

(a) If-Then

(b) If-Then-Else

(c) Do-While

(d) Case Type

ANSWER (AI-derived — no official key exists for this paper)

(c) Do-While

EXAM SHORTCUT (AI-derived explanation)

Repetition based on a condition is a loop. Among the options only Do-While is a loop; If-Then, If-Then-Else and Case are selection constructs. Answer (c).

TIPS & TRICKS (AI-derived explanation)

- The three structured-programming control constructs are sequence, selection (if / case) and iteration (while / do-while / for).
- Keywords in the stem give it away: 'executed several times' means iteration; 'based on a condition' means the loop is conditional rather than counted.
- Do-While tests the condition after the body, so it runs at least once; While tests before, so it may run zero times.
- Case Type (switch) is multi-way selection, not repetition - the most tempting distractor if you read only 'several'.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The construct in which the same block of statements is executed repeatedly so long as a specified condition holds is called a loop or iteration construct.
2. If-Then (option a) executes a block at most once, depending on a condition - selection, not repetition.
3. If-Then-Else (option b) chooses between two blocks, each executed at most once - selection.
4. Do-While (option c) executes the body and then tests the condition, repeating while the condition remains true - this is repetition.
5. Case Type (option d) is multi-way selection, choosing one branch among many - not repetition.
6. Hence option (c) is correct.

C10 · COMPUTER APPLICATION AND DATA PROCESSING

Data, Information, Database & DBMS; Frontend/Backend

11 questions · 2018: 0 2019: 0 2020: 0 2021: 2 2022: 2 2023: 2 2024: 2 2025: 3 2026: 0

Weight. 11 questions (6.1% of Computer Application and Data Processing) · appeared in 5 of 9 papers · peak year 2025 (3)

Examiner's pattern. Databases plus file-format literacy. The hierarchical (tree) data model was asked in both 2021 and 2024. Also: DBMS advantages, SQL capabilities, ACID properties, client-server architecture, and classifying raster image, audio and multimedia formats.

Must-know. ACID = Atomicity, Consistency, Isolation, Durability (security is NOT one of them). Hierarchical model = tree; network model = graph. Raster: JPEG, GIF, PNG, BMP. Vector: SVG. Audio: WAV, MP3, WMA (TIFF and MPEG-video are not audio).

1. 2021 · Q51 Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

Which DBMS data model arranges data files in a tree structure?

OPTIONS

(a) Network model

(b) Relational model

(c) Hierarchical model

(d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(c) Hierarchical model

EXAM SHORTCUT (AI-derived explanation)

A tree structure with one parent per child is the HIERARCHICAL data model.

TIPS & TRICKS (AI-derived explanation)

- Model shapes: hierarchical = tree (one parent, many children); network = graph (many parents allowed); relational = tables linked by keys.
- The hierarchical model supports only one-to-many relationships; representing many-to-many requires the network model, which is the key distinction between options (a) and (c).
- IBM's IMS is the classic hierarchical DBMS; CODASYL/IDMS is the classic network DBMS; almost all modern systems are relational.
- Navigation in a hierarchical database always starts at the root and follows parent-child pointers - hence its speed for tree-shaped data and its rigidity otherwise.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. In the hierarchical data model, records are organised as a tree: each child record has exactly one parent, and there is a single root.

2. Data files are therefore arranged in a tree structure and are navigated from the root downwards through parent-child links.
3. The network model generalises this to a graph in which a record may have several parents, so it is not a tree.
4. The relational model stores data in flat tables related by keys, with no inherent hierarchy at all.
5. Hence the tree-structured model is the hierarchical model, option (c).

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2. 2021 · Q60 Database, Information & DBMS · Factual

QUESTION

Which is *not* a Raster image format?

OPTIONS

(a) JPEG

(b) TIFF

(c) GIF

(d) SVG

ANSWER (AI-derived — no official key exists for this paper)

(d) SVG

EXAM SHORTCUT (AI-derived explanation)

JPEG, TIFF and GIF store a grid of pixels (raster). SVG stores mathematical shape descriptions and is a VECTOR format.

TIPS & TRICKS (AI-derived explanation)

- The acronym gives it away: SVG = Scalable Vector Graphics. 'Vector' is in the name.
- Raster images are made of pixels and lose quality on enlargement; vector images are made of paths and scale without any loss - that is the practical distinction.
- Raster formats to recognise: JPEG, PNG, GIF, TIFF, BMP. Vector formats: SVG, EPS, AI, and PDF for the vector parts of its content.
- Photographs are naturally raster; logos, icons and diagrams are best kept as vectors, which is why SVG dominates on the web for those.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A raster (bitmap) image stores a rectangular grid of pixels, each with its own colour value; resolution is fixed and enlargement causes pixellation.
2. JPEG is a lossy raster format designed for photographs.
3. TIFF is a flexible, usually lossless raster format used in printing and archiving.
4. GIF is a raster format limited to 256 colours and supporting simple animation.
5. SVG stands for Scalable Vector Graphics: it describes images as mathematical paths, curves and shapes in XML, so it can be scaled to any size without loss of quality.

6. Hence SVG is not a raster image format, option (d).

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3. 2022 · Q36 Database, Information & DBMS · Factual

QUESTION

Which of the following are raster image formats? 1. JPEG, 2. GIF, 3. PNG, 4. SVG

OPTIONS

(a) 1, 2 and 3

(b) 1, 2 and 4

(c) 1, 3 and 4

(d) 2, 3 and 4

ANSWER (AI-derived — no official key exists for this paper)

(a) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

JPEG, GIF and PNG all store grids of pixels (raster). SVG stores mathematical paths and is a vector format.

TIPS & TRICKS (AI-derived explanation)

- SVG = Scalable Vector Graphics - the word 'vector' is in the name, so any option containing item 4 can be struck out immediately.
- Raster formats to recognise: JPEG, PNG, GIF, TIFF, BMP. Vector formats: SVG, EPS, AI.
- Raster images pixellate when enlarged; vector images redraw perfectly at any size - the practical test of which family a format belongs to.
- This repeats 2021 Q60 in the opposite direction ('which is NOT raster'), so one fact answers items in two papers.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A raster (bitmap) image is stored as a rectangular grid of pixels, each with its own colour value.
2. Item 1, JPEG: a lossy raster format designed for photographs. RASTER.
3. Item 2, GIF: a raster format limited to 256 colours, supporting simple animation. RASTER.
4. Item 3, PNG: a lossless raster format supporting transparency. RASTER.
5. Item 4, SVG: Scalable Vector Graphics describes images as XML-encoded paths, curves and shapes, so it is a VECTOR format and scales without loss.
6. Hence items 1, 2 and 3 are raster formats, option (a).

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4. 2022 · Q80 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Factual

QUESTION

Which properties of the transaction mechanism in a database system are correct? 1. Atomicity, 2. Durability, 3. Isolation, 4. Security

OPTIONS

(a) 1, 2 and 3

(b) 1, 2 and 4

(c) 1, 3 and 4

(d) 2, 3 and 4

ANSWER (AI-derived — no official key exists for this paper)

(a) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

Transactions obey the ACID properties: Atomicity, Consistency, Isolation and Durability. Security is not one of them, so items 1, 2 and 3 are correct.

TIPS & TRICKS (AI-derived explanation)

- ACID is the mnemonic: Atomicity, Consistency, Isolation, Durability. Any item outside those four is a distractor.
- Consistency is the member NOT listed among the four items here, and Security is the intruder - identify the intruder and the answer follows.
- Meanings: Atomicity = all or nothing; Consistency = the database moves from one valid state to another; Isolation = concurrent transactions do not interfere; Durability = committed changes survive failures.
- Security is a broader access-control concern handled by authentication and authorisation, not by the transaction manager.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A database transaction is required to satisfy the four ACID properties.
2. Item 1, Atomicity: a transaction is indivisible - either all of its operations take effect or none do, with rollback on failure. CORRECT.
3. Item 2, Durability: once a transaction commits, its effects survive subsequent system crashes, being recoverable from the $\log$. CORRECT.
4. Item 3, Isolation: concurrently executing transactions do not interfere with one another; each sees a state as if it ran alone. CORRECT.
5. Item 4, Security: protecting data against unauthorised access is achieved by authentication and authorisation mechanisms, not by the transaction manager, and is not one of the ACID properties. INCORRECT.
6. Hence items 1, 2 and 3 are the transaction properties, option (a).

5. 2023 · Q62 Database, Information & DBMS · Factual

QUESTION

Which of the following are audio file formats? 1. MPEG 2. WAV 3. TIFF 4. WMA Select the correct answer using the code given below:

OPTIONS

(a) 1, 2 and 3

(b) 1, 2 and 4

(c) 1, 3 and 4

(d) 2, 3 and 4

ANSWER (AI-derived — no official key exists for this paper)

(b) 1, 2 and 4

EXAM SHORTCUT (AI-derived explanation)

MPEG, WAV and WMA are audio (or audio-carrying) formats; TIFF is an image format.

TIPS & TRICKS (AI-derived explanation)

- TIFF = Tagged Image File Format - the word IMAGE is in the name, so any option containing item 3 is out at once.
- Audio formats to recognise: MP3 (MPEG-1 Layer 3), WAV, WMA, AAC, FLAC, OGG.
- Image formats: JPEG, PNG, GIF, TIFF, BMP, SVG. Video: MP4, AVI, MOV, MKV.
- MPEG is a family of compression standards covering both audio and video; MP3 audio is its best-known member, which is why it counts here.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Item 1, MPEG: the Moving Picture Experts Group standards cover audio as well as video, and MPEG-1 Audio Layer $3(MP3)$ is the most widely used audio format of all. AUDIO.
2. Item 2, WAV: the Waveform Audio File Format, the standard uncompressed audio container on Windows. AUDIO.
3. Item 3, TIFF: the Tagged Image File Format, a raster IMAGE format used in printing and archiving. NOT audio.
4. Item 4, WMA: Windows Media Audio, Microsoft's compressed audio format. AUDIO.
5. Hence items 1, 2 and 4 are audio formats, option (b).

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6. 2023 · Q71 Database, Information & DBMS · Conceptual

QUESTION

Consider the following statements in respect of SQL: 1. SQL can execute queries against a database. 2. SQL can update records in a database. 3. SQL query automatically eliminates duplicates. Which of the statements given above are correct?

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(a) 1 and 2 only

EXAM SHORTCUT (AI-derived explanation)

SQL can certainly query and update, but it does NOT remove duplicates automatically - the DISTINCT keyword is required.

TIPS & TRICKS (AI-derived explanation)

- SQL SELECT returns ALL matching rows, duplicates included; DISTINCT must be requested explicitly. That is the false statement here.
- The SQL sub-languages: DDL (CREATE, ALTER, DROP), DML (SELECT, INSERT, UPDATE, DELETE), DCL (GRANT, REVOKE). Statements 1 and 2 are DML capabilities.
- Contrast with relational algebra, where the projection operator DOES eliminate duplicates by definition - the divergence between theory and SQL is exactly what is being tested.
- UNION removes duplicates by default while UNION ALL keeps them - a useful reminder that duplicate handling in SQL is always explicit.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: SQL's SELECT statement executes queries against a database, retrieving rows that satisfy stated conditions. TRUE.
2. Statement 2: SQL's UPDATE statement modifies existing records, and INSERT and DELETE add and remove them. TRUE.
3. Statement 3: a SQL query returns every row satisfying the WHERE clause, including duplicate rows. To eliminate duplicates the programmer must write SELECT DISTINCT.
4. This differs from relational algebra, where projection removes duplicates automatically - SQL works with multisets rather than sets. FALSE.
5. Only statements 1 and 2 are correct.
6. Hence option (a) is correct.

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7. 2024 · Q57 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Conceptual

QUESTION

Consider the following statements regarding Database Management System (DBMS): I. It helps the user in maintaining data integrity while performing the various operations on the database values. II. It helps the user in maintaining consistency in the information stored in the database. III. It allows the user to maintain the security of the confidential information by password-protecting the database. Which of the statements given above are correct?

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER **(AI-derived — no official key exists for this paper)**

(d) I, II and III

EXAM SHORTCUT **(AI-derived explanation)**

Data integrity, consistency and security are three of the standard advantages of a DBMS over file-based storage, so all three statements hold.

TIPS & TRICKS **(AI-derived explanation)**

- Standard DBMS advantages: controlled redundancy, data consistency, integrity constraints, security, concurrent access, backup and recovery, data independence.
- Integrity is enforced through constraints (primary keys, foreign keys, check conditions) that the DBMS validates on every operation.
- Consistency follows from controlled redundancy: because a fact is stored once, it cannot disagree with itself across files.
- Security is enforced through authentication, user accounts and privilege grants - password protection being the simplest form.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. A Database Management System sits between users and the stored data and enforces rules that a plain file system cannot.
2. Statement I: integrity constraints - primary keys, foreign keys, domain and check constraints - are validated by the DBMS on every insertion, update and deletion, so invalid data cannot enter. TRUE.
3. Statement II: by controlling redundancy so that each fact is stored once, and by propagating updates centrally, the DBMS keeps the stored information consistent. TRUE.
4. Statement III: the DBMS provides authentication and authorisation, allowing access to be restricted by user account and privilege, of which password protection is the basic form. TRUE.
5. All three are standard advantages of a DBMS.
6. Hence option (d) is correct.

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8. **2024 · Q60** Network - LAN, WAN, Internet, Intranet · Factual

QUESTION

In which of the following data models are the records arranged in the form of a tree?

OPTIONS

(a) Entity model

(b) Relational model

(c) Hierarchical model

(d) Network model

ANSWER **(AI-derived — no official key exists for this paper)**

(c) Hierarchical model

EXAM SHORTCUT **(AI-derived explanation)**

Records arranged as a tree, each child having exactly one parent, define the HIERARCHICAL data model.

TIPS & TRICKS **(AI-derived explanation)**

- Model shapes: hierarchical = tree (one parent per child), network = graph (many parents allowed), relational = tables joined by keys.
- The hierarchical model supports only one-to-many relationships; many-to-many requires the network model - the key discriminator between options (c) and (d).
- IBM's IMS is the classic hierarchical DBMS and CODASYL/IDMS the classic network one; almost all modern systems are relational.
- The entity(-relationship) model is a DESIGN notation used before implementation, not a storage model - which places option (a) in a different category.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. In the hierarchical data model, records are organised as a tree: there is a single root, and every child record has exactly one parent.
2. Navigation proceeds from the root downwards along parent-child links, which makes access fast for naturally tree-shaped data but rigid otherwise.
3. The network model generalises this to a graph in which a record may have several parents, so it is not a tree.
4. The relational model stores data in flat tables related by keys, with no inherent hierarchy.
5. The entity model (entity-relationship diagram) is a conceptual design notation rather than a storage structure.
6. Hence records are arranged as a tree in the hierarchical model, option (c).

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9. 2025 · Q40 Computer Security, Virus, Antivirus, Firewall, Spyware, Malware · Conceptual

QUESTION

In a client-server DBMS architecture, which is primarily responsible for managing database access, security, and data integrity?

OPTIONS

(a) Query processor

(b) Client

(c) Server

(d) Database administrator

ANSWER **(AI-derived — no official key exists for this paper)**

(c) Server

EXAM SHORTCUT **(AI-derived explanation)**

In client-server DBMS architecture the SERVER holds the database and enforces access control, security and integrity constraints; the client only issues requests and displays results.

TIPS & TRICKS **(AI-derived explanation)**

- Split the roles: the client provides the user interface and sends queries; the server stores the data and enforces all the rules.
- Centralising security and integrity on the server is the whole point of the architecture - clients cannot be trusted to police themselves.
- The query processor is a COMPONENT of the server software, not the responsible entity - a part-versus-whole distinction.
- The database administrator is a PERSON who configures policies; the question asks which architectural component enforces them at run time.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. In client-server DBMS architecture the workload is divided between two tiers.
2. The client provides the user interface, formulates requests and displays the results returned to it.
3. The server hosts the database itself and runs the DBMS engine: it authenticates users, enforces access privileges, validates integrity constraints, manages concurrency and performs backup and recovery.
4. Centralising these functions on the server is what allows a consistent security and integrity policy to be applied to every client.
5. The query processor is one component within the server software, and the database administrator is a person who defines policy rather than an architectural component.
6. Hence the server is primarily responsible, option (c).

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10. 2025 · Q73 Operations of a Computer & Basics · Conceptual

QUESTION

Which are characteristics of virtual reality? I. It is created using multimedia. II. It enables users to move and react in a computer-simulated environment. III. It stimulates senses like sight, hearing, and touch.

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(d) I, II and III

EXAM SHORTCUT (AI-derived explanation)

All three statements are standard textbook properties of virtual reality - it is multimedia-based, interactive within a simulated environment, and multi-sensory. Answer (d).

TIPS & TRICKS (AI-derived explanation)

- In UPSC computer-awareness sets, when every listed statement is a benign textbook property, 'all of the above' is usually right; look for a statement that is factually wrong before rejecting it.
- VR is defined by three $I's$: Immersion, Interaction, Imagination - which map onto statements III, II and I respectively.
- VR uses multimedia elements (3-D graphics, audio, animation, sometimes haptics) as its building blocks, so statement I is true by construction.
- Contrast with augmented reality, which overlays computer-generated content on the real world rather than replacing it.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement I: virtual reality environments are built from multimedia components - 3-D graphics, animation, sound, video and haptic feedback. True.
2. Statement II: the defining feature of VR is interactivity - the user moves within and reacts to a computer-simulated environment in real time. True.
3. Statement III: VR stimulates multiple senses - sight through head-mounted displays, hearing through spatial audio, and touch through haptic gloves or controllers. True.
4. All three statements are correct.
5. Hence option (d) is correct.

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11. **2025 · Q74** Operations of a Computer & Basics · Definition

QUESTION

Multimedia combines different media types. How many of the following are part of multimedia? I. Text II. Video III. Audio IV. Graphics and Images V. Animation

OPTIONS

(a) Only two

(b) Only three

(c) Only four

(d) All five

ANSWER (AI-derived — no official key exists for this paper)

(d) All five

EXAM SHORTCUT (AI-derived explanation)

The standard definition lists exactly five media types: text, graphics/images, audio, video and animation. All five listed items qualify. Answer (d).

TIPS & TRICKS (AI-derived explanation)

- Mnemonic for the five elements of multimedia: TAGVA - Text, Audio, Graphics, Video, Animation.
- 'Multi' + 'media' means many media, so a definition question of this shape rarely excludes a genuine media type.
- Distinguish video (captured motion of real scenes, frame sequences) from animation (synthetically generated motion) - both count separately.
- Interactivity is often added as a sixth defining attribute of multimedia systems, but it is not itself a media type.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Multimedia is defined as the integration of two or more media types in a single application.
2. The universally listed constituent media are: text, graphics and images, audio (sound), video, and animation.
3. Checking each listed item: I Text - yes; II Video - yes; III Audio - yes; IV Graphics and Images - yes; V Animation - yes.
4. All five are part of multimedia.
5. Hence option (d) is correct.